

Echoes Across the Sea: Reassessing Japonic Prehistory through Austronesian Comparanda

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Abstract

Investigations into proto-Japonic and proto-Austronesian etyma reveal a highly systematic set of phonetic and semantic correspondances. Several words are manually compared across the Japonic and Austronesian languages from the *Oxford-NINJAL Old Japanese Corpus*, *Eastern Old Japanese Dictionary*, *Online Dictionary of Japonic Languages*, *Nikolay Nevskiy's Miyakoan dictionary* and the *Austronesian Comparative Dictionary*, including data from *Baxter-Sagart Old Chinese Reconstructions*. Both languages heavily adhere to: bisyllabic phonotactics with medial nasal-obstruent clusters; regular approximations of non-Japonic phonemes into existing proto-Japonic phonemes of similar features; numerous lexical items relating to food, family, and borrowing-resistant vocabulary; sets of reduplicated root to stem correspondances; as well as grammatical parallels. From this data, previous scholarship, historical, and genetic data, it can be gleaned that Austronesian peoples must have arrived around the Late Jōmon Period in Kyūsyū and/or the southern Korean peninsula. These linguistic discoveries may elucidate this otherwise mysterious period of migration in Japonic prehistory.

Keywords

proto-Japonic – Old Japanese – Austronesian – Malayo-Polynesian – etymology – language contact

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I dedicate this manuscript to my partner, whose support was indispensable to the completion of this manuscript.

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1 Previous Scholarship

Although the idea that Japonic could have an Austronesian adstratum or descendance is not new and has been justly criticized due to flawed use of the Comparative Method, this paper will push forward the former of these theories with many unique comparisons and a new regular set of phonetic correspondances between proto-Japonic and proto-Austronesian and proto-(Western-)Malayo-Polynesian etyma. Existing hypotheses will be explored for their flaws and merits to allow for comparison between those hypotheses and the one presented here.

Murayama is one of the earliest scholars to publish about potential Austronesian connections in Japonic, even observing some parallels within the dataset presented here. Martin (1994. p.101), however, reviews this hypothesis saying, "the unduly large number of Japanese examples proposed as having Malayo-Polynesian cognates that are either hapax legomena in the Old Japanese corpus, and hence of anything but certain semantic significance, or else imaginative, interpretative etymologies of god-names and other highly specialized lexical items concerning the original, precise meaning of which very little probably ever can (or should) be said". That said, Murayama cites several valid examples: a proto-Austronesian lemma *gigi (Murayama, 1988. p.218), better replaced by 3.1.15 pMP *kitkit, which becomes WOJ *ki* 'canine tooth' in NJ *kiba* 'fang'; pAN *bilah (id. p.207) becoming WOJ *pira* 'flat' (EOJ 'flat, ordinary'), replaced here by pWMP 3.2.21 *bilaq 'part split off' to become NJ *hira* 'counter for thin, flat things (petals, leaves, etc.)', instead 3.2.24 pWMP *bidan 'unit of measure for cloth, sails, etc.' is treated here as the source for WOJ *pira*; including a surprising but unfortunately flawed comparison pAN *bijak (id. p.195), which should be 3.2.93 pWMP *pihak/piqak, becoming the exceptionally common Japonic word *pye* 'counter for layers; place; direction; shore', which even Benedict and other scholars did not seemingly discover; Javanese *sila* 'radiance' and Melanesian *zila* 'shine (of a heavenly body)' (id. p.41) with WOJ *sira*-/sirwo 'white', replaced here by 3.2.129 pMP *silaw 'dazzling, of light; and lastly, compares Indonesian *galap* 'dark' (id. p.35) with WOJ *kura*-/kurwo, replaced here by 3.2.60 pPh *kulap 'dim, not well-lit'.¹ The observation of two very primary color terms in Japonic should not have gone neglected but were ignored due to unsound correspondances (pAN *ə :: pJ *u?). This dataset improves several comparisons by using pAN/pMP terms that do safely correspond (pPh *kula(p) :: pJ *kura-). Interestingly among the comparisons is pMP *apuy (sic; should be pMP *hapuy) which closely resembles Ainu *ape* 'fire' which Vovin (1993) reconstructs as *apOy.

Kawamoto, T. (1977), as observed by Vovin (1994), claims that "seven PAn phonemes: *d, *j, *l, *r, *R, *z, and *Z may correspond to OJ /s/ or /r/ (without specification of reflexes) (Kawamoto 1997:2)". Needless to say, this would severely violate the Comparative Method and will not be further discussed or analyzed.

Benedict's *Japanese/Austro-Tai* (1990) compiles a list of Austronesian Japonic cognates with ad-hoc rules such as canonical right-reduction: "COOK/ROAST P-Austro-Japanese [p-AJ]

¹ Murayama also compares pAN *buja 'flower' to pJ *pana 'id.' but there is a clear problem with the vocalism. He does include a Burmese example *pan* 'id' which certainly holds up, especially with epenthesis to conform to a bisyllabic phonotactic structure (Burmese *pan* > pJ *pan.a (flower-EP)).

*talak... Jp. yak-i 'burn, roast, bake, broil' (p.177)," with total loss of p-AJ *ta to yield *lak > pJ *yak. Many of the cognates do not have an entry in the ONCOJ and are not considered, such as "BOIL/: *luwag > wak-i" (p.27)". A thorough re-examination of this work may be beneficial but would require extensive knowledge and records of Austronesian descendants which I do not possess. For example, "BORE/GIMLET P-Austro-Japanese *girik P-Austronesian- *girik: P-Hesperonesian id. 'bore'. Jp. kiri, OJ kyiri = kiri 'gimlet/awl'. Ryukyuan: Shuri ?iri, Yonaguni iri 'id.', perhaps from *giri (Martin 1979), (p.171)" which does not cite any Austronesian words to support *girik, but would fit the sound change rules proposed here (initial devoicing, liquid approximation, and coda loss). Vovin (1994) explores at length how Benedict and Kawamoto lack regular phonetic correspondence, present lexical and semantic ghosts, include flawed reconstructions and segmentations. All this said, those comparisons which were observed there first are cited within their respective entries.

Kumar (2000) has also published comparisons between Old Javanese and Japanese to support her view that Yayoi period immigrants were elites from ancient Indonesia. Unfortunately, Bentley (2011) has observed that there are lexical ghosts, such as for Javanese *matur* 'to celebrate' that do not appear in the Indonesian dictionary Kumar cites, "[e]xamining the entries in P. J. Zoetmulder's Old Javanese-English Dictionary, we find that the form is actually OV hatur- or atur-. There are variations of this such as makatur-atur or mahatur-hatur." Although most of her comparisons are flawed, there are traces of plausible comparisons but not with Javanese. Javanese: *sosok* (#3.2.46), *sawa* (#3.2.110), *paḍaṅ* (#3.2.80), *lěsuṅ* (#3.2.71), *gilaṅ-gilaṅ* (#3.2.40), *tapih* (#3.2.27) all have valid comparisons between proto-Japonic and branches of proto-Austronesian. The rest of her data may be individually analyzed by a scholar of Old Javanese to avoid flawed comparisons, but this falls outside the scope of this examination. Her archeological evidence will be explored in more detail in section 5.2.2.

A more recent attempt by Itabashi, Y. (2011) correlates voiced consonants to pitch accent in Japonic but includes many questionable comparisons such as: "OJ tana (HH) 'heaven, sky' < PJ *tana (HH): PWMP *lanit (p.15)," without including an explanation for how pWMP *-it would become pJ *a. Unfortunately, no correspondence between these features was observed within the dataset presented here, and so Itabashi's comparanda are not included.

Kupchik (2021) finds a number of celestial vocabulary in common with Austronesian languages, such as: OR *tēda* (< pR *tēda) 'sun; (king, Okinawan)' with Amis *cidal* //tsidal/ 'sun', OR *sina* 'sun' with pMP *sinaR 'ray of light, sunbeam', OR *sino* with Eastern (Old) Javanese *sinar* [sinɔr] ~ [sinɔr] 'light, radiance', *kawa* 'heaven' with pAN *kawaS 'sky, heaven, season, year', and OR *ora* 'moon' and Amis *folad* 'moon'. He also cites Kerr (1958), who notices North Ryukyuan and Formosan similarities: thatched huts, communal storehouses, stone embankments that surround individual homes, women carrying heavy loads with support bands on the head, women tattooing their fingers but not the face, and a similar hairstyle called *katakashira*. Kupchik concludes, "[i]t remains unclear as to how the Amis came in contact with Proto-Ryukyuan speakers, but the linguistic evidence indicates this contact occurred in the time of PR. Were Amis travelling to the Northern Ryukyus or Southern Kyushu for trade 900–1300 years ago? Was there an Amis diaspora somewhere in these regions during this time? At present,

I cannot provide anything more than speculation to answer such questions, but future research into potential Amis loanwords in other areas of the OR lexicon may offer further clues.” The link between Amis and the Malayo-Polynesian branch of Austronesian will be explored in 5.2.1., and is treated as the most likely vector of Austronesian in the Japanese archipelago as well.

Vovin (2021) also finds several words in common with Austronesian languages: OJ *tara-* ‘blood (in makura kotoba) < pAN *daRaq ‘blood’; OJ *taratisi* ‘blood and flesh’ < *tara-t-isi < pAN *isi ‘flesh (of humans, animals, fruits, tubers), *t- honorific; OJ *taratinay* ‘blood mother’ < *tara-t-ina-y < pAN *ina ‘mother’ and -y (form of address); OJ *nama yami-nə* ‘deep river darkness’ < pMP *namaw ‘sheltered water, deep place in a river, cove, harbor, lagoon’; OJ *take* ‘peak’ < proto-Batanic *takəi ‘mountain, wilderness’; OJ *a(-si)* ‘foot’ < Batanic *ai* < pAN *qaqay ‘foot, leg’; OJ *isu/isa* ‘whale’ < Amis *Piso* ‘whale’; OJ *-sa* ‘dative-locative case marker’ < pAN *sa ‘locative marker’ (Phillipines) ; OJ *i-* ‘directive-locative focus verbal prefix i-’ < Amis *i-* locative. The inclusion of a fossilized kinship term is very striking as it demonstrates the Austronesian stratum is quite ancient and its appearance with an old word for blood to differentiate one’s birth mother hints at an admixed kinship system with Austronesian peoples. For more body and other basic terminology, see section 4.3. Vovin’s observation that “Yami *takəi* ‘mountain’ looks suspiciously more like PJ *takay rather than its descendant forms [...] Proto-Batanic *takəi was apparently borrowed into insular PJ as *takay due to Arisaka’s constraint on the combination of /a/ and /ə/ within the same morpheme,” and the fact that “in WOJ *take* < PJ *takay functions as a single morpheme and does not have a compounding form taka- resulting from a *-y loss in preconsonantal position. Cf. WOJ *take* < PJ *takay ‘bamboo’, for which both a free form *take* and a compounding form *taka-* are well attested,” serve as the basis for two sound changes: schwa assimilation and coda retention. Vovin posits the Batanic languages as the source of borrowing because “*takəy* ‘mountain’, ‘wilderness’ is not attested anywhere else in the Austronesian language family,” which could still have had an earlier lost source in pAN or pMP, although this cannot be ascertained. Considering the importance of Amis and Batanic languages, it can be assumed that the migration into Japan occurred as a result of the Austronesian expansion (~4-5kya), likely before the Yayoi period.

2 Methodology

2.1 Phonological Correspondances

Several correspondances between pAN (and descendant branches) are proposed that demonstrate remarkable consistency across the dataset,

- 1) pAN *N /ɤ/, *ñ /ɲ/, *ŋ > pJ *n (nasal approximation)
- 2) pAN *l, *R /r/ > pJ *r /r/ (liquid approximation)
- 3) pAN *C /ts/, *c /tʃ/ > pJ *t/s (affricate approximation)
- 4) pAN *s /s/ > pJ *s (sibilant approximation)
- 5) pAN *h, *q > pJ *ø (guttural deletion)
- 6) pAN *e /ə/ > pJ *a [/C(-labial)] (schwa change)

- 7) pAN *e /ə/ > pJ *ə [/C(+labial)_, _C(+labial)] (labial schwa retention)
- 8) pAN *e /ə/ > pJ *a [/CaC_] (Arisaka's Law, schwa assimilation)
- 9) pAN *a > pJ ə [/CəC_] (Arisaka's Law, open vowel assimilation)
- 10) pAN *d > pJ *r [/V_V] (lenition)
- 11) pAN *b > pJ *p; pAN *g > pJ *k [/#CV, /VCV]
(devoicing (of voiced stops outside nasal clusters))
- 12) pJ /j/ is the only final consonant allowed (coda retention)
- 13) pJ no initial liquids are allowed, (pAN *l, r, R > pJ *ø [/#_]) (initial liquid loss)
- 14) pAN nasal clusters (NC) become pJ prenasalized voiced consonants (prenasalization)
- 15) Unreduplication, borrowing the un-reduplicated form of a word as a monosyllable

Many comparisons involve monosyllabic segments, however, the probability of a CVC segment in Austronesian matching a Japonic one by chance is very low at 0.2%.² Considering proto-Japonic already has a small phonemic inventory, these segments matching phonetically and semantically suggests inheritance, not coincidence. Additionally, most of the monosyllabic segments in Japonic match doubled roots in Austronesian and have consistent sound correspondences, supporting Austronesian as the donor language for the presence of these roots in Japonic. Many of these roots also have an Old Chinese parallel, perhaps indicating a shared common origin, although almost certainly areal. Some examples also appear to have fossilized consonantal codas, especially due to morphological conditioning from agglutination (ex. *pakpak > *pa(k) 'wing' (noun), *saksak > *sak-u, *sak-i 'cut' (verb)).

Bisyllabic lemmas are the majority of loans. Many of them support an origin of word-medial nasal obstruent clusters from Austronesian (pAN *(C)VNCVC :: pJ *(C)VNCV). The remaining characteristics show modifications expected of Japonic phonotactics such as: a prohibition on voiced initial consonants (ex. pAN *b/#_ and *p/#_ :: pJ *p/#_); loss of liquid initials (ex. pAN *l/#_ and *r/#_ :: pJ *ø); loss of consonantal codas except *y; and Arisaka's law conditioning CaCə and CəCa sequences into CaCa and CəCə, respectively. These examples also retain some fossilized final codas (ex. *sopot > WOJ sopot-i, *amat > WOJ amat-a). Correspondences are listed with their degree of fit, labeling relevant changes across both languages.

2.2 *Etymology Sources*

The pAN etymologies are available thanks to the Austronesian Comparative Dictionary (Blust & Trussel, 2013), which catalogues words across Austronesian languages under their collective cognate sets. The pJ reconstructions are my own, and idiosyncratic in some respects, hence any faults here are entirely my own. Reconstructions from Samuel Martin's monograph "The Japanese Language through Time (1987) and Francis-Ratte's dissertation "Proto-Korean-Japanese: A New Reconstruction of the Common Origin of the Japanese and

² This is calculated by the following: possible initial consonants in proto-Japonic (1/8 = 12.5%) x possible vowels (1/7 = 14.3%) x possible consonants (1/9 = 11.1%).

Korean Languages” (2016) are included for reference and comparison against my own reconstructions. Unique segmentation will also be supported by attested paradigmatic morphology across descendant lexemes.

These etyma also collate several descendant terms in the Oxford-NINJAL Corpus of Old Japanese (Frellesvig, Horn, et al, 2023) (ONCOJ), the Eastern Old Japanese Corpus and Dictionary (Vovin & Ishisaki-Vovin, 2022), the Online Dictionary of the Japonic Languages (Salvatore et al, 2024), and Nikolay Nevskiy's Miyako dictionary, finished and edited by Aleksandra Jarosz (Jarosz, 2015). Each pJ entry compared to pAN will be accompanied, when possible, by its Western Old Japanese cognate, Eastern Old Japanese cognate, and an Okinawan (North Ryukyuan) and Miyako (South Ryukyuan) cognate. Any attempt at discerning which etyma extend to proto-Japonic rather than proto-Japanese is outside the scope of this paper. Each pAN etymon will also be accompanied by a modern Austronesian (or Old Javanese) cognate, usually Ilokano, Malay, or Tagalog, which have the most cognates in common.

3 Dataset

3.1 Reduplicated Roots

A large number of Austronesian-Japonic parallels involve reduplicated roots in Austronesian that are simplified into monosyllabic stems in Japonic. Some of these may appear to have been borrowed as bisyllables with reduced consonant clusters ($CVC_1-C_2VC > CVC_2VC$ (e.g. 3.1.53 pAN *tuktuk > WOJ *tutuk*), but more likely, they were borrowed as monosyllables that experienced reduplication in Japonic after being borrowed. Javanese *tutuk* ‘mouth [Krama Inggil speech level]’ is the only descendant that reduced the medial consonant cluster (e.g. Yami *toktok* to peck, Bontok *tuktuk* ‘the beak of a bird; to peck’). The example #54 WOJ *tuk* ‘pound’ but EOJ *tutuk* ‘to hit’, shows this reduced reduplication in process (pJ *tuk > tuktuk# > pJ *tu(k)-tuk). Interestingly, quite a few of these etyma have phonetically and semantically similar monosyllabic Old Chinese parallels, which lend further credence to the idea of these reduplicated syllables in Austronesian existing in a monosyllabic stem form in Japonic.

1. pMP *barbar ‘spread out, unfurl’ > pJ *par ‘to spread/stretch’ > WOJ *par* ‘stretch’, EOJ *par* ‘to put forth, to stretch’
 pJ *par-p? (‘spread-ITER’) > WOJ *pap* ‘stretch (i), crawl’, EOJ *pap* ‘to crawl’
 Okinawan (Iheya) [*hain* ‘to stretch’
 Miyako (Hirara) *pa:z* ‘stretch’
 Miyako (Irabu, Sawada) *pal* ‘stretch’
 Malay (MP) *babar* ‘expansion, opening out (of booming out a sail, spreading road-metal over the roadway, etc.)’
 Sound Changes: Devoicing
 Synonyms: WOJ *nobe* ‘extend (t)’, NJ *nobiru* ‘to stretch, to extend, to lengthen’

The analysis of *pap* as *par-p links the close phonetic form with their shared meaning of ‘stretch’. Consonantal finals have been posited by Frellesvig (2021. p.42) as being replaced by the second consonant in a cluster (e.g. WOJ *nar* and WOJ *nas* < pJ *nar-s).

Martin (1987. p.685) reconstructs “haru (A) < faru < *para- (? *pa-ra-; cf. hayu, haeru). ‘stretch/paste it’,” segmenting *pa- instead of *par. Note that attaching -ra- (NTR, -(a)r in this paper) would not result in the transitive *par*. Instead, I posit a doublet pJ *pay ‘to grow’ from alteration of *y* and *r*, which is also seen in “the imperatives of [EOJ] vowel base verbs attach -ro, as opposed to central OJ yo.” (Frellesvig, 2010. p.154). Having these two forms makes the bigrade forms of both morphemes result from directly attaching WOJ *eru* ‘to get’ (< pJ *ay) instead of from the lenition of OJ *e* > *y*). WOJ *pare* ‘swell’ < pre-OJ *par-e (spread-get) < pJ *par-ay and WOJ *paye* ‘grow’ < pre-OJ *pay-e < pJ *pay-ay; compared to WOJ *okwi* < pre-OJ *oko-y < oko-e# < pJ *əkə-ay. Suffixing *eru* after pJ *ə > OJ *o* would explain why *okwi* is the result and not *oke* from *əkə-y, *əkə-ay, nor *ək-ay.

Francis-Ratte (2016. p.395) cites Whitman (1985) MK *polo*- ‘pastes, spreads it’ ~ OJ *par*- ‘pastes, spreads it’ to reconstruct pKJ *pəra-, which is less phonetically close but still possibly related; perhaps pMP *barbar spread to proto-Koreanic and proto-Japonic separately if pJ did not loan the word to pK. OC 蕃 *[b]ar ‘luxuriant (growth)’, OC 繁 *[b]ar ‘abundant, numerous’, and OC 播 *pʰar-s ‘to sow’ show how *barbar may be related in an agricultural sense, like 18th c. English *broadcast* ‘spreading seeds’. Curiously, pIE *bhel-³, bhlē ‘to grow, spread, swell’ (Pokorny, 2007. p.364) has a similar phonetic and semantic shape, perhaps as a long-range result of the language-farming dispersal hypothesis. See also pAA *be:l ‘spread a mat (v.)’ (Sidwell, 2024. p.xvi) and proto-Dravidian *par-a ‘to spread’ (Krishnamurti, 2003. p.529).

2. pMP *bekbek ‘fine particles, powder which results from pounding grains, etc; pulverize grains’ > pJ *pək ‘dust/crush to powder’
 pJ *pək.ə-r-i (crush.EP-NTR-NMZ) > NJ *hokori* ‘dust’
 pR *pok-i (dust-NMZ) > Miyako (Hirara) *puki* ‘dust, dandruff’ (Jarosz, p.621)
 Kankanaey (MP) *bekbek* ‘to beat; to break; to cut to pieces (something more or less hard)’
 Sound Changes: Devoicing, Schwa Retention, Epenthesis
 Synonyms: WOJ *tiri* ‘dust’

Martin (1987. p.414) reconstructs *po?-okor[a-C]i (3.1) with an unknown *po segment and NJ *okor* ‘to occur, to happen’, but this would not explain this Ryukyuan alternative. This etymon is part of a set of these reduplicated roots that use an epenthetic echo vowel to attach suffixes, which was posited by Frellesvig (2021. p.43). For example, “yorosi- (adj) < *yər.ə-si” vs. “yorasi (adj) < *yər-asi”. There is an alternative Ryukyuan form without the additional intransitive suffix.

3. pWMP *binbin ‘twist or tweak the ear’ > pJ *pin-ay > NJ *hineru* ‘to twist, to wrench, to turn (a switch on or off, etc.)’
 Cebuano (MP) *binbin* ‘tweak a part of the body with the thumb and forefinger, usually the ears’
 Sound Changes: Devoicing, Nasal Approximation

Synonyms: NJ *nejiru* ‘to twist, to screw, to wrench, to sprain’
 Martin (1987. p.689) reconstructs “hiner-u (B) < finer-u ? *fyiner-u < *pi(-) nera- (< ?). ‘twist; pinch’”.

4. pMP *biRbiR ‘lip’ > pJ *kutuy-ⁿ-pir.u (mouth-GEN-lips.EP?) > NJ *kuchibiru* ‘lips’
 Ilokano (MP) *bibír* ‘lip; more especially underlip’

Sound Changes: Devoicing, Liquid Approximation, Epenthesis?

Martin (1987. p.467) reconstructs “*kuti[y]-byiro (4.2) < kutu-Ci-n-piro,” also segmenting ‘piru’ as a separate morpheme but differing on the identity of the final vowel. The uncertain final is likely to resolve the coda final without loss but is flawed due to lacking other non-echo vowel comparisons. The expected form is *pir.i rather than *pir.u.

5. pMP *budbud ‘sprinkle seed, sow seed in dibble holes’ > pJ *pur ‘shake, sprinkle’ > OJ *pur* ‘rain, fall (precipitation)’, WOJ *pur* ‘shake’
 EOJ *pur* ‘wave’

Miyako (Hirara) *fu:z* ‘to rain’ (Jarosz, p. 538)

Yaeyama (Funauki) *φu:* ‘to rain’

Tagalog (MP) *budbód* ‘sprinkling of particles of powdered substances as salt, pepper, etc.’

Sound Changes: Devoicing, Lenition

Martin (1987. p.695) reconstructs “huru (B) < furu < *pu-ra- ... Cf. huku; husu; huseru; hurasu; ne^b/_muru,” but most of the words included are dubitably related: to blow, to bend down, to lie down, to send (rain), and sleep, respectively. Instead, these are all more likely pJ *puk, *pus, *pur, and possibly *ner-pus-r (?) (sleep-lie.down-NTR).

Tagalog *budbód* may imply that the sense of OJ *pur* ‘shake’ is related. Considering the significance of agriculture related vocabulary, the meaning ‘to fall (of precipitation)’ might be from analogy of rain to planting rice, hence the secondary sense.

6. pMP *bukbuk ‘bubbling, boiling’ > pJ *puk > NJ *fuku* ‘to emit, to spout, to spurt, to boil over’

Malay (MP) *bobok* ‘to gurgle (as a bottle dipped empty into water)’

Sound Changes: Devoicing

Synonyms: WOJ *ni* ‘boil’

Martin (1987. p.689) reconstructs “huku (B) < fuku < ?*puku- < *puka- (the second vowel assimilated to the first, as also in hukureru, hukumu, ...). ‘blow; emit’”. Many of Martin’s reconstructions include an additional verbal -a, which is usually used to explain the bigrade forms of those stems from crasis of pJ *ay > OJ *e*. Just as explained in #3.1.1, the bigrade often arises from directly attaching WOJ *e* to the stem. This hypothesis may explain why the lower bigrade is more common than the upper bigrade, since many of the bisyllabic entries resulting in a final -a also yield lower bigrades (c.f. WOJ *agar* < pJ *anka-r < #3.2.4 pMP *anka(t), WOJ *age* < pre-OJ *anka-e < pJ *anka-ay). Monosyllabic entries therefore rebut the final -a here and henceforth.

7. pMP *cukcuk ‘skewer’ > pJ *tuk ‘poke through’ > WOJ *tuk* ‘poke; pierce’

Okinawan (Iheya) *tēiteiutusun* ‘pierce’

Malay (MP) *cucok* ‘pricking; stabbing; driving a point into anything’

Sound Changes: Sibilant Approximation

Synonyms: WOJ *nuk* ‘pluck, poke’

Martin (1987. p.773) reconstructs “tuku (A) < *tuka- ‘poke, thrust; spit out’”. Francis-Ratte (2016. p.409) reconstructs “pKJ *cuka- ‘thrusts in’,” through comparison with “MK *chó-* ‘kicks’”. See also OC 刺 *[ts^h]ek ‘pierce, stab’ (Baxter-Sagart, 2014.) and pAA *cu:kʔ ‘poke, pierce (v.)’ (Sidwell, 2024. p.xxx). All of these preserve an original /ts/ that became /t/ in Japonic, suggesting /ts/ was phonemic in pJ when *cuk was loaned to OK, later merging with /t/. Benedict (1990. p.254-255) compares “THRUST/: *(ń)tsuk(ń)tsuk > Jp. tuk-i ‘thrust, pierce, spear, prick’”.

8. pWMP *degdeg ‘crowded together, packed in tightly’ > pJ *tak ‘to pack together’ > WOJ *tak* ‘comb and bind up’

pJ *su-ⁿ-tak (nest-GEN-pack together) > WOJ *sudak* ‘(of birds) be many, flock, gather together’

pJ *tak-ar (pack.together-NTR) > NJ *takar* ‘to gather, to crowd round, to swarm, to flock’

Malay (MP) *bər-dədak* ‘to be massed or crowded together, as ships at a good anchorage’

Sound Changes: Devoicing, Schwa Change

Synonyms: WOJ *atumar* ‘collect (i)’

Martin (1987. p.762) reconstructs “takaru (A) < (Nr [Nara?]) takare[y]- < *taka(-)ra-Ci-. ‘swarm, flock, gather; rob; cadge’ ... Cf. sudaku”. The form *takare* Martin cites is more likely from pJ *tak-ar than *taka-r (tall-NTR)ʔ. See also OC 庶 *s-tak-s ‘numerous; commoner’ (Baxter-Sagart, 2014).

9. pWMP *dukduk ‘to sit, be seated’ > pJ *tuk ‘sit, assume role’ > NJ *tuk* ‘to take (seat, position, course, office, etc.), to assume, to be hired, to be employed, to ascend (the throne)’

Tagalog (MP) *mag-luklók* ‘to set on a throne’

Sound Changes: Devoicing

Synonyms: WOJ *suwe* ‘make sit’

Note that Tagalog shares the sense of ‘set on a throne’ and ‘ascend (the throne)’, semantically extending taking a seat as assuming a role.

10. pPh *gidgid ‘rub something off’ > pJ *kir- (use.friction) > MJ? *kir* ‘to make fire’

(<https://kobun.weblio.jp/content/きる>)

pJ *kir.i-kir.i (use.friction-EP-RED [+EMP, +Voice]) > NJ *girigiri* ‘grinding (sound), grating, gnashing’

Tausug (MP) *gidgid* ‘to rub something vigorously against something else, rub something off’

Sound Changes: Devoicing, Lenition

Francis-Ratte (2016) reconstructs “pKJ *kir- ‘scrapes, grinds’,” through comparison with “MK *kól-* ‘whets, grinds,’ MK *kulk-* ‘draws, scrapes’”.

A small group of comparisons involve fire (#3.1.51 *temtem, #3.2.112 *sadab, #3.2.30 *dakél, #3.2.46 *hátuŋ), possibly related to cooking food considering the large number of eating terms.

11. pAN *kaŋkaŋ ‘resounding sound’ > pJ *kan.a-(i)nt(a)-e (make.sound-EP-come.out-get)> NJ *kanaderu* ‘to play an instrument (esp. string instruments)’

Kankanaey (MP) *kaŋkáŋ* ‘to ring; to cause to sound; to clank; to clang; striking a sonorous body’

Sound Changes: Nasal Approximation

Martin (1987. p.703) reconstructs “kanaderu < kanade[y]- < ?*ka[ki]nade[y]- < *kak[a-C]i_B, nanta-Ci_B. ‘dance moving one's hands; grasp (a whip etc.), play (a harp etc.)’”

12. pMP *ke(k)kek ‘to cluck’ > pJ *kak-kiak (cluck-squack) > WOJ *kakye* ‘chicken’

Ilokano (MP) *kekék* ‘the cry of the hen when calling her chicks’

Sound Changes: Schwa Change

Martin (1987. p.434) reconstructs “kake (2.4) (< kakye (?< *kakiCa?; ?< Ch *.ka-.k(y)ey ‘house fowl’),” presumably referring to the Sino-Japanese reading *ka* for 家 and *ke* for 鷄. This explanation is also phonetically sound, OC 家 *kʰra - OC 鷄 *kʰe > pJ *ka-ke > OJ *kakye*. This loan may be a compound of onomatopoeic pMP *kek which is borrowed as pJ *kak and #3.3.5 pMP *kiak which becomes an open syllable pJ *kia > WOJ *kye*. Both words from pMP can be seen in Ilokano as well, so it is not doubtful the compound arose outside Japonic. Alternatively, Vovin (2014b) compares his reconstruction *kake to a possible proto-Kra-Dai (pKD) *katke in Fāngyán, leaving three possible origins.

Any of these possibilities suggests a southern/continental origin for the word for chickens, which is also observed genetically, “Japanese indigenous chickens were largely introduced via land routes through China and partially established from chickens introduced via sea routes from Southeast Asia,” according to (Hata, et al. 2020.).

13. pAN *kemkem ‘bite down hard’ > pJ *kam ‘to chew’ > WOJ *kam* ‘to chew (rice to make sake)’ [combined two entries]

Okinawan (Iheya) *kamu*[_N] ‘to bite’

Okinawan (Iheya) *kamu*[_N] ‘to eat’

Amami (Okinoerabu) *kamiN* ‘to eat’

Miyako (Irabu, Sawada) (Jarosz, p.566) *kam* ‘to bite’

Ilokano (MP) *kemkem-én* ‘to chew; clench the teeth; bear a grudge’

Sound Changes: Schwa Change?

Martin (1987. p.703) reconstructs “kamu (B) < *kama-. ‘chew, bite, eat; (= kamosu) brew’,” NJ *kamosu* ‘to brew (alcohol)’ is likely related but the segment -os is unclear.

Francis-Ratte (2016. p.193) reconstructs this term as pKJ *kami- with the descendants MK *ke:melí* ‘leech’ and MK *kam-spol-* ‘licks it up, sucks up food’ which may be related but less likely to be the ultimate source due to the missing *i segment in proto-Japonic.

This etymon is clearly an exception to the labial-adjacent schwa change (*ə > *a), but is included as related to the more likely pT *gam^A ‘bite (n.), speech’ (Pittayaporn, 2009. p.359). The inclusion of this proto Tai etymon also supports the list of comparanda observed by Vovin (2014b). See also: Vietnamese *ngoàm* ‘bite’ and OC 銜 *[g]ʰr[a]m ‘a [horse] bit/[mouthpiece]’ (Baxter-Sagart, 2014), further supporting the origin of this basic lexeme from a southern sprachbund.

14. pAN *kiskis ‘to shave, scrape off’ > pJ *kis.i-m (grind-EP-VBZ) > NJ *kisimu* ‘to jar, to creak, to grate’

pJ *kis.i-r (grind-EP-NTR) > NJ *kisiru* ‘to creak, to squeak, to grate, to grind, to rasp’

Ilokano (MP) *ag-kiskís* ‘to shave; scrape’

Sound Changes: Sibilant Approximation, Epenthesis

Martin (1987. p.709) reconstructs “kisimu (B) ?< *kyisimu < *kisi_{mim}-ma-. ‘creak, squeak’” and “kisir-u (B) ?< *kyisir-u < *kisi_{mim}-ra-. ‘creak, rasp’”.

15. pMP *kitkit ‘to bite, gnaw’ > pJ *ki ‘canine tooth’ > WOJ *ki* ‘fangs, teeth’

pJ *ki-ⁿ-pa (canine-GEN-tooth) > NJ *kiba* ‘tusk, fang’

Miyako (Hirara) *ksi:ba* ‘fang’

Ilokano (MP) *kitkit-én* ‘to nibble, gnaw’

Sound Changes: Coda Loss (Nominal)

Martin (1987. p.449) reconstructs “ki (1.2) < kiy < *ku/oCi. ‘fang’ (=kiba HL) ... (Probably *kuCi rather than *koCi...[.]),” this is flawed, since the ancestor is *ki* not *kwi*. OJ *ki* and *pa* appear to distinguish sharp, biting teeth (canines) and flat, chewing teeth (molar) respectively. Perhaps OJ *pam* ‘to eat’ (of animals) is related (*pa-m? (molar.tooth-VBZ)) as Martin (id. p.684) cites as well, “hamu (B) < famu < *pama- (?= *pa_{1.3b} “teeth” -ma-, cf. Yoshida 1979:85).” Benedict (1990. p. 256) “TOOTH/: *gigit > OJ kyi = ki ‘fang/ tusk’; J p. kiba ‘canine tooth (-ba)’”.

16. pMP *kulkul ‘bend the limbs, curl up the body’ > pJ *kur ‘twist, spin’ > WOJ *kur* ‘reel, haul’

pJ *kur.u-p (spin-EP-ITER) > WOJ *kurup* ‘go mad’

pJ *kur.u-m-ar (spin-EP-VBZ-NTR) > NJ *kurumaru* ‘to be wrapped up (e.g. in a blanket), to cover oneself’

Malay (MP) *tər-kokol* ‘huddled up, as of a man shivering with ague’

Sound Changes: Liquid Approximation, Epenthesis

Martin (1987. p.716) reconstructs “kuru (B) < *kuru- (cf. kurumu, kuruu, kuru-si-) ?< [*]kura- ... ‘wind, reel’”. See also OC 輾 *[g]ʰ<r>or? (dialect: *-r > *-j) ‘turn around (as a wheel)’ and OC 渾 *[g]ʰ[u]r ‘chaotic’ (Baxter-Sagart, 2014) with similar sense and phonetics as the Japonic

examples, and may be a more likely source for both the pJ and pMP given the poor semantic match in pMP.

17. pWMP *kumkum ‘fold the arms over the chest; fold the wings, of a bird’ > pJ *kum ‘cross, link’ > WOJ *kum* ‘group (t)’
pJ *kum-a (cross-TR.OBJ) > OJ *kuma* ‘bend’
pJ *ku(m).kum (cross-RED) > WOJ *kukum* ‘wrap, tuck’
pJ *pa-ⁿ-ku(m).kum (feather-GEN-cross.RED) > WOJ *pagukum* ‘wrap in feathers’
Tagalog (MP) *kumkóm* ‘held against the chest’
Sound Changes: Exact

Martin (1987. p.716) reconstructs “kumu (B) < *kuma-. ‘braid; assemble it; ... (sic)’... kumi ‘set’,” the sense ‘to braid’ is related to crossing two things. This example has a reduplicated form without *rendaku* but the monosyllabic form suggests *kukum* is derived from *kum* (c.f. WOJ *tutuk* ‘continue’/NJ *tuduku* ‘continue’, with optional *rendaku* from reduplicating WOJ *tuk* ‘adhere’). It is also worth highlighting the similarity between Tagalog *kumkóm* and the usage of WOJ *pagukum* ‘wrap in feathers’ in the Manyoshu.

- (28.a) opobune ni, imo noru mono ni, aramaseba, *pagukumimotite*, yukamasi monowo
‘If you came on this great ship, I would travel *holding you under my wings*! Alas...’
(MYS 15.3579)

18. pMP *kupkup ‘to clutch, hold tightly’ > pJ *kup ‘hold tight’ > WOJ *kup* ‘hold in the mouth’
pJ *kup-(r)asi (hold.tight-ADJZ) > WOJ *kupasi* ‘fine, beautiful, exquisite’
Tagalog (MP) *kupkóp* ‘kept and protected under one’s care (literally as a hen covers her chicks); held against one’s breast or chest, as a mother holding a child’
Synonyms: OJ *mudak* ‘embrace’

Comparing *kupkup to several verbs helps explain the strangeness of the sense ‘hold in the mouth’ which could be from semantic infection by WOJ *kup* ‘eat’. The adjective *kupasi*, ending in *-asi*, appears to be derived from a verb *kup* but (eat-ADZ) does not have as much explanatory power as an unattested OJ verb *kup ‘hold tight’, later becoming ‘hold (in the mouth)’. See also Tagalog *kupkóp* and the semantics relating largely to holding with love and care. This word may have been replaced by its synonym OJ *mudak* ‘embrace’.

The phonetically similar WOJ *kupape* ‘add’ more likely comes from OC 合 *k^hop ‘together; put together; combined’ (Baxter-Sagart, 2014) and is related to MK *kwop-* ‘doubles, increases it two-fold’, showing a north to south path from Chinese through Koreanic to Japonic instead.

19. pAN *kurkur ‘dig in the soil’ > pJ *kur > NJ *kur* ‘to gouge, to hollow out, to bore, to excavate, to scoop out, to greatly perturb’
Amis (Formosan) *korkor* ‘to dig in the soil’

Tagalog (MP) *kulkól* ‘shallow digging; poking the earth with a pointed stick or similar tool’

Synonyms: WOJ *por* ‘scoop out’

Sound Changes: Exact

20. pWMP *kuskus ‘to scrape out’ > pJ *kus-i (scrape.out-NOM) > WOJ *kusi* ‘stake, post’

pJ *kus-i-ⁿ-kay (scrape.out-CONN-hair) > WOJ *kusige* ‘comb’

Okinawan (Iheya) *kusi*: ‘skewer’

Ilokano (MP) *ag-k<an>uskús* ‘to rub the head with something rough’

Sound Changes: Affricate Approximation

The meaning of the pWMP etymon and the Ilokano descendant both support the reconstruction of a verb pJ *kus ‘to scrape out’ being used in regard to the head in *kusige*. See also pAA *kru:c (Sidwell, 2024. p.xxiv).

21. pWMP *kuskus ‘rising of smoke or steam’ > pJ *kus.u-(i)mpu-r

(smoke-EP-smell.strongly-NTR) > NJ *kusuburu* ‘to smoke, to smoulder, to smolder, to sputter, to be sooty’

pJ *kay-impu-r-i (feeling?-smell.strongly-NTR-NMZ) > WOJ *keburu* ‘smoke’

Malay (MP) *kukus* ‘steaming; cooking in steam; distilling (rice is usually boiled in water in Malaya)’

Sound Changes: Sibilant Approximation, Epenthesis

Martin (1987. p.717) reconstructs “kusuburu (B) < *kusunpura- < *pusunpa-ra- ... ‘get smoky/sooty; smolder’” and (p.696) “iburu (?A/B) < *inpu-ra- (< ?). ‘smolder, be smoky’”.

Positing *kusuburu* as a compound instead with #3.2.52 pMP *imbu explains both: the segment *bur (since the verbal suffix WOJ -*bwi*- ‘verbal formant’ does not appear to have this allophonic variant) and the reading for 煙る NJ *iburu* shared with *kusuburu*. Consider that WOJ *keburu* also has the same segment *bur, the segment *ke* being comparable to WOJ *poke* ‘smoke’ (fire-feeling).

22. pPh *layláy ‘to wilt, wither, droop, as plants; dangling, hanging loosely’ > pJ *ay-ay ‘droop-get’ > WOJ *aye* ‘fall off’

Ilokano (MP) *i-pa-layláy* ‘to expose to the heat in order to wilt’

Sound Changes: Liquid Loss

Synonyms: WOJ *oti* ‘fall’

Martin (1987. p.674) reconstructs “aeru (?A) < aye[y]- < *ada-Ci- ‘fall, spill, drip’ (vi.)”.

However, OJ y from pJ *d is flawed according to Vovin, “The hypothesis that OJ y < pJ *d is widely supported, but it is entirely based, I am afraid, on a mistaken theory that the Yonaguni dialect which indeed has initial d- corresponding to OJ y- (though OJ medial -y- corresponds to Yonaguni -y-, not -d-) preserves, alone among all temporal and geographical varieties of Japanese, a pJ *d which shifted to y elsewhere (JLTT:20)”. Both this and OJ w < pJ *b will be reanalysed as original pJ *y and *w.

WOJ *aye* is much rarer at 3 uses recorded by the ONCOJ compared to WOJ *oti* at 49 uses, which may suggest that the latter gradually replaced the former.

23. pAN *metmet ‘to hold tightly’/pAN *mesmes ‘grasp, grip, squeeze’ > pJ *mæt/məs > WOJ *mot* ‘to grasp’, EOJ *mæt/məs/mæt* ‘to hold, to take, to bring’
Common Ryukyuan *muci* ‘to hold’ (Jarosz, p.363)
Ilokano (MP) *na-metmēt* ‘avaricious, greedy’ < *mætmat
Amis (Formosan) *mecmec* ‘to massage’ < *məsməs
Sound Changes: Schwa Retention
Synonyms: WOJ *kisum* ‘hold, keep’, WOJ/EOJ *tor/tər* ‘take, hold in the hand’
Martin (1987. p.728) reconstructs “motu (B) < *mota- ‘hold, have’ ... Cf. moteru”. The similarity in pAN *metmet and *mesmes explains the presence of the EOJ reflex *məs*. Francis-Ratte (2016. p.286) reconstructs “pKJ *mætə- ‘holds’,” through comparison with “MK *motó-*, *moti-* ‘is long-lasting, is durable; keeps things’”. The existence of both pAN *metmet and *mesmes with WOJ *mot* and EOJ *məs* means the morpheme in Japonic was borrowed from Austronesian, rather than being cognate with Koreanic.

24. pPh *ŋalŋál ‘cry out, bellow, moan’ > pJ *nar > OJ *nar* ‘make sound’
pJ *nar-s (make sound-TRZ) > WOJ *nas* ‘make sing, cry’ (Frellesvig, 2021. p.42)
pJ *nar.a-s (make sound.EP-TRZ) > NJ *naras* ‘to sound, ring’ (id.)
Miyako (Hirara) *na:z* ‘to cry’
Tagalog (MP) *ŋalŋál* ‘prone to grumble and sob; wailing; loudly lamenting; weeping aloud’
Sound Changes: Nasal Approximation, Liquid Approximation
Martin (1987. p.733) reconstructs “naru (A) < *na(-)ra-. ‘sound, ring’ ... Cf. naku, ne.” Two different sources are offered in this paper for WOJ *nak* (#3.2.26 *ŋakŋak) and *ne* (#3.2.27 *ŋa(ŋ)ŋaŋ). Francis-Ratte (2016. p.361) reconstructs “pKJ *nəra- ‘resounds, makes a sound’,” through comparison with MK *nwo:l-* ‘plays, frolics,’ nwol(G)áy ‘song’”. Considering the patterns observed in the data here, Arisaka’s Law would simplify pJ #nəra > pJ *nəra. See also OC 歎 *ŋ⁶ar ‘to sigh’ (Baxter-Sagart, 2014) which supports a nuclear -a- instead of -ə-.

25. pAN *ñamñam ‘tasty, delicious (?)’ > pJ(P) *nam-oy? (‘taste-get’) > EOJ *namwi* ‘offering’, NJ *name* ‘to lick, to lap, to suck, to taste’
pJ(P) *nipu-nam-oy? (new-taste-get) > WOJ *nipunamwi* ‘offering (and tasting) of the first fruits’
Okinawan (Iheya) *nami:/N* ‘to lick’
Ilokano (MP) *nánam* ‘taste, flavor, savor’
Sound Changes: Nasal Approximation
The WOJ example *namwi* is difficult to segment, but may be an old phonetic variant of the NJ form, *name*. See also pAA *ŋa:m ‘sweet’ (Sidwell, 2024. p.xii).

26. pMP *ŋakŋak ‘raucous sound’ > pJ *nak ‘animal/human cry’ > OJ *nak* ‘to cry’

Okinawan (Iheya) [*načun* ‘to cry’

Miyako (Irabu, Sawada) *nafu* ‘to cry’

Kankanaey (MP) *ŋakŋák* ‘howl, yell, as dogs do’

Sound Changes: Nasal Approximation

Synonyms: WOJ *muse* ‘sob’

Martin (1987. p.731) reconstructs “*naku* (A) < *na(-)ka-. ‘cry’ ... Cf. *naru*, *nasu*; *ne*,” positing a base morpheme *na with the verbal formant pJ *-k; Frellesvig (2021. p.42) reconstructs OJ *nak* as pJ *nar-k (cry.out-VBZ) but the pMP etymon *ŋakŋak is the closest source phonetically.

Kankanaey *ŋakŋák* shows how this word was used for animals, Bontok *ma-ŋakŋák* ‘to laugh loudly; the sound of loud laughter’ shows its use with people; in Japonic the distinction is still made with the characters 泣< (human) and 鳴< (animal). See also OC 𪛗 *[\ŋ]ʰak ‘beat a drum’ (Baxter-Sagart, 2014).

Benedict (1990. p.174) compares “CALL (ANIMAL): *ŋakŋak > nak-i ‘call (of animals, birds, insects), cry, weep’”.

27. pMP *ŋa(ŋ)ŋaŋ ‘inarticulate sound’ > pJ *na-(a)y (sound, get.NMZ) > WOJ *ne* ‘sound, cry’

pJ *na (?) > WOJ *na* ‘name’

Kankanaey (MP) *ŋaŋáŋ* ‘pronouncing inarticulate sounds, as mute persons’

Sound Changes: Coda Loss (Nominal)

Martin (1987. p.495) reconstructs “*ne* (1.1) < *ne[y] < *naCi (? < na[r]i*_{2.1} < *nar[a-C]iA; ? < na[k]i_{2.1} < *nak[a-C]iA; ? < *na-Ci (cf. *naku*, *naru*, *nasu*). ‘sound; crying, weeping’”. WOJ *na* could instead be from another source than pMP *ŋa(ŋ)ŋaŋ, but WOJ *ne* fits both semantically and phonologically as a noun formed from attaching the bigrade in its stem form (e.g. WOJ *nagare* ‘flow’ (v) and *nagare* (n)). See also OC 言 *ŋa[n] ‘speak, speech’ (Baxter-Sagart, 2014) which may help support thinking of ‘name’ as related to ‘sound’.

28. pPh *pagpág ‘brush or shake off dust or dirt’ > pJ *pak ‘hit dust’ > WOJ *pak* ‘sweep’

pJ *pa(k)-tak (hit.dust-strike) > NJ *hataku* ‘to dust, to remove dust’

pJ *pa(k)-pak-i (hit.dust-RED-NMZ) > WOJ *papaki* ‘broom’ (NJ *hōki* ‘id.’)

pPh *i-pagpág ‘to brush or shake off dust or dirt’ > Ibaloy (MP) *i-pagpag* ‘to brush something off with hard blows (as dust from a coat)’

Sound Changes: Devoicing

Martin (1987. p.414) reconstructs “*hooki* (3.5a/b) < *fauki* < *fawaki* < *fafakyi* < *papaki* < *pa[ka]_Bpa[k]a-C]i_B. ‘broom’,” but the reduction of the second syllable *paka-paka > *pa-pak is harder to explain than a consonant cluster simply simplifying from *pakpak > *papak.

The segmentation here for *hataku* is idiosyncratic but has great explanatory power for the phonetics of *hataku* which is otherwise etymologically opaque: *pa-tak?, *pata-k?, *pat-ak?. See also OC 搏 *pʰak ‘strike’ and 撲 *pʰok ‘beat (v.)’ (Baxter-Sagart, 2014).

29. pAN *pakpak ‘to clap, flap the wings; sound of clapping or flapping’ > pJ *pa(k) > WOJ
‘feather, leaf’, EOJ ‘wing’

pJ *pa-nay > NJ *pane*

Okinawan (Iheya) [*ɸʷani* ‘wing’

Miyako (Hirara) *pani* ‘a feather, a wing’

Ilokano (MP) *pakpák* ‘bamboo rattle’

Sound Changes: Coda Loss (Nominal)

Synonyms: WOJ *tubasa* ‘pair of wings’

Martin (1987. p.394) reconstructs “ha (1.2?) ha < fa < *pa. ‘feather’ (= hane).” Tagalog (MP) *pakpák* ‘wing’ also appears to be deverbal. Vovin (2014b) compares proto-Kam-Sui *pwa, another competing or possibly related etymon. Benedict (1990. p.165-166) “BEAT/: *pakpak > Jp. Fa ‘feather’”.

30. pWMP *palpal ‘trimmed short, as grass; bare (of a field)’ > pJ *par ‘to clear’ > WOJ *par*
‘open land’, EOJ *par* ‘to clear’

Miyako (Hisamatsu) *pariz*

Amami (Okinoerabu) *harijun*

Yaeyama (Funauki) *pʰqri*

Okinawa (Iheya) *hari:[N]*

Malay (MP) *papal* ‘short; cropped; trimmed’

Sound Changes: Liquid Approximation

Martin (1987. p.685) reconstructs “haru (B) < faru < *para- (?< *paru-). ‘open ground, clear land (for cultivation)’... Cf. haruku, hareru; hara.” WOJ *hara* is more likely from #3.2.80 *padan instead. Francis-Ratte (2016. p.405) reconstructs “pKJ *pəra- ‘sweeps away’,” through comparison with “MK *poli*- ‘throws out, abandons’”. This would more likely result in WOJ *por*.

31. pWMP *paŋpaŋ ‘steep slope (as of riverbank or mountain)’ > pJ *pa-ma
(land.edge-place) > OJ *pama* ‘beach’

Okinawan (Iheya) *haa[ma]* ‘beach’

Miyako (Hirara) *pama* ‘beach’

Miyako (Irabu, Sawada) *pama* ‘beach’

Ilokano (MP) *paŋpáŋ* ‘furrow, ridge (thrown up by a plow)’

Sound Changes: Coda Loss (Nominal)

Martin (1987. p.397) reconstructs “hama (2.3) hama < fama < *pama ‘beach’”. Vovin (2014b) compared pJ *pa ‘edge’ (c.f. *pasi) to proto-Tai *faŋ ‘shore, bank’, both of which may be explained by pWMP ‘steep slope’. See also OC 阪 *pa[n]ʔ ‘slope, bank’ and 坊 *[b]aŋ ‘dike’ (Baxter-Sagart, 2014).

32. pMP *paqpaq ‘pre-masticate food for infant’, ‘chew’ (PAN) > pJ *pa ‘chewing tooth’ > OJ
pa ‘tooth’

Okinawan (Iheya) *ɸʷa[ʔ]*: ‘tooth’

Miyako (Hirara) *pa*: ‘tooth’

Amis (Formosan) *pa?pa?* ‘chew; the action of the jaws and teeth in chewing’

Sound Changes: Coda Loss (Nominal)

Martin (1987. p.394) reconstructs “ha (1.3b) ha < fa < *pa. ‘tooth’”. Vovin (2014b) compares pJ *pa to pT *van (from pAN *nipen (Ostapirat, 2018. p.113)) which is less similar, although the consistently identical accents in Vovin’s comparanda makes them very compelling as being related in some way. Yaeyama (Hateruma) has *pan* (Martin, X. id.) which may point more closely to proto Tai as well.

33. pPh *paspás ‘do something quickly, in a hurry’ > pJ *pas-ir (hurry-enter?) > WOJ *pasir* ‘run’

pJ *pas-as-ay (hurry-TRZ-get) > WOJ *pasase* ‘make run’

pJ *pas-as-anka-(a)y (hurry-TRZ-rise-get) > WOJ *pasasage* ‘make gallop, EOJ *pasasa*”ge ‘to make someone/something run into’

Ilokano (MP) *paspas-án* ‘to do quickly’

Sound Changes: Affricate Approximation

Synonyms: WOJ/EOJ *iswog/iso*”g ‘hurry’

Martin (1987. p.685) “hasir-u (B) < fasir-u < *pasira- ?< *pas[a-C]i-ra- (cf. haseru). ‘run’”.

Francis-Ratte (2016. p.369) reconstructs “pKJ *pasi- ‘runs’,” through comparison with “MK *paspo*- ‘is busy’”. NJ *haseru* ‘to run, to hurry (when going somewhere)’, WOJ *pasase*, WOJ/EOJ *pasasage/pasasa*”ge, all run into conflict with reconstructing final *i. WOJ *pasir* seems to include some unknown verbal suffix *-ir* instead, perhaps a suffix with OJ *ir* ‘enter’ (c.f. NJ *hashiri-dasu* ‘to take off running’, with the opposite suffix *-deru* ‘take out’)

34. pWMP *patpat ‘cutting, slashing, as in clearing land’ > pJ *pat-a (slash-TR.OBJ),
*pat-ak-ay (slash-open-get) > NJ *hata*, *hatake* ‘field (for growing wheat, fruit, vegetables, etc). cultivated land, vegetable plot, kitchen garden, plantation’

Sound Comparisons: Exact

Ibatan (MP) *patpat* ‘slash down vegetation to clear the land’

Martin (1987. 401) reconstructs “hata (2.4) < fata < *pata. ‘(dry) field’,” (id.). The kokuji 畑 including the radicals for fire and paddy field suggest the burn and slash method of agriculture was practiced, providing reason to assume the morpheme *pat did have the sense of ‘slashing’.

Martin (id.) reconstructs “hatake (3.7a/b) < fatakey < *pata_{2,4}-ka-Ci_{1,x} (‘place’ + suffix; or is pata a truncation of pata-ka?). ‘field’.” It would be curious if *hatake* was the only WOJ word affixing some unknown suffix to the pJ place suffix *-ka. Instead, it would make sense if WOJ *ake* ‘open (t)’ is attached, as it appears to be in WOJ *pirake* ‘open [flat] (i)’ with a shift in transitivity. This would make *patake* a noun derived from the stem of a bigrade that would also mean ‘to open land (by slashing)’.

Francis-Ratte (2016. p.253) reconstructs “pKJ *pata ‘farm field’,” through comparison with “MK *path* ‘farm field’... the Korean form incorporates the velar locative marker *kə > -h”. The Korean word is borrowed through Japonic once again rather than having diverged from pKJ.

OJ *pata* ‘paddy field’ may be a different homophone to NJ *hata* ‘(dry) field’, derived from #3.2.16 *bates > #patə(s) > pJ *pata ‘edge (of a paddy field)’. See also OC 伐 *m-pat ‘cut; strike; attack’ (Baxter-Sagart, 2014).

35. pPh *petpet ‘piece of something that has been cut’ > pJ *pət-a (cut.wood?-TR.OBJ) > NJ *hota* ‘kindling, small bits of scrap wood used to fuel a stove’

Itbayaten (MP) *ni-petpet* ‘cut’

Sound Changes: Schwa Retention

This one has few descendants to support it but is an uncommon word, possibly suggesting an origin in an early strata (c.f. the hapax legomenon *ukaŋ (#3.2.165) to WOJ *ukare*). Phonetically, semantically, and morphologically, this word appears to hold water.

36. pWMP *petpet ‘stuffed, crowded; blocked, clogged’ > pJ *pət.ə-n-pasir (clogged-GEN-run) > NJ *hotobasiru* ‘to surge, to well up, to gush out’

Kankanaey (MP) *petpét* stuffed up; bunged up, as the shank of a pipe

Sound Changes: Schwa Retention

Martin (1987. p.693) reconstructs “hotobasir-u (A?) < fodofasir-u/fotobasir-u <

*poto_{mim}-n-pasira-B. ‘leap, spring; gush, spurt’”. Similar to the word before, this rare word is largely supported by its degree of fit rather than number of descendants, and the rendaku with the transparent *pasir* present internally support segmentation, *hoto-basiru*.

37. pWMP *perper ‘to urge, press on insistently’ > pJ *pər > WOJ *por* ‘desire’

Old Javanese (MP) *a-məpər* ‘to urge, press, push; to be insistent (in questioning), press for further particulars’

Sound Changes: Schwa Retention

Martin (1987. p.692) reconstructs “horu < foru < *pora- (? *pwora, cf. ho-si_{adj B}). ‘desire’... Cf. horeru.)” Unfortunately, there is a dearth of evidence in Japonic to reconstruct it to pJ but the phonetics are identical with semantic similarity, so it should not be disregarded altogether.

38. pMP *pispis ‘to squeeze’ > pJ *pis.i-nk? (press.between-EP-VBZ) > NJ *hisigu* ‘to crush, to squash, to overpower, to beat, to crush’

pJ *pis.i-pis.i (press.between-EP-RED-EP) > NJ *hisihisi* ‘acutely (feel), keenly, severely’

pJ *may-pis.i-nk (eye-press.between-EP-VBZ) > *mehisigu* “close one’s eyes” (Martin, X.)

pJ *pis.i-mek? (crush.between-EP-appear) > NJ *hishimeku* ‘to clamour, to crowd, to jostle’

Keley-i (MP) *pihpih* to squeeze

Synonyms: WOJ *kudak* ‘smash’

Sound Changes: Epenthesis

Martin (1987. p.690) reconstructs “hisigu (A?) < fisigu < fisiku < ?*fyisiku < *fyi[kyi]siku <

*pik[a-C]i_A sika-(? < saka-_B with assimilation to the preceding vowel). ‘crush’”. Both the

semantic and phonetic comparisons are unduly complex, where a simple morpheme *pis

‘squeeze’ with a voiced variant *-nk of the verbalizer *-k suffix is readily understood. Although

the segment *-nk is of uncertain origin, the reduplication NJ *hisihi* supports segmentation as *hisi* and, due to the semantic similarity, into an echo vowel bisyllabification. pMP *pipis ‘to crush or mash between two hard surfaces’ and its descendant Malay (MP) *pipis* ‘mashing between two hard surfaces; grinding spices for curry’ also show how this root *pis means to crush (between two surfaces).

39. pAN *pitpit ‘pinch, squeeze, press between two surfaces; pluck, pick off’ > pJ *pit-ari (close.together-ADVZ) > NJ *pittari* ‘tightly, closely, exactly, precisely; perfectly (suited), ideally’

Ilokano (MP) *pitpit* ‘flat, flattened, crushed’

Sound Changes: Exact

This example may be considered weak since it depends on a single modern mimetic word but has great phonetic, semantic, and morphological explanatory power. Consider the similarity of two things pressing against each other with pMP *pispis ‘to squeeze’ and Malay *pipis* ‘mash between two hard surfaces’. See also OJ 閉 *p[i]t-s ‘close (v.); obstruct’. (Baxter-Sagart, 2014).

40. pPh *pudpud ‘worn out by overuse; broken; blunt’ > pJ *pur.u-ku (old-ADVZ) > WOJ *puru*(-ku) ‘old’

pJ *pur.u-s (old-EP-TRZ) > WOJ *purus* ‘make old’

Ilokano (MP) *p<um>udpód* ‘to wear out (shoes, etc.)’

Sound Changes: Lenition, Epenthesis

Synonyms: WOJ *oi* ‘age’

Martin (1987. p.695) reconstructs “*hurusu* (B) < *furusu* < **puru*_{adj B}-ma-. ‘get old’”. Since this word also appears to only be used in reference to objects, it would explain very well why NJ *furui* is considered unnatural to use in reference to aging people.

41. pWMP *punpun ‘bunch, cluster, as of fruit’ > pJ *pu(n) ‘place where fruit grows’ > WOJ *pu* ‘lawn, growing place’

Sundanese (MP) *punpun* ‘to appear in large quantity’

Sound Changes: Coda Loss (Nominal)

Synonyms: WOJ *nipa* ‘garden’, NJ *shiba* ‘lawn, sod, turf’

Martin (1987. p.416) reconstructs “*hu* (1.x) < *fu* (?< *[o]fu_{B pred} < ... ; ?< *[f]ay]u_{B pred} < ...; ?< *p[ar]u ‘field’) ‘place where things grow’ (cf. sonou): *siba-hu* ‘turn, greensward, lawn’”.

Although Martin relates it to “*p[ar]u_B” (WOJ *paru* ‘open land’), “*[f]ay]u_{B pred}” (WOJ *paye* ‘grow’) and “[o]fu_{B pred}” (WOJ *opwi* ‘grow’). See #3.1.1 *barbar for *par* and *pay*, and #3.2.63 for *opwi*. See also pAA *bu:ŋ? ‘bunch, collection’ (Sidwell, 2024. p.xxx).

42. pAN *seksek ‘to stuff, cram in; be crowded’ > pJ *sek ‘to block’ > WOJ *sek* ‘dam (v)’, WOJ *seki* ‘dam (n); border’

EOJ *seki* ‘barrier’

Ilokano (MP) *seksek-an* ‘to stuff; fill up with’

Sound Changes: Schwa Change?, Affricate Approximation

Synonyms: NJ *fusagu* ‘to stop up, to close up, to block (up), to plug up’

Martin (1987. p.749) reconstructs “seku (B) (?< *saku < *saka-; ?< *soku < *soka-). ‘obstruct, check’.” This word appears to have a strange schwa change also seen in #3.1.44 *selsel, where pAN *se corresponds with pJ(P) *se (and expected WOJ *sa* ‘narrow’). This change could be explained as fronting of schwa, adjacent to the palatal pAN *s /ʃ/ (> pJ *syə# > pJ *syə/se > WOJ *se*). See also OC 塞 *[s]ʰək-s ‘frontier’ which also contains /ə/ like *seksek /ʃəkʃək/ (Baxter-Sagart, 2014).

43. pAN *saksak ‘hack, chop into pieces’ > pJ *sak > WOJ *sak* ‘split’

pJ *sak-ar (split-NTR) > WOJ *sakar* ‘be apart’

pJ *sak-ay (split-get) > EOJ *sakε* ‘be split in two’

Okinawan (Iheya) *sateu*/_N ‘to split’

Miyako (Hirara) *saksi* ‘to split’

Amis (Formosan) *cakcak* ‘cut into pieces with a machete, as sweet potatoes; hoe ground until it is well tilled’

Sound Changes: Affricate Approximation

Martin (1987. p.746) reconstructs “saku (B?) < *saka- (cf. sakeru) / *saku- (cf. sakumu). ‘rip it, split it’”. Francis-Ratte (2016. p.395) reconstructs “pKJ *sak- ‘splits it by cutting’,” through comparison with “MK *kask-* ‘trims, cuts it’... The comparison takes MK *kask-*... as *kal-sak- *trims-splits”. Unless pKJ donated the term to Austronesian, the directionality should instead be pAN *saksak > pJ *sak > pK *sak. Vovin (2014b) compares “*čaak ‘be separated’ (tone: D1L, š- in NT)”. See also OC 析 *[s]ʰek ‘split (v.)’ (Baxter-Sagart, 2014).

44. pAN *selsel ‘insert, cram in’ > pJ *sa/sye? > WOJ *sa/se* ‘narrow (n)’

pJ *se-ma (narrow-space) > WOJ *seba* ‘narrow (adj)’

pJ *se-ma-r (narrow-space-NTR) > NJ *semar* ‘to approach, to draw near, to be imminent’

Amis (Formosan) *celcel* ‘slip through a crowded place’

Sound Changes: Schwa Change, Affricate Approximation, Coda Loss (Nominal)

Martin (1987. p.520) reconstructs “se (1.1) (?< *se[y] < *sa-Ci_{adj A} or *[a]sa-Ci_{adj A}). ‘shoal; rapids’”. WOJ *asa* ‘shallow’ is suggested here to be from #3.2.7 *asat instead. Refer to #3.1.42 pAN *seksek.

45. pAN *supsup ‘sip; suck’ > pJ *sup ‘suck in’ > NJ *su(w)* ‘to smoke, to breathe in, to inhale; to suck, to sip, to slurp’

Okinawan (Iheya) *su:/in* ‘to breathe’

Amis (Formosan) *copcop* ‘to suck (as a mosquito sucking blood)’

Sound Changes: Affricate Approximation

Martin (1987. p.760) reconstructs “suu (A) < sufu < *supa-. ‘suck; inhale; absorb’”. Francis-Ratte (2016. p.400) reconstructs “pKJ *sup- ‘sucks it’,” through comparison with “MK *spól* ‘sucks it,’ *swu:m* ‘breath’”. Vovin (2014b) compares pT *ču[u]p to pJ *sup. The presence of a few pT words

resembling pAN/pMP reduplications and Tai's monosyllabic structure presents it as a possible candidate for the ultimate donor of these words (pAN *supsup > pT *ču[u]p > pJ *sup). However, pT or pKD cognates are much fewer in comparison to the data presented here, so the intricate interplay between these language families should be further investigated. See also OC 嗜 *tsʰ[ə]p 'sting and suck (mosquito)' (Baxter-Sagart, 2014). This word may appear onomatopoeic but is shared throughout pT, pAN, and OC, showing it likely migrated north through Japonic to Koreanic. Benedict (1990. p. 250) compares "SUCK *(n)tsup(n)tsup > Jp. su-i, OJ suF-i 'suck, sip, inhale'".

46. pAN *SuSuq 'drip or drain out' > pJ *susu-(n)-k (drip.water.over-VBZ) > OJ *susuk*, WOJ *susug* 'wash'
 pJ *səsəʔ-k > WOJ *sosok* 'pour, wash'
 Amis (Formosan) *sosoʔ* 'to let drip out, to drain'
 Sound Changes: Coda Loss
 Synonyms: WOJ *arap* 'wash'

Martin (1987. p.759) reconstructs "susugu (A) < (Nr) susuku < ?*susu_{mim}-ka-. 'wash, rinse,' positing the root as mimetic. This word is the only one with pAN *S /s/, which naturally correlates to pJ *s /s/. It has a few variants but WOJ *sosok* is very interesting considering that Amis *sosoʔ* has /o/ instead of /u/. The existence of both forms does suggest either an earlier borrowing from pAN then a later one from pMP (which changes pAN *S to pMP *h). See also OC 漱 *s[ʰ]rok-s 'wash' (Baxter-Sagart, 2014) which also has nuclear /o/ that could initially raise in pJ (OC *srok > pJ *sok > pJ *su(k)-suk?). Benedict (1990. p. 258) compares "WASH/: *(n)[ts,tʃ]u(n)[ts,tʃ]uk > susug-i 'wash, rinse, pour on'".

47. pAN *taktak 'clatter, clack' > pJ *ta(k)tak/tak-ⁿ-tak > WOJ *tatak/tadak* 'hit, strike'
 pJ *ta(k)-tak-ap (hit-ITER/hit.RED-ITER) > WOJ *tatakap* 'fight'
 Okinawan (Iheya) *tatačun* 'to strike'
 Miyako (Hirara) *tatakʷi* 'strike'
 Javanese (MP) *t<r>a tak* 'making light, hollow tapping sounds'
 Synonyms: WOJ *ut* 'hit'
 Sound Changes: Exact

Martin (1987. p.764) reconstructs "tatau (B) ?< *ta_{1.3a}tuka-_A ('hand poke'... 'strike, beat, tap; mince'," but vowel assimilation is not established like reduplication is. Francis-Ratte (2016. p.406) reconstructs "pKJ *takə- 'takes in, receives'," through comparison with "MK *tothwó*- 'fights,' *thi*- 'strikes,'" *tho*- 'takes in, receives'". WOJ *tadak* looks like a typical case of *rendaku* from reduplication (c.f. WOJ *tomar* 'stop', WOJ *todomar* 'stop; be left behind' (< *to(m)-ⁿ-tom-ar) and WOJ *pik* 'pull; pluck', WOJ *pibik* 'reverberate' (< *pi(k)-ⁿ-pik)); therefore, there may have also been an unattested verb pJ *tak. See also OC 斫 *tak 'cut, hack' and OC 鬥 *tʰok-s 'quarrel, wrangle' (Baxter-Sagart, 2014). HIT': Benedict (1990. p.208) compares "P-Austro-Tai *(n)tak(n)tak > Jp. tatak-i 'hit, beat, pound'".

48. pAN *taktak ‘to fall, of many things at once’ > pJ *tak-i (fall.at.once-NMZ) > WOJ *taki* ‘waterfall, cataract’

Okinawan (Iheya) [*taci* ‘waterfall’

Miyako (Hirara) *taki* ‘a waterfall’

Tagalog (MP) *takták* ‘act of shaking out the contents of a container by pounding the open mouth against some hard object’

Synonyms: WOJ *tarumi* ‘waterfall’

Sound Changes: Exact

Martin (1987. 539) reconstructs “taki (2.1) < takyi (< ?, cf. tagir-u, tagitu). ‘waterfall < rapids’”. The semantics for this word may seem distant but consider Yami *taktak* ‘to sprinkle, to water’. See also pT *tok^D ‘to drop’ (Pittayaporn, 2009. p.352).

49. pAN *tamtam ‘smack the lips’ > pJ *tam-ay-s(?) > NJ *tamesu* ‘X’

Chamorro (MP) *tamtam* ‘test, try out, sip, taste’

Sound Changes: Exact

Martin (1987. p.763) reconstructs “tamesu (B) <? *ta-myesu < *ta_{1.3a}miCa-sa-B (‘let hand see’). ‘try’”.

50. pAN *tastas ‘sever, cut through, rip out stitches’ > pJ *tas-i/tat-i > EOJ *tasi/tati* ‘long sword’, WOJ *tati* ‘id.’

Ilokano (MP) *satsát* ‘to unravel; tear, rip garments; demolish; undo crochet work by pulling the string’

Sound Changes: Affricate Approximation

Synonyms: WOJ *kir* ‘cut’, WOJ *taye* ‘be cut off’

Martin (1987. p.766) reconstructs “tatu (B) < *tata-. ‘cut off, sever; abstain from; exterminate’”.

Francis-Ratte (2016. p.217) reconstructs “pKJ *tat- ‘closes it’,” through comparison with “MK *tat-* ‘closes it’”. See also OC 制 *tet-s ‘cut out, prepare’ (Baxter-Sagart, 2014) and pT *tac^D ‘to cut’ (Pittayaporn, 2009. p.354). The existence of allophonic variants *tasi* and *tati* and both pT and OC comparanda demonstrate another southern to northern loan.

51. pPh *temtem ‘to burn something in a bonfire’ > pJ *təm.ə-s (kindle-EP-TRZ) > WOJ *tomos* ‘kindle’

pJ *təm.ə-r (kindle-EP-NTR) > NJ *tomoru* ‘to be lit (e.g. candle, lamp, light bulb), to be lighted, to be burning’

Itbayaten (MP) *ka-temtem* ‘cremation’

Sound Changes: Epenthesis, Schwa Retention

Synonyms: WOJ *kwogare* ‘burn, smolder’, WOJ *mwoye/moye* ‘burn, feel passion’

Martin (1987. p.769) reconstructs “tomoru, toboru (B) < *tonpo-ra- (?< ...; see tobosu). (vi.) ‘burn, be lit, alight’... Cf. hotoboru,” adds an additional *o instead which can be explained instead from echo vowel epenthesis.

52. pMP *titi ‘breast; suck the breast’ > pJ *ti ‘breast, bodily fluid?’ > WOJ *ti* ‘milk, breast’, ‘blood’?

pJ *ti-ti (milk/breast-RED) > NJ *titi* ‘milk, breast’

Okinawan (Iheya) *tei*/: ‘milk’

Amami (Okinoerabu) *tei*: ‘milk’

Miyako (Hisamatsu) *citci* ‘milk’

Malay (MP) *tetek mamilla* ‘teat; breast in relation to sucking only’

Sound Changes: Exact

Martin (1987. p.545) reconstructs “ti (1.3a) ?< *ti[y] < *tu-Ci. ‘milk’... Cf. tu_{1.3a} ‘spit’ (?= ‘liquid’, cf. mizu); titi_{2.4},” assuming an extended sense of ‘liquid’, here assumed to originally be milk then extended to ‘bodily fluid’. WOJ *ti* ‘blood’ may be related in this manner, whereas NJ *tuba* is likely not related since *ti* ‘milk’ comes from pJ *ti, not pJ *tuy, according to this reconstruction based on pMP *titi.

Francis-Ratte (2016. p.201) reconstructs “pKJ *cece ‘breasts’,” through comparison with “MK *cyéc* ‘breasts’”. Since WOJ *ti* comes from pJ *ti, *te, and *tuy, any of these may be likely and possibly appear similar from convergence. Benedict (1990. p.173) compares “BREAST: tšitši > Jp. *titi* ‘breast, milk’”.

53. pAN *tuktuk ‘beak of a bird; to peck’ > pJ *tu(k)tuk > WOJ *tutuk* ‘peck at’

Yami (Formosan) *toktok* ‘to peck’

Sound Changes: Exact

Martin (1987. p.776) reconstructs “tutuku (?A/B) < *tutuka- ?< *tu[ka-]_Atuka-_A (Unger); ?< *tutu-ka- (Yoshida, Ōno)”. WOJ *tutuk* appears to be an example of reduplication of a monosyllabic *tuk without rendaku, which would result in *tuduk*. Benedict (1990. p.228) compares “PECK: P-Austro-Japanese *tuktuk > Jp. *tutuk-i* ‘poke (at), pick (at), peck’”. See also OC 啄 *mə-tʰ<r>ok ‘to peck’ (Baxter-Sagart, 2014.)

54. pAN *tuktuk ‘to knock, pound, beat; crush’ > pJ *tuk > WOJ *tuk* ‘pound; hull’

pJ *tu(k)-tuk (pound-RED) > EOJ *tutuk* ‘to hit’

Ilokano (MP) *tuktók* ‘sound of pecking; knocking’

Sound Changes: Exact

Martin (1987. p.774) reconstructs “tuku (B) < *tuka- ‘pound, husk’”. Francis-Ratte (2016. p.347) reconstructs “pKJ *tuxi- ‘pounds it with mortar’,” through comparison with “MK *tih-* (*titha*) ‘pounds it with mortar’... *tuxi- > *tuhi- > tih-”. This should probably be compared to #3.2.162 *tutuh, which would explain *tuhi, since the glottal consonant appears to have been present in Japonic too, WOJ *tutui/tuti*. Another agricultural word likely relating to the hulling of rice. EOJ also has *tutuk* which looks like another reduplicated descendant. Benedict (1990. p.231) compares “POUND: *truktruk > Jp. *tuk-i* ‘pound (rice), hull (barley)’”.

55. pAN *tuRtuR ‘resonant sound’ > pJ *sapiⁿ-tur (harmonize-GEN-make.a.resonant.sound)
> WOJ *sapidur* ‘chirp’

Ilokano (MP) *togtóg* ‘knock (on a door, etc.); beat (a drum, rumbling sound of waves, people, or animals running, etc.)’

Sound Changes: Liquid Approximation

Martin (1987. p.745) reconstructs “*saezuru* (A) < *saf^e/i**duru* < *safyiduru* < ?**sapi_{mim}*-n-tura-A. ‘sing, chirp, twitter; prattle’,” also supports splitting into two morphemes **sapi* and **tura*. The first morpheme of this compound is contained later in this list (#3.2.123 **sapiq*) but only appears once in the ONCOJ. The two Austronesian morphemes provide sources for *sapi* and *dur* which are also clearly segmentable due to the length of the word (Japonic trisyllabic morphemes being mostly borrowings) and *rendaku* voicing of *dur*, together accounting for ‘chirp’ as a compound. See also OC 鶉 *[d]ur ‘quail’ (Baxter-Sagart, 2014).

3.2 Bisyllables

The next section of words is the largest out of the data, showing that Japonic’s bisyllabic phonotactic structure (C)V(N)CV is likely inherited or adapted from Austronesian. Both prevent initial or final consonant clusters (CC) and vowel clusters (VV) but allow medial nasal-obstruent clusters (nt, njk, mp) which highlights the nature of Japonic’s assimilative nasal mora *h̥* N.

1. pWMP **abá* ‘exclamation: ah!, oh!, alas!’ > pJ **apa-ray* (exclamation-?) > *apare*
apa-r-ay-m*/apa-r-ay-b* (exclamation-NTR-get-VBZ) > *aparem/apareb*
Tagalog (MP) *abá* ‘emphatic denial; sudden reproach; simple surprise; hail!’

Sound Changes: Devoicing

Martin (1987. p.679) reconstructs “*awaremu* (?A/B) *awaremu* < *afare^b/m**u* < **apare[y]_{intj}* (< *apa-ra-Ci*[-*Ci*]_{B inf})-n[a]-pa- (or-ma). ‘pity’”. The segmentation of *apare* is simple, since Japonic morphemes tend to be disyllabic and *-re* appears in other words such as: *a-re* ‘I’ (*a* ‘id.’) (pronoun); *naga-re* ‘flow’ (*nagas* ‘make flow’) (verb); *saza-re* ‘small, minute’ (*sasara* ‘small, minutely divided’, *sasa-nami* ‘small wave’) (adjective). Many of the words that this suffix appears in are also in this list: *apure/abure* (#3.2.5 **apun*), *kopore/kobore* (#3.2.37 **gebáq*), *padara/padare* (#3.2.84 **pantek*), *panar/panat/panare* (#3.2.83 **panaq*), *pukure* (#3.4.2 **buk(ə)law*), *sasara/sazare* (#3.2.52 **isa*), *tubure* (#3.2.163 **tumpuk*), and *ukare* (#3.2.165 **ukan*). The few exceptions: *arare*, *kwogare*, *nagare*, *sudare*, *sumire*, *tabure*, and *tapare* could have originated from disyllabic loanwords as well.

2. pWMP **amat* ‘continuous, unceasing’ > pJ **amat.a?* (excess.EP?) > WOJ *amata* ‘many, extremely’
pJ **ama-r* (excess.NTR) > WOJ *amar* ‘be in excess’
pJ **ama-s* (excess.TRZ) > WOJ *amas* ‘leave excess’
Miyako (Hirara) *ama-ta* ‘a lot’ (Jarosz, p.30)
Malay (MP) *amat* ‘very; exceedingly; after an adjective it means excess’
Sound Changes: Coda Loss

Martin (1987. p.676) reconstructs “amaru (B) < *ama-ra. ‘be excessive, too much; remain (left over)’... Cf. amasu, ahureru.” Francis-Ratte (2016. p.321) also reconstructs “pKJ *ama- ‘is many, more than’,” through comparison with “MK *a:matwo* ‘perhaps,’ *a:mwo* ‘any,’ *a:molán* ‘any sort, any way,’ [and] ENK *emeli* ‘very much’”. If these words are related, they are more likely borrowed from Japonic which lost the final /t/ in *amat > pJ *ama > pK *ama.

The final /t/ in the pAN etymon appears to be retained with an echo vowel in the WOJ adverb *amata*. However, in the verbs *amar/amas*, it appears to be deleted, perhaps from a cluster reduction (*amat-r?). In most examples however, final /t/ also appears to be lost altogether (#3.2.59 keRet, #3.2.63 lebe(t), #3.2.4 aŋka(t), etc.). This enigma is an inconsistency, but the rest of the phonetics and semantics cannot be disregarded. Additionally, this inconsistency does appear to explain one more morpheme (#3.2.126 *sepet > WOJ *sopoti*).

3. pAN *amax ‘father’ > pJ *t-ama-sipi? (HON-father-?) > WOJ *tama*, *tamasipi* ‘soul’
 Okinawan (Iheya) [*tama*]/*ei* ‘soul’
 Miyako (Hirara) *tamasī* ‘soul, spirit’
 Ilokano (MP) *amá* ‘father. Also said of persons that are old enough to be one's father’
 Sound Changes: Coda Loss

Martin (1987. p.539) reconstructs “tama (?2.3) (?< ‘jewel’; ?< atama ‘head’)”. Vovin (2021) finds an old synonym for mother pJ *tinay from pMP *t-ina (HON-mother), so an old synonym for *titi* from the same process is not far-fetched. Benedict (1990. p.187) compares “FATHER/SPIRIT P-Austro-Japanese *t-ama P-Austronesian *(t-)ama ‘father’ (Blust 1980). Jp. *tama* < t-ama (‘(deceased father =) spirit, soul, ghost’... For the semantics... it appears that the departed ‘fathers’ are the ‘recent dead’, still hovering about as the ‘spirits of the dead’= ‘ghosts’ (*tama*), whereas the departed ‘ancestors’, the great-grandparents and such, are the ‘long dead’, venerated as ‘gods’ (*kami*)”.

4. pWMP *aŋkat ‘lift, raise, pick up’ > pJ *aⁿka-r (rise-NTR) > WOJ *agar* ‘rise’
 pJ *aⁿka-(a)y (rise-get) > WOJ *age* ‘give, raise, go up’
 Okinawan (Iheya) *aga/in* ‘raise’
 Miyako (Hirara) *aga:z* ‘rise’
 Tagalog (MP) *aŋkát* ‘consignment of merchandise’
 Sound Changes: Coda Loss, Prenasalization

Martin (1987. p.674) reconstructs “*agaru* (A) < *anka-ra-. ‘rise’”. With these bisyllabic pAN/pMP cases it is more likely that the pJ root also possesses an underlying bisyllabic form *anka; however, the possibility of echo vowel epenthesis *ank.a-r could still explain the derived forms. The Tagalog meaning may also provide an explanation for WOJ *akap* ‘atone; buy’, WOJ *aki* ‘deal; trade’, NJ *aganau* ‘to atone for; to buy, to pay for’. The NJ example appears to link the two WOJ words and shows an optional nasal.

5. pMP *apuŋ ‘float’ > pJ *apu/*a^mpu-r-ay (float-NTR-get) > WOJ *apure/abure* ‘overflow’
 Malay (MP) *apoy* ‘floating on water; driftwood’

Sound Changes: Coda Loss, (Prenasalization)

Martin (1987. p.675) reconstructs “ahureru (?A/B) < abure- (Mg[?]) = abure[y]- < *anpu(-)ra-Ci- ‘overflow’”. This is one of several entries that shows optional prenasalization in pJ but not pAN (see also #3.2.37 *gebáq, #3.2.45 *isaŋ, #3.2.128 *sikaŋ).

6. pWMP *asag ‘incite, urge on, stir up’ > pJ *asa-(a)y (hurry-get) > NJ *aseru* ‘to be in a hurry, to be impatient, to be anxious (to do), to fret’

Yonaguni (Ozu) *aserasii*

Kapampangan (MP) *asag* ‘incite’

Sound Changes: Coda Loss

Martin (1987. p.677) reconstructs “aser-u (B) ?< *ase_{2.5}-ra; ?< a[fi_B-] ser_B (cf. ser-u, seku; isogu). ‘fret over; be hasty, impatient, overzealous’”. Instead of positing a common origin with NJ *seru* ‘to approach, to draw near, to be imminent’, a single morpheme here, pWMP *asag provides a simpler explanation.

7. pMP *asat ‘dry out, become dry’ > pJ *asa-ku (shallow-ADJZ) > WOJ *asa-ku* ‘shallow’

pJ *asa-(a)y (shallow-get) > WOJ *ase* ‘get shallow’

pJ *as(a) (be.shallow) > WOJ *as* ‘dilute, weaken; make shallow’

Okinawan (Iheya) *asa]han* ‘shallow’

pMiyako *asa ‘shallow’

Javanese (MP) *asat* ‘dried, become dry’

Sound Changes: Coda Loss

Martin (1987. p.677) reconstructs “aseru (?A/B) < ase[y]- *asa_{adj A}-Ci-. ‘get shallow; fade, discolor’... Cf. useru.” The discrepancy between the initial vowel in *useru* ‘to disappear, to vanish, to fade away’ makes it unlikely to be related.

8. pWMP *ataq ‘unhusked rice grain left among the husked grains’ > pJ *ata-ra

(wasted.rice-PROP) > WOJ *atara* ‘in vain, be a waste’

Malay (MP) *antah* ‘husk; sheath (of grain); The pieces of dry husk shaken off in the process of winnowing’

Sound Changes: Coda Loss

Martin (1987. p.387) reconstructs “atara (? < *atara-A). ‘alas; regrettably’ (adv).” There is no independent evidence for *ata as a morpheme but *-ra appears several times in this cognate set (*padara* (#3.2.84 *pantek), *sasara* (#3.2.52 *isa), *tubara* (#3.2.159 *tumpaŋ), *tubura* (#3.2.164 *tumpuk), and *tapira* (#3.3.4 *taŋ(ə)pil)), supporting segmentation. The meaning in pJ can be explained as a semantic extension of an originally descriptive lexeme.

9. pAN *aya/*ayat ‘exclamation of annoyance, surprise, etc’, ‘threaten, confront violently’ >

pJ *aya-po (threaten-ADJZ) > EOJ *ayapo* ‘dangerous’

pJ *aya-si (surprise-ADJZ) > WOJ *ayasi* ‘mysterious, strange’

pJ *aya-ni > (surprise-ADVZ) > WOJ *aya ni* ‘incredibly; inexplicably’

pJ *aya-amat? (surprise-excess) > WOJ *ayamat* ‘err’

EOJ *aya* ‘strange, unusual’

pMiyako (Hirara) *ajasi ‘suspicious, ominous (sic)’

Ilokano (MP) *ayá* ‘an adverb of interrogation implying an additional notion of wonder’

Sound Changes: Exact/Coda Loss

Martin (1987. p.679) reconstructs ayabumu (A) ?< *ada_{2,3}-n-pu(-)ma-; ?< *ada_{2,3}-n[i] puma-A; ?< ayafumu < *ada-pu_{adj A}-ma-. ‘feel anxiety over; suspect’”. The reconstruction here for *ayamat* is idiosyncratic, but explains the final consonant in the verb as being the same as *amat*; the meaning also makes sense as being derived from excessively worrying someone.

10. pWMP *ayam ‘plait, weave’ > pJ *aya > WOJ *aya* ‘weaving; woven pattern’

Okinawan (Iheya) *aja*/: ‘woven pattern’

Common Ryukyuan *aja* ‘pattern, ornament’

Minangkabau (MP)? *ayam* ‘to plait’

Sound Changes: Coda Loss

Martin (1987. p.387) reconstructs "aya (2.3) < *ada. ‘design’”. Francis-Ratte (2016. p.235) reconstructs “pKJ *aja ‘design, pattern’,” through comparison with “NK *uy ‘design, pattern’ (NK *mwunuy* ‘design, pattern’ ~ SK *mwun* 紋, 文 ‘pattern’). Why the vocalism of *aya would shift to *uy- is left unexplained and seems unlikely as a sporadic change.

The MP example here is one of the only remaining in Austronesian but is present in WOJ and several Ryukyuan languages. However, the noun could be from an unattested pJ *ay ‘weave’ with the transitive object suffix *-a (*ay-a ‘that which is woven’ > ‘woven pattern’), which may disqualify this entry.

11. pWMP *ayun ‘to swing, oscillate’ > pJ *ayu-ək (swing-put) > WOJ *aywok* ‘quake; move’, EOJ *ayok* ‘to waver’

pJ *isa-(a)yu-ap (small/one-swing-ITER) > WOJ *isaywop* ‘drift, waver’

Ifugaw (MP) *ayóh* ‘cradle, or what is used as a cradle; act of rocking’

Sound Changes: Coda Loss

Martin (1987. p.680) reconstructs “ayugu (?B) < ayuku < aywoku < *adwoka- (< ?; cf. asi_{2,3}). ‘shake, move’”. The segment -ywo- is explained by crasis of pJ *yuə* > *ywo*, and the meanings of quaking and wavering are explained by the idea of shaking in place, *ayu-* ‘swing’ and *ok* ‘settle, put’.

12. pWMP *bagak ‘foolish’ > pJ *n(ə)-paka (GEN-idiot) > NJ *baka* ‘idiot’

Tagalog (MP) *bangák* ‘foolish (said of young women)’

Sound Changes: Coda Loss, Devoicing

Although this word is only present in Modern Japanese, the register in Tagalog seems to show a change that would explain the usage distinction between Standard Dialect (NJ *hyoujungo*) NJ *baka* ‘dummy’ (potentially non-offensive) and NJ *aho* ‘moron’ (quite offensive).

13. a. pMP *banaqaR ‘radiance, as of rising sun’ > pJ *pana(a) ‘radiant’

pJ *pana-panta (radiant-lasting) > WOJ *panapada* ‘extremely’

Miyako (Hirara) *pana* (?) ‘an upper part, an edge; time, the moment when; a breakthrough moment’

Tagalog (MP) *banda?ag* ‘soft ray, glimmer’

Sound Changes: Devoicing, Deletion, Coda Loss

b. pWMP *bantas ‘lasting, enduring’ > pJ *paⁿta ‘lasting’

pJ *pana-paⁿta (radiant-lasting) > WOJ *panapada* ‘extremely’

pHirara *panta ‘busy’ (?) (Jarosz, 2015. p.614)

Javanese (MP) *bantas* ‘lasting, durable, of tissue (Pigeaud 1938)’

Sound Changes: Devoicing, Prenasalization, Coda Loss

Martin (1987. p.827) does not provide a reconstruction for “[hanahadasi-] (B) < fanafada_{4.2} (< ...) -si-. ‘extreme’”.

Many of the examples in the Manyōsyū involve the sky: 7.1370 puranu ame, 10.2322 puranu yuki, 10.2336 ywo pukete na yuki, 11.2400 X, 11.2692 simo onikikyeri, 13.3250 ametuti no. Most of these involve snow, perhaps comparing the white color to that of the sun. Therefore, ‘radiant’ is a better guess than pJ *pana ‘flower’ or ‘nose’, especially when paired with the otherwise unknown morpheme pJ *panta, which cannot be ‘skin’ (flower/nose-skin > extremely??).

14. pMP *bakes ‘belt, anything that encircles tightly’ > pJ *pak(a) > WOJ *pak* ‘string (bow); wear’, EOJ *pak* ‘to put, to wear, to carry’

Okinawan (Iheya) *hateu* [_N ‘to put on’

Miyako (Hirara) *pakama* (pak-əmə? (put.on-surface)) men's underwear trousers worn under elegant outfit’

Ilokano (MP) *barkésan* ‘an inflammatory disease of the skin, often originating at the waist; to tie into one bundle’

Sound Changes: Devoicing, Coda Loss, Schwa Assimilation

Martin (1987. p.684) reconstructs “haku (?A) < faku < *paka-. ‘slip (something) on, wear’”.

Francis-Ratte (2016. p.293) reconstructs “pKJ *pak- ‘puts it through’,” through comparison with “MK *pak*- ‘inserts it, fills it’”. Vovin (2014b) compares pT *pak (D1S) ‘insert’, which does not explain the sense of wearing on the lower part of the body as well as pMP *bakes, but they could both be related. See also OC 佩 *[b]^hək-s ‘to wear at girdle, pendant’ (Baxter-Sagart, 2014).

Seeing the number of southern loans (pT, pMP, and OC), this loan likely spread north to Koreanic through Japonic.

15. pMP *bantun ‘pull up, as out of a pit or out of water’ > pJ *paⁿtu-s (pull.out-TRZ) > NJ *hazusu* ‘to remove’

pJ *paⁿtu-r-ay (pull.out-NTR-get) > NJ *hazure* ‘to be disconnected’

pJ *paⁿtu-ka-(a)si (pull.out-ADJZ-ADJZ) > WOJ *padukasi* ‘embarrassed, ashamed’

Miyako (Hirara) *pazī* ‘shame, dishonor’

Sound Changes: Devoicing, Prenasalization, Coda Loss

Martin (1987. p.687) reconstructs “hazusu (A) < fadusu < *pantu-sa-. ‘disconnect, remove, unfasten; miss, let it go by; avoid, dodge’”. The sense of ‘shame’ is derived as a form of ostracization, or social removal.

16. pWMP *bates ‘border, boundary’ > pJ *pata > WOJ *pata* ‘side, edge’

pJ *pata-s (edge-TRZ) > OJ *patas* ‘accomplish, finish (EOJ)’

pJ *pata-(a)y (edge-get) > WOJ *pate* ‘end, dock (of a boat)’

pJ *pata-te/pata-de (edge-place) > WOJ *patate/patade* ‘limit, edge, side’

Okinawan (Iheya) *hata/ki* ‘field’

Miyako (Hirara) *pada* ‘the edge, the end’

Malay (MP) *batas* ‘boundary; limit; boundary dyke to a ricefield; rice-plot as shut in by these dykes’

Sound Changes: Devoicing, Coda Loss, Schwa Assimilation

Martin (1987. 401) reconstructs “hata (2.1) < fata < *pata. ‘side, rim’”. Francis-Ratte (2016. p.246) reconstructs “pKJ *pataŋ ‘end’,” through comparison with MK *patáng* ‘sole, bottom’”. I suspect this may be a false cognate considering MK *patáng* comes from *pataŋ but WOJ *pata* appears semantically closer to pWMP *bates, with final /f/ not final /ŋ/. Malay *batas* ‘boundary; limit; boundary dyke to a ricefield; rice-plot as shut in by these dykes’ shows that this word may be the source of WOJ *pata* ‘paddy field’ and would explain the polysemy with WOJ *pata* ‘side, edge’, pJ *pata (edges (of a rice paddy)). NJ *hata(ke)* ‘field, cultivated land’ instead refers to dry fields as Martin (1987. 401) reconstructs “hata (2.4) < fata < *pata. ‘(dry) field’”. It also shows analogy with WOJ *pirake* ‘open (i.)’ < *pira-(a)kay (flat-open).

17. pMP *beŋkas ‘untie, undo’ > pJ *pəŋkə-s (untie-TRZ) > NJ *hogosu, hogusu* ‘to unravel, to untie, to untangle, to loosen’

Cebuano (MP) *bun̄kás* ‘undo something sewn’

Sound Changes: Devoicing, Prenasalization, Coda Loss, Schwa Retention, Schwa Assimilation

Martin (1987. p.691) reconstructs “hogusu < fog^o/u[?] < (*)fagasu < *panka-sa-. ‘undo, untie, unsew, unravel, disentangle’,” why *panka would result in *hogusu* instead of *hagasu* is left unexplained. The pMP etymon here would match *hogosu* due to Arisaka’s law, from *beŋkas /bəŋkaf/ > pəŋka# > *pəŋkə, conditioning the second syllable according to the first.

18. pMP *benteŋ ‘extended, stretched taut, put under tension’ > pJ *pəŋtə-k (pull.apart/loose-VBZ) > WOJ *podok* ‘loosen’

Miyako (Irabu, Sawada) *putuci* ‘unstitching (of clothes)’ (Jarosz, p. 621)

Cebuano (MP) *buklād* ‘spread something (as a mat) out flat; for a girl to grow to maidenhood (poetic); for a blossom to open’

Sound Changes: Coda Loss, Devoicing, Prenasalization, Schwa Retention

Martin (1987. p.691) reconstructs “hodoku (B) < fodoku ?< *pon[o]_(2.3) toka-B (‘slightly undo’). ‘loosen, untie’”.

Even though Irabu *putuci* appears to reflect *poto rather than *podo (< pJ *pə⁽ⁿ⁾tə), there are other optional nasals in the pAN comparanda, so the conditions for this are uncertain but present. The final verbal suffix *-k might be an approximation of the velar pAN *ŋ or the verbal formant *-k (c.f. WOJ *itu* ‘pure’ > WOJ *ituk* ‘worship the spirits; perform acts of worship’ and WOJ *tuna* ‘rope’ > WOJ *tunag/tunak* ‘bind’).

19. pMP *beraŋ ‘width, extent across’ > pJ *pərə > NJ *horo* ‘canopy (esp. the cloth or canvas used for it), awning, top (of a convertible)’

Kelabit (MP) *beraŋ* ‘width (of river, house, field, etc.)’

Sound Changes: Coda Loss, Schwa Retention, Schwa Assimilation

NJ *horo* being used especially in reference to the spanning cloth for a canopy helps to cement this comparison.

20. pMP *betus ‘burst open, as an overfilled sack’ > pJ *pə(n)tu-s (unravel-TRZ) > NJ *hotusu*, *hodusu* ‘to unravel, to untie, to untangle, to loosen’ (identical to *hogosu*, *hogusu*)

pJ *pə(n)tu-r-ay (unravel-NTR-get) > NJ *hotsureru* ‘to become frayed, to become loose’

Sundanese (MP) *betus* ‘tear, burst open, as an overfilled pillow under pressure’

Sound Changes: Devoicing, Coda Loss, Schwa Retention

Martin (1987. p.693) reconstructs “hotureru (B) < foture[y]- < fature[y]- < *patu(-)ra-Ci (<?). ‘ravel fray’ (v.i) (=hatureru, hogureru)”. See also #3.2.17 *beŋkas. Once again, the schwa retention adjacent to a labial demonstrates original *ə instead of *ə < *a.

21. pWMP *bidaŋ ‘unit of measure for cloth, sails, etc.’ > pJ *pira-(C)ə?/pirə? >

WOJ *piro* ‘wide; unit of measurement (length)’, EOJ *pirə* ‘to be wide’

pJ *pirə-r > WOJ *piror* ‘be widespread’

pJ *pira > WOJ *pira* ‘flat’ (EOJ ‘flat, ordinary’)

Malay (MP) *bidan* ‘spacious; expansive; broad; a numeral coefficient for sails, mats, hides, awnings, screens, and so on’

Ainu *pirasa* ‘to spread out.’ (Pellard, 2017. p.58)

Sound Changes: Devoicing, Coda Loss, Open Assimilation?

Martin (1987. p.408) reconstructs “hira- (adj stem) (2.1) ‘flat, ordinary’: see following compounds, hira_{adj A}.” and “hira (?2.2b) hira < fyira < *pira (< *pira_{adj A}). ‘sheet; petal’”.

OJ *pira* has been compared to Middle Korean *pilé* ~ *piléy* ‘cliff, bank,’ (id.) and Ainu *pira* ‘cliff’ (Pellard, 2017. p.58), which could be related but less clearly.

The loan *horo* has a problem with the final vowel which would result in the borrowing *pira* rather than *piro*. However, the word semantically and phonetically otherwise is nearly identical and may be considered an irregularity. There does appear to be the matching WOJ *pira* with the more general sense, ‘flat’, which may be derived from the use of pWMP *bidaŋ for wide, flat things as seen in Malay *bidan*.

22. pWMP *bi(ŋ)kaŋ ‘spread apart, as the legs’ > pJ *pik(a) > WOJ *pik* ‘pull’ (EOJ ‘to pull’, ‘to invite’, ‘to drag’, ‘to draw’)

pJ *pika-(a)y (pull-get) > ‘be pulled’

Miyako (Hirara) *p^hik^h* ‘pull’

Tagalog (MP) *biŋkán* ‘arched wide (said of shanks, legs, fork, or the like)’

Sound Changes: Devoicing, Coda Loss

Martin (1987. p.689) reconstructs “hiku (A) < fyiku < *pika-/*piko- (cf. hikozurau). ‘pull, ... (vt.); ‘withdraw, retreat, subside’ (vi.)”. Francis-Ratte (2016. p.352) reconstructs “pKJ *pika- ‘pulls it out, pulls it open’,” through comparison with MK *phyé-*, *phú-* ‘unfolds, spreads it out, opens it up’”. The semantic shift is ‘wide apart’ as stretching from one end to another ‘pull, drag, draw’. It is also visible in #3.2.128 *sikaŋ ‘wide apart, of the legs, which appears phonetically similar to this pWMP example.

23. pMP *bilak ‘shine, glitter’ > pJ *pira-mek (flash-appear) > NJ *hirameku* ‘to flash (e.g. lightning), to flicker, to glitter, to sparkle, to gleam’

Kankanaey (MP) *bilák* ‘glitter, shine (for instance, gold)’

Sound Changes: Devoicing, Coda Loss

Mentawai (MP) *bila* ‘to flash, of lightning’

Buginese (MP) *billa?* ‘lightning’

Sound Changes: Devoicing, Liquid Approximation, Coda Loss

Martin (1987. p.689) reconstructs “hirameku (?A/B) < *fyira-myeku < *pira_{mim}-miCa-ka_{bad v.}[?] ‘flash; flutter’”.

24. pWMP *bilaq ‘part split off’ > pJ *pira > NJ *hira* ‘counter for thin, flat things (petals, leaves, etc.)’

NJ *hanabira* (< *pana-ⁿ-pira (flower-GEN-flat.piece))

Malay (MP) *bilah* ‘thin strip or lath’

Sound Changes: Devoicing, Liquid Approximation, Coda Loss

Martin (1987. p.398) reconstructs “hanabira (?4.5) < fana-byira < *pana_{2,3}-n-pira_{2,2b} ‘petal’,” with *bira* related to WOJ *pira* ‘flat’.

25. pMP *buket ‘thick, viscous’ > pJ *puka-ku (deep-ADJZ) > OJ *puka-ku* ‘deep’

Okinawan (Iheya) [*fuka*]/[*san* ‘deep’

Miyako (Hirara) *puka:puka*

Malay (MP) *bukat* ‘full of dregs; dirty, of well-water; muddy, of a stream’

Sound Changes: Devoicing, Schwa Change, Coda Loss

Martin (1987. p.694) reconstructs “hukeru (B) < fukey- < *puka_{adj B}-Ci- ‘deepen; get late, grow old’; < *pukar-Ci- ‘get steamed’”. The example in Malay shows a possible path from ‘viscous’ to ‘deep’, a river that is muddy may feel deeper than it is when trudging through.

26. pAN *CaNem ‘grave; to bury’ > pJ *sana-/*san(a)-a (buried.thing(-TR.OBJ)) > *sana* ‘seed, pit, stone’
 pJ *tana/sana-(a)y (buried.thing-get) > OJ *tane* ‘seed’, WOJ *sane* ‘seed, pit, stone’
 Okinawan (Iheya) [*sani*] ‘seed’
 Miyako (Hirara; Irabu, Sawada) *tani* ‘seed’
 Ilokano (MP) *taném* ‘tomb, grave’
 Sound Changes: Affricate Approximation, Nasal Approximation, Schwa Assimilation, Coda Loss

Martin (1987. p.541) reconstructs “tane (2.4) < *tane[y] < *tana-Ci (? < *ta_{1.3a}na-Ci_{1.3a} ‘paddy-field root’). ‘seed (small?)’. Cf. sane”. There is pMP *tanem ‘grave; to bury; to plant’, which does include descendants with the meaning of planting, pAN *C, would better explain why there is both *tane* and *sane*. It may be that ‘to plant’ was a sense in pMP, of which there are no further descendants. Though it is more reasonable to reconstruct pJ *san(a) ‘to bury’, with pJ *-a > *san(a)-a ‘that which is buried’. WOJ *tane/sane* could be deverbals formed from suffixing WOJ *e* (pJ *ay). Although this is the only case of pAN *C, *tane* and *sane* support making this unique comparison.

27. pAN *Capel ‘a patch; to patch’ > pJ *tapa-(a)y (patch-get.NMZ) > WOJ *tape* ‘cloth made from the bark of the mulberry tree’
 Tagalog (MP) *tápal* ‘mending patch (of cloth); poultice, plaster’
 Sound Changes: Affricate Approximation, Schwa Assimilation, Coda Loss

Martin (1987. p.537) reconstructs “tae (2.x) < tafey < *tapaCi. ‘bark cloth’. Polynesian tapa (proto-Austronesian *tambang*); *nigita*”. The pAN form cited **tambang* according to the ACD means ‘ambush, surprise attack’ and it is unclear which Polynesian language *tapa* refers to. Instead Tagalog provides the closest phonetic and semantic match.

28. pAN *CekeS ‘kind of very slender bamboo’ > pJ *taka(h)?/*taka-(a)y (bamboo-get.NMZ) ‘(slender) bamboo’ > WOJ *taka/take* ‘bamboo’, EOJ *take* ‘bamboo’
 Common Ryukyuan *taki* ‘bamboo’
 Okinawan (Iheya) [*daki*] ‘bamboo’
 Irabu *a3 i-daki* ‘bamboo used for the a3 i (cf.) in Irabu’
 Amis (Formosan) *tkes* ‘small species of bamboo’
 Sound Changes: Schwa Change, Schwa Assimilation, Coda Loss, Affricate Approximation
 Synonyms: WOJ *sinwo/sino* ‘thin-culmed dwarf bamboo’ and WOJ *sasa* ‘broad-leaf bamboo’

Martin (1987. p.539) reconstructs “take (2.1) < takey < *taka-Ci. ‘bamboo’”. The final pAN *S here is the only other example besides *SuSuq, which may have influenced the covert alteration between *taka* and *take*, assuming it underwent the pAN *S to pMP *h change; but examples of final pAN *h seem to simply delete, so this loan is not phonetically perfect.

29. pAN *Cukul ‘huddle, bow the head’ > pJ *una-ⁿ-tuk (neck-GEN-bow.head) > NJ *unaduku* ‘to nod, to bow one's head in assent’

Sound Changes: Affricate Approximation, Coda Loss

Martin (1987. p.779) reconstructs “unazuku < unaduku < *una_{2,3}-n-tuka-A. ‘nod, bow head’”.

Much like *ayamat* (#3.2.9 *aya(t) and *ayatur* (#3.2.104 *qayam), this word can be segmented due to its length (and *rendaku*).

30. pPh *dakél ‘big, large (in size or quantity)’ > pJ *taka-ku (high-ADJZ) > OJ *taka-ku* ‘tall, high’

pJ *taka-(a)y (high-get.NMZ) > WOJ *take* ‘peak’

Okinawan (Iheya) *taka*/[sɔn ‘tall’

Common Ryukyuan *taka* ‘high, tall’ (Jarosz, p.641)

Yami (MP) *ráko* ‘very big, large’

Sound Changes: Devoicing, Schwa Assimilation, Coda Loss

Martin (1987. p.538) reconstructs “taka (?2.3) (< adj B). ‘height, high; amount, quantity’”.

Francis-Ratte (2016. p.286) reconstructs “pKJ *takar ‘height’,” through comparison with “MK *talak* ‘loft, attic’”. The metathesis is curious but not impossible.

Vovin (2021) compares pJ *takay to Yami *takəi* ‘mountain’, which does have a remarkable similarity, including schwa assimilation from Arisaka’s Law. Since this word is not a part of the Austronesian Comparative Dictionary, it was not chosen as the candidate for this set, which pPh *dakól sufficiently accounts for semantically and phonetically.

31. pMP *dapet/dápat ‘to place in contact with one another, come in contact, space closely’ >

pJ *tayu-tap (swing-be.close.together) > WOJ *tayutap* ‘be unsettled by movement’

Bontok (MP) *dapət* ‘to place objects side by side, so that they are touching; to have the edges of two objects together; to meet’

Sound Changes: Devoicing, Coda Loss, (Schwa Assimilation)

This word only appears in combination with 3.2.151 pWMP *tayug ‘swing, oscillate’ but has great explanatory power for the uncertain segment *tap. A few examples using this word so include things that often come densely packed or layered: *tayutapu nami* ‘waves swinging closely together’ MYS 7.1089 *amakumono tayutapikureba* ‘rain clouds hovering and swinging closely together’ MYS 15.3716.

32. pWMP *deket ‘ignite, set aflame’ > pJ *tak(a) > OJ *tak* ‘kindle’

Okinawan (Iheya): *tatəu*/[N ‘to kindle’

Miyako (Hirara): *tak-ri* ‘to kindle’

Tagalog (MP) *dikit* ‘act of kindling; a setting on fire’

Sound Changes: Devoicing, Schwa Change, Coda Loss, Schwa Assimilation

Martin (1987. p.762) reconstructs “taku (A) < *taka-... ‘burn it; boil/cook it’”.

Francis-Ratte (2016. p.284) reconstructs “pKJ *taka- ‘heats it with fire’,” through comparison with “MK *tahi-* ‘makes a fire, heats with fire’”. See also OC 炙 *tAk-s ‘roast, broil’ (Baxter-Sagart, 2014).

33. pAN *duma ‘someone; other people; be different’ > pJ *tuma (significant.other)? > OJ *tuma* ‘spouse’, EOJ ‘spouse, wife’
Ilokano (MP) *ag-dúma* ‘to differ’
Sound Changes: Devoicing

Martin (1987. p.556) does not include a reconstruction for “tuma (?2.2a)”. The semantic shift here is admittedly rather speculative.

34. pWMP *dura ‘to shy away, of animals’ > pJ *tura-ku (uncomfortable-ADJZ) > NJ *turai* ‘painful, bitter’
Malay (MP) *dura* ‘anxiety, disquiet’
Sound Changes: Devoicing

Martin (1987. p.842) does not include a reconstruction for “tura (A). ‘tough’”. The pWMP meaning is definitely far and this word is poorly attested, but is included due to the meaning in Malay and close phonetic shape.

35. pWMP *embej ‘bind the waist; band around the waist’ > pJ *əmp(ə) > WOJ *ob* ‘gird on’
pJ *əmp(ə)-i (gird.on-CONN) > OJ *obi* ‘belt’
Miyako (Irabu, Sawada) *ubzi* ‘a belt, a sash’ (Jarosz, p.651)
Tboli (MP) *m-ebed* ‘tie knots, wind about’
Sound Changes: Schwa Retention, Prenasalization, Coda Loss

Martin (1987. p.503) reconstructs “obi (2.4) < *obyi < *onp[a-C]i_B. ‘belt, girdle’”.

36. pAN *garan ‘fierce, intense, as of appetite or temper’ > pJ *kara-ku (strong.taste-ADJZ) > WOJ *kara-ku* ‘salty’
EOJ *kara* ‘pungent’
pMiyako *kara ‘spicy’
Malay (MP) *garan* ‘fierce; turbulent; (of color) loud or glaring’
Sound Changes: Devoicing, Coda Loss, Liquid Approximation

Martin (1987. p.830) does not include a reconstruction for “kara- (B). ‘spicy, salty’”. The difference in meanings across Japonic all relate to the pAN sense of ‘intense, as of appetite’ being narrowed into different intense flavor profiles.

37. pWMP *gaRaw ‘hoarse, raspy (of the voice)’ > pJ *kara-(a)y (hoarse-get) > WOJ *kare* ‘get hoarse’
Malay (MP) *garau* ‘deep (of the voice), harsh’
Sound Changes: Devoicing, Coda Loss, Liquid Approximation

Martin (1987. p.704) reconstructs “kareru (A) < kare[y]- < *kara-Ci. ‘wither; get dry/parched, get hoarse’”. Although the word *kare* ‘get hoarse’ can be compared to WOJ *kare* ‘wither, dry’, pWMP *gaRaw ‘hoarse, raspy (of the voice)’ is a clearer match semantically. pPh *keRaw ‘dry

up' may be related in some manner seeing their similar phonetic shape with a velar initial and identical *-Raw final.

38. pPh *gebáq 'to destroy, demolish' > pJ *kəpə/kəmbə-t (shatter-TRZ?) > WOJ *kopot/kobot* 'shatter (t)'

pJ *kəpə/kəmbə-r-ay (shatter-NTR-get) > WOJ *kopore/kobore* 'be shattered'

Tagalog (MP) *gibá?* 'demolished; destroyed; wrecked; fallen down; in a shambles'

Sound Changes: Devoicing, Schwa Retention, Open Assimilation, Deletion

Martin (1987. p.710) reconstructs "kobotu (B) < kofotu (Mr [?]) < *kopo-ta-. 'shatter it' (=kowasu), later confused with kobosu 'spill it'." Although the voiced alternative *kobot* is more common, unvoiced *kopot* shows the optional nasal is intrusive, not original.

39. pAN *gemel 'take in the hand, clasp, grasp' > pJ *kəmə-r (grasp-NTR) > WOJ *komor* 'be wrapped, be secluded'

pJ *kəm(ə)-ay/kəmə-(a)y (grasp-get) > WOJ *kome/komwi* 'enclose'

Miyako (Hirara) *kumu:z* 'to lock oneself away in one's house' (Jarosz, p.576)

Ilokano (MP) *gammál* 'grip, grasp, clasp with the hand or fingers'

Sound Changes: Devoicing, Schwa Retention, Coda Loss

Martin (1987. p.712) reconstructs "komoru (B) < komaru (the second vowel assimilated to the labial nasal and/or the first vowel) < *koma-ra-. 'be confined; be filled (with)'. Francis-Ratte (2016. p.381) reconstructs "pKJ *kəmə- 'shuts it in'," through comparison with "MK *kóm-* 'shuts (the eyes), closes (the eyes)". The final schwa from *gemel being retained to follow Arisaka's law (> kəma# > *kəmə) would explain the alternation between *kome* and *komwi* as two ways of resolving the vowel cluster /ə-ay/.

40. pWMP *genep 'complete; even (of numbers)' > pJ *kana-(a)p (complete-join) > *kanap* 'be appropriate'

pJ *kana-(a)y (complete-get) > WOJ *kane* 'combine'

Malay (MP) *gənap* 'completing; rounding off; making up a pair; even, of numbers'

Sound Changes: Devoicing, Schwa Change, Coda Loss, Schwa Assimilation

Martin (1987. p.703) reconstructs "kanau < kanafu- < *kana_{72.3} "one"-pa- 'be a match (for); be suitable (for); be possible (for), be realized". The semantics from pWMP has strong explanatory power over the resulting meanings *kanap*, joining two even things together, and *kane*, becoming complete as in being paired or becoming whole.

41. pMP *gilap 'radiance' > pJ *kira-kira-si (flash-RED.ADJZ) > WOJ *kirakirasi* 'pretty'

NJ *kirameku* 'to glitter, to sparkle, to twinkle'

Ilokano (MP) *gilap* 'flash'

Sound Changes: Devoicing, Liquid Approximation, Coda Loss

Martin (1987. p.451) reconstructs "kirara (3.6 < ?4.6) ? < *kyirara < *kira-[a]ra ('glitter stuff')." See also OC 燐 *[ɕ]^w(r)[a]p 'gleam' (Baxter-Sagart, 2014). Benedict (1990. p.239) compares

“SHINE/: P-Austro-Tai $*(\eta)kilaŋ(kilaŋ) \sim *(\eta)gilaŋ(gilaŋ) > \text{Jp. kira-kira- (compound) kira-}$ ‘glittering(ly), brilliant(ly)’”.

42. pPh $*gimpis$ ‘crush between weights; press out’ > pJ $*ki^m pi-si$ (crush-ADJZ) > NJ *kibishii* ‘severe, strict, rigid’

Bontok (MP) *gípis* ‘to squeeze out the juice from sugarcane, either by hand or in a mill; the squeezed out pith of sugarcane’

Sound Changes: Devoicing, Prenasalization, Coda Loss

Martin (1987. p.831) reconstructs “*kibisi-* (B) < *kibi* (?/kibu) (< ?) -*si-*. ‘strict; tight(ly arrayed)’”. This word is not well attested but is included due to the nasal cluster in pPh accounting for the voicing (from prenasalization) in the NJ word. The sense ‘strict’ in Japonic likely developed from semantic broadening.

43. pWMP $*gu(m)pit$ ‘pinch together; narrow’ > pJ $*ku^m pi$ (narrow.part) > WOJ *kubi* ‘neck; collar’

Javanese (MP) *η-gupit* ‘narrow (of a path, alley)’

Sound Changes: Devoicing, Prenasalization, Coda Loss

Martin (1987. p.461) reconstructs “*kubi* (2.1) < *kubyi* < $*kunpi$. ‘neck; head’”. Consider NJ *tekubi* ‘wrist’ which demonstrates the original sense of ‘narrow part’.

44. pPh $*gunay$ ‘movement; to move’ > pJ $*kunay$ (wriggle) > NJ *kuneru* ‘to bend loosely back and forth, to wriggle, to be crooked’

pJ $*kunay-kunay$ (wriggle-RED) > NJ *kune-kune* ‘winding, meandering, wriggling’

Ilokano (MP) *gunáy* ‘movement; action, activity’

Sound Changes: Devoicing, Coda Retention

Martin (1987. p.716) reconstructs “*kuner-u* (B) (?< *kunner-u*) ?< *ku[ri -]ner-* < $*kur[a-C[i_B ner-B[...]$ ‘turn and twist, be crooked/zigzag; wind, meander; (=suneru) sulk’”. Since there is no word *kunar*, so *kune* may not actually be derived from suffixing -e, but rather, the result of the final vowel cluster pPh $*ay > pJ *ay > OJ e$.

45. pWMP $*guyud$ ‘pull or drag a heavy object along, as on a sled’ > pJ $*kuy(u)-i$ (be.burdened-NMZ) > WOJ *ku(y)i* ‘regret’

pJ $*kuy(u)-asi$ (be.burdened-ADJZ) > WOJ *kuyasi* ‘regrettable’

Ilokano (MP) *gúyod* ‘pull, jerk, tug’

Sound Changes: Devoicing, Coda Loss

Martin (1987. p.715) reconstructs “*kuiru* (B) < *kuyi[y]-* < $*kuyo-Ci-$ (cf. *kuyo-kuyo*) < $*kuda-Ci-$. ‘regret’”. The pWMP word seems to explain the final $*-y$ in the WOJ verb, arising from deletion of the final $*u$, since it clearly is not the passive/spontaneity suffix pJ $*-y(e)$ (c.f. WOJ *kiko-ye* ‘be audible’, WOJ *mi-ye* ‘be visible’).

46. PMP $*hasaŋ$ ‘gills’ > pJ $*asa-r$ > WOJ *asar* ‘look for; hunt for food’, NJ *isar* ‘id.’

PWMP *isaŋ ‘gills’ > pJ *i(n)sa-r > WOJ *izar* ‘to fish’

Miyako (Hisamatsu) *isaz* ‘fishing’

Malay (MP) *asaŋ* ‘gills of fish’ (Brunei, Sarawak)

Malay (MP) *isaŋ*, *iŋsaŋ* ‘gills of a fish’

Martin (1987. p.677) reconstructs “asaru (B) < *asa-ra ‘scavenge, forage, hunt, fish for’... Cf. isaru; Middle Korean: az- < a'zo- < *a'sV- ‘grab’” and (p.699) reconstructs “isaru (A) < izaru (?< *i[bo]_{2.1 a}]sa-ra-B). ‘fish’”.

The alternate forms of this word, NJ *isar* and WOJ *izar* show: one, an odd vowel alteration a :: i; two, an optional voicing. Given the existence of both these forms in Malay and the closer semantic match (to ‘gill’ a fish, similar to English *to hook (a fish)*), the MP etyma more likely correspond to this word than MK *az*.

47. PPh *hátuŋ ‘fuel for fire, anything used to start a fire’ > pJ *atu ‘hot’ > WOJ *atu* ‘warm, hot; thick; deep, extreme’

Miyako (Hisamatsu) *acumunu* ‘hot’

Bontok (MP) ?atuŋ ‘to have a burn (as a burned hand)’

Sound Changes: Deletion, Coda Loss

48. pAN *heban ‘carry a child with a carrying cloth’ > pJ *əp(ə) > OJ *op* ‘bear, carry; cover’
Itbayaten (MP) *ahban* ‘cloth or blanket for carrying a baby or an infant, hanging down from mother’s neck in order to tuck it up’

Sound Changes: Deletion, Devoicing, Open Assimilation, Coda Loss, Schwa Retention

Synonyms: NJ *hakobu* ‘to carry, to transport, to move, to convey’

Martin (1987. p.743) reconstructs “ou (?B) < ofu < *opo- ‘cover’”. Francis-Ratte (2016. p.207) reconstructs “pKJ *əp- ‘carries on back’,” through comparison with “MK *ep-* ‘bears, carries on the back’”. Since pAN *heban is the most phonetically complex, it was likely simplified through pJ then passed onto pK, since Koreanic does not appear to have as strong a connection with Austronesian. The meaning in pAN shows how *op* developed both senses of ‘carry’ and ‘cover’.

49. pPh *idal ‘to fuss, complain; to rebel, cause a disturbance’ > pJ *ira-ira (irritate-RED) >

NJ *iraira* ‘to get irritated, to get annoyed, to be on the edge’

Casiguran Dumagat (MP) *məg-idel* ‘to dislike something; to refuse to accept something’

Sound Changes: Lenition, Coda Loss

Martin (1987. p.698) reconstructs “iradatu (?B) < *ira_{2.3}-n-tata-B (‘thorn stand’). ‘get irritated/exasperated’”. Instead of relating the morpheme *ira to ‘thorn’, it has a clear semantic parallel in *idal ‘fuss, complain’.

50. pWMP *insan ‘union, unity’ > pJ *iⁿsa-na-p (unite-EMP-ITER) > WOJ *izanap* ‘invite’

Iban (MP) *insan/isan* ‘relationship between the parents of those who have married’

Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.700) reconstructs “izanau (?A/B) < izanafu < *is/za_{2.4} (< ?) -na-pa-. ‘entice; invite’”. Francis-Ratte (2016. p.299) reconstructs “pKJ *ninsa- ‘joins’,” through comparison with “MK *ni:z*- ‘joins, pieces it together’. What happened to the initial *n in Japonic is unfortunately left unexplained. The following -na in the word *izanap* can be compared to other verbs derived from this emphatic suffix (c.f. WOJ *udu* ‘rare, wonderful’ and WOJ *udunap* ‘prize, value’). It may instead be a morphologically preserved coda in medial position (*insan-ap), but this is less certain. Vovin (2012) suggests the deities Izanami and Izanagi come from “pre-WOJ *inzanan < Old Korean archetype *yenc-ana-n ‘place.on-INF-go.out.-PAST/ATTR’,” rather than sharing a root with WOJ *izanap*.

51. pWMP *i_ɲsiD ‘move slightly, budge’ > *iⁿsi-r (push.with.finger-NTR) > NJ *ijir* ‘to finger, to touch, to fiddle with’

Iban (MP) *insit* ‘move, move slightly, budge, shift’

Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.700) reconstructs *izir-u* (?B) < *idir-u* < *inti-ra-. ‘finger, fumble/tamper with, touch; tease; importune’”.

52. pMP *imbu ‘stench, strong unpleasant odor’ > pJ *impu-r (smell.strongly-NTR) > NJ *iburu* ‘to smoke, to smoulder, to smolder, to sputter, to be sooty’ (identical to *kusuburu*)

Malagasy (MP) *imbo* ‘a strong and offensive odor, but less strong than hantsina’

Sound Changes: Exact

See (#3.1.21 *kuskus).

53. pAN *isa ‘one’ > pJ *isa-(i)sa-ka (one-RED-ADJZ) > WOJ *isasaka* ‘be for a little while’

pJ *(i)sa-(i)sa-ra (one-RED-PROP) > WOJ *sasara* ‘small, minutely divided’

pJ *isa-(i)sa-m-ay-ni (one-RED-VBZ-get-ADVZ) > WOJ *isasameni* ‘perfunctorily, a little while’

pJ *isa-ra/*isa-(i)sa > MJ? *isara/isasa* ‘a little/small’

Tagalog (MP) *isá* ‘one; another’

Sound Changes: Exact

Martin (1987. p.426) reconstructs “*isasaka* (4.x) (?< i[_{to}]_{2.2b}sasa[_{ya}]ka*_{4.11} < ...). ‘a little, somewhat; slight’”. JMDict lists *isara/isasa* as archaic, although they are not attested in OJ so they are listed as MJ. These two words show that *isa* is segmentable (*isa-ra) and that *isasa/sasa* is a result from a shortening (*isa-(i)sa/*(i)sa-(i)sa).

54. pWMP *kabal ‘invulnerable, invulnerability’ > pJ *kapa (protective.cover) > OJ *kapa* ‘pelt, hide’

Okinawan (Iheya) *kaŋ*: ‘hide’

Miyako (Hirara) *ka*: ‘skin, bark, surface layer’

Ilokano (MP) *kabál* ‘breastplate; armor; bulletproof vest’

Sound Changes: Devoicing, Coda Loss

Martin (1987. p.445) reconstructs “kawa (2.3) < kafa < *kapa. ‘skin, fur’”. Francis-Ratte (2016. P. 384) reconstructs “pKJ *kapa ‘skin’,” through comparison with “MK *kaphól* ‘sheath,’ kepcil ‘bark’”. Once more, I would suggest that the directionality is from Japonic to Koreanic, which would make Francis-Ratte’s analysis of *kaphól* < pre-MK *kap ‘skin’ + *kól* ‘reed’ as *kap(a) ‘protective covering’ + ‘reed’ > ‘sheath’. The Ilokano meaning compared to the difference between OJ *kapa* ‘pelt’, Hirara *ka*: ‘skin, bark’ and OJ *pada* ‘skin’ shows how pJ *kapa can refer to a protective covering over the skin.

See also: the Austronesian synonym pMP *kebel ‘invulnerable, invulnerability’; its descendant Malay (MP) *kəbal* ‘making the skin slippery so that weapons glance off it; protecting the flesh by rubbing quicksilver into the body so as to get a subcutaneous metallic armor’; and pAN *kabal ‘clothing’, suggesting an original meaning of a protective covering for the skin.

55. pMP *ka(m)pet ‘plugged, stopped, blocked’ > pJ *kampa-(a)y (block.off-get.NMZ) > WOJ *kabe* ‘wall’

pJ *kampa-(a)p (block.off-ITER) > NJ *kabau* ‘to protect (someone), to look after (e.g. an injured leg), to defend, to cover for, to stand up for’

Palauan (MP) *kaud* ‘dam; end of menstruation’

Sound Changes: Schwa Assimilation

Martin (1987. p.431) reconstructs “kabe (2.1) < *kabey < *kanpaCi. ‘wall’... the first vowel has perhaps assimilated to the following labial but Miyako (Hirara) kuubi implies proto-Ryukyuan *ko(o)be. Cf. kaba[f]i < *kanpap[a-C]i ‘shelter, protect’”.

56. pMP *kampit ‘to adhere, stick to’ > pJ *ka^mpi > NJ *kabi* ‘mold, mould, mildew’

Tagalog (MP) *kápit* ~ *kapít* ‘adhering; sticking; stuck; glued’

Sound Changes: Prenasalization, Coda Loss (Nominal)?

Martin (1987. p.431) reconstructs “kabi (2.1) < kabi < *kanpu/o-Ci (? < verb inf ‘sprout’). ‘mildew, mold’”. Although this word may have been formed from a deverbal unattested verb pJ *kamp- ‘to mold; to stick (?)’, it may instead be a direct borrowing *kampit > *kampi.

57. pWMP *kembay ‘swollen, expanding outward’ > pJ *kəmpə-s (spill-TRZ) > NJ *kobos* ‘to spill, to drop, to shed (tears)’

pJ *kəmpə-r-ay (spill-NTR-get) > NJ *koboreru* ‘to spill, to fall out of, to overflow, to peek through’

Malay (MP) *kəmbay* ‘expansion; blossoming out’

Sound Changes: Schwa Retention, Schwa Assimilation

Martin (1987. p.710) reconstructs “koboreru (B) < kobore[y]= < kofore[y]- < *ko(?n)po-ra-Ci-. ‘fall, drop, be scattered’”.

58. pMP *kembuy ‘swollen, filled out’ > pJ *kə^mpu > NJ *kobu* ‘bump, lump, protuberance, swelling’

Balinese (MP) *kembuy* ‘swelling, blister’

Sound Changes: Schwa Retention, Prenasalization

Martin (1987. p.453) reconstructs “kobu (?2.5, ?2.3) kobu < *ko(n)pu. ‘gnarl, lump’ (?< kobu[si]).”. Benedict (1990. p.251) compares “SWELLING: P-Austro-Tai *kəmpuŋ ~ kə(m)bun > Jp. kobu ‘swelling, wen, lump, bump’”.

59. pPh *keRaw ‘dry up’ > pJ *kara-s (dry.up-TRZ) > WOJ *karas* ‘dry’

pJ *kara-(a)y (dry.up-get) > OJ *kare* ‘wither’

Miyako (Hirara) *kari:z* ‘to wither’

Bontok (MP) *kəlaw* ‘to be dried out, of a pondfield’

Sound Changes: Coda Loss, Schwa Change, Liquid Approximation

Martin (1987. p.704) reconstructs “karasu (A) < *kara-sa-. ‘blight, kill, make/let wither’ ... Cf. kareru; korusu”. See also OC 乾 **[k]ʰar* ‘dry’ (Baxter-Sagart, 2014).

60. pAN *keRet ‘cut something slender or soft; reap crops’ > pJ *kar(a) ‘reap (crops)’ > WOJ *kar* ‘cut; harvest; split’, EOJ *kar* ‘to cut’

Okinawan (Iheya) [*hain*] ‘to crop’

Miyako (Hirara) *ka:z* ‘to crop, to cut down (grass)’ (Jarosz, p.563)

Amis (Formosan) *klet* ‘use a sickle to cut off twine’

Sound Changes: Liquid Approximation, Schwa Change, Coda Loss, Schwa Assimilation

Martin (1987. p.704) reconstructs “karu (A) < *kara. ‘mow; cut’”. Francis-Ratte (2016. p.326) reconstructs “pKJ *kara- ‘mows it’,” through comparison with MK *ka:l*- ‘plows it, cultivates it’. Proto-Japonic appears semantically closer, suggesting it to be the donor rather than recipient language for the morpheme *kar ‘cuts, splits it’.

61. pPh *kulap ‘dim, not well-lit’ > pJ *kura-ku (dark-ADJZ) > WOJ *kura-ku* ‘dark’

pJ *kur.o? (black) > WOJ *kurwo* ‘black’

pJ *kura-(a)y (dark-get) > OJ *kure* ‘get dark’

pJ *kura-s (dark-TRZ) > WOJ *kuras* ‘let time pass’

Okinawan (Iheya) [*kuru*]/[*san*] ‘black’

Miyako (Hirara) *kurug’a* ‘black hair (of a horse)’

Cebuano (MP) *kuláp* ~ *kúlap* ‘dim, not affording much light; or for the eyes to be dim’

Sound Changes: Schwa Change, Coda Loss, Liquid Approximation

Black and white are the two color words that tend to develop first, demonstrating how this Austronesian stratum may be considered a substratum in the Japonic language. As will be discussed, the Austronesian stratum likely appeared in the southern Korean peninsula and/or Kyūsyū during the late Jōmon period. The superstratum remains unknown but is likely related to the Mumun culture and the northern immigrants that initiated the Yayoi period.

62. pMP *laban ‘oppose, go against someone’ > pJ *apa-(a)y-te/tay? (oppose-get-gerund) > WOJ *apete* ‘resolutely’

Ilokano (MP) *lában* ‘fight, quarrel; feud; competition’

Sound Changes: Liquid Loss, Devoicing, Coda Loss

Martin (1987. p.674) reconstructs “[aeru] (B) < afey- (*apa-Ci-. ‘endure; dare, boldly do’”. The sense ‘resolutely’ must have come from an unattested WOJ *ape* ‘challenge’.

63. pMP *laŋaq ‘gape, open wide’ > pJ *ana > WOJ *ana* ‘hole’

Miyako (Hirara) *ana* (Jarosz, p.33)

Malay (MP) *laŋah* ‘open, agape, ajar’

Sound Changes: Liquid Loss, Coda Loss

Martin (1987. p.381) does not provide a reconstruction for “ana (2.3)”.

Francis-Ratte (2016. p.201) reconstructs “pKJ *ana- ‘brings it in’,” through comparison with “MK *a:n-* ‘holds inside, hugs’”. The discrepancy in semantics rules in favor of ‘gape, open wide’ over ‘holds inside’ as a potential cognate with ‘hole’.

64. pMP *lebet ‘set close together, dense, luxuriant’ > pJ *əpə-ku (dense, big, many-ADJZ) > WOJ *opo(-ku)* ‘big, grand; plentiful, great in amount or degree’

pJ *əpəy-s (grow.much-TRZ) > WOJ *opos* ‘cultivate’

pJ *əpə-(a)y (grow.much-get) > WOJ *opwi* ‘grow’

pMiyako *up‘u ‘big, great’

Ilokano (MP) *labbét* ‘flock together, gather in crowds’

Sound Changes: Liquid Loss, Devoicing, Coda Loss, Schwa Retention

Martin (1987. p.507) reconstructs “oo(-) (2.3 = B) < ofo(-) < *opo(-) ‘big; much’”. pMP *lebet has great explanatory power for the pJ morpheme *əpə developing the senses: ‘large; many’ and ‘cultivate; grow’. Ilokano also shows the word referring to large numbers of something.

65. pAN *likaC ‘lightning’ > pJ *ikat-uti (lightning-inside)/*ika-tu-ti

(lightning-GEN-venerable.male)/*ika-ⁿ-tuti (lightning-GEN-ground)? > WOJ *ikatuti/ikaduti* ‘thunder’

Amis (Formosan) *likat* ‘light from a source’

Sound Changes: Liquid Loss, Coda Loss?

Martin (1987. p.422) reconstructs “ikazuti (4.1) < ikaduti < *ika_{2.2b}-n- -tu ti_{1.1} (‘might spirit’). ‘lightning’”. Although coda loss has been systematic, the example #3.2.126 *sepet shows how -t may have been for some time, or perhaps allowed in some dialects. This explanation would better fit the analysis of ‘thunder’ as *ikat-uti*, ‘inside the lightning’; when final -t was lost, the morpheme was reanalysed as *ika* and segmented as *ika-tu-ti/ika-(n)tuti*.

66. pWMP *li(n)dis ‘crush, roll over’ > pJ *iⁿti-m-ay (crush-VBZ-get) > NJ *ijimeru* ‘to ill-treat, to bully, to torment, to pick on, to tease’

Kankanaey (MP) *lidis-én* ‘crush, squash’

Sound Changes: Liquid Loss, Prenasalization, Coda Loss

Martin (1987. p.700) reconstructs “izimeru (?A) < idimey- < *inti-ma-Ci-. ‘tease, torment’”.

67. pWMP *liŋaR ‘look from side to side, turn the head in this way’ > pJ *ina
(shake.head.in.denial) > OJ *ina* ‘no’

pJ *ina-bVi? (deny-VRBZ) > WOJ *inabwi* ‘refuse’

Malay (MP) *mə-liŋar* ‘to glance sideways (furtively, or in fear or warning)’

Sound Changes: Liquid Loss, Nasal Approximation, Coda Loss

Martin (1987. p.424) reconstructs “ina (2.4) ‘nay; disagreement’”. Head shakes as denial and head nods as approval are not necessarily universal, but commonly observed (Kettner & Carpendale, 2013). The example *unaduku* ‘to nod, to bow one's head in assent’ shows the positive contrast vertically, so the horizontal movement may have been negatively contrasted.

68. pWMP *liŋaŋ ‘to sway back and forth, as plant stems in the wind’ > pJ *i^hka-m-ay
(sway-VBZ-get) > NJ *yugameru/igameru* ‘to bend, to curve, to warp, to distort, to disfigure’

Iban (MP) *liŋaŋ* ‘rocking, swaying, crank, unsteady (as a boat rocking in the water)’

Sound Changes: Liquid Loss, Prenasalization, Coda Loss

Martin (1987. p.788) reconstructs “yugameru (A) < yugame[y]- < yu[re -ka]gamey- < *dura-Ci[-Ci]_Akan-kama-Ci-_A (< ...). ‘warp/bend/distort it’”.

69. pWMP *li(ŋ)get ‘gnash the teeth in anger or impatience’ > pJ *ika-r (feel.angry-NTR) > WOJ *ikar* ‘be angry’

Bontok (MP) *ligət* ‘angry’

Sound Changes: Liquid Loss, Schwa Change, Coda Loss

Martin (1987. p.696) reconstructs “ikaru (A) < *ika-ra-. ‘get angry’”.

70. pPh *lípay ‘joyful, happy’ > pJ *ipa-(a)p (rejoice-ITER) > WOJ *ipap* ‘worship, keep pure’
pJ *ipa-k (rejoice-VRBZ) > MJ? *ipak* ‘act childish’ <https://kobun.weblio.jp/content/いばく>

Cebuano (MP) *lípay* ‘glad; be happy’

Sound Changes: Coda Loss?

Martin (1987. p.700) reconstructs “iwau (B) < ifafu < *ipapa- (<?). ‘congratulate, celebrate’”. Consider also Cebuano *paha-lípay* ‘to congratulate, felicitate’. Coda loss is unexpected for pPh *y but otherwise makes sense semantically.

71. pMP *lumenj ‘smeared, smudged, dirty with’ > pJ *um(ə)-ay (covered.in.dirt-get) > NJ
umeru ‘to bury (e.g. in the ground), to fill up (e.g. audience fills a hall), to cause to be packed’

pJ *umə-r-ay (covered.in.dirt-NTR-get) > WOJ *umore* ‘[unknown; buried, submerged]’,
NJ *umoreru* ‘to be buried, to be covered, to be hidden’

Malay (MP) *be-lumanj* ‘besmeared all over, e.g. with mud as a buffalo after a wallow’

Sound Changes: Liquid Loss, Schwa Retention, Coda Loss

Martin (1987. p.778) reconstructs “umeru (A) < umey- < *uma-Ci-. ‘bury; fill up, plug (a gap)’”.

72. pAN/pMP *lusuj ‘rice mortar’ > pJ(P) *usu > WOJ *usu* ‘mortar’

Okinawan (Iheya) *u:ʃei* ‘mortar’

Yami (MP) *osoŋ* ‘rice mortar’

Sound Changes: Liquid Loss, Coda Loss

Martin (1987. p.564) reconstructs “usu (2.4) < ‘mortar’... Cf. *utu*_{72.1} ‘hollow’”. This important loan has been observed before by both Benedict (1990) and Robbeets (2017). Benedict (1990. p.221) compares “MORTAR: *lutsuj > Jp. usu”.

73. pAN *mamah ‘parent’s younger sibling, junior uncle; a young child’s term of address for his father, vocative of *ama’ > pJ *mama (step-relative) > NJ *mamako* ‘stepchild’

Miyako (Irabu, Sawada) *mamasa* ‘a stepfather’

Tagalog (MP) *máma-ʔ* ‘term used in referring to, or in addressing a man unknown to the speaker’

Sound Changes: Coda Loss

Martin (1987. p.471) reconstructs “mama (4.2 < mama-fawa (JP) < mama-bafa (Mg [ʔ]) < *mamaʔ(2.1) (-n-) papa_{72.1}”. Although the morpheme pJ *mama does not appear in OJ, the Sawada example supports its reconstruction *mama-saʔ (step-father?). This word being borrowed reflects a long period of contact between Japonic and Austronesian populations and perhaps some type of intermarriage.

74. pMP *mamaq ‘chew without intending to swallow, as betel nut; premasticate food to give to an infant; premasticated food’ > pJ *mama (chewed.rice) > NJ *mama* ‘cooked rice (children’s word)’

Ilokano (MP) *mamá* ‘a concoction for chewing made of betel leaf’

Sound Changes: Coda Loss

This word unfortunately has few internal attestations. The phonetic and semantic correspondences, and Sundanese *mamah* ‘word used to order a child to eat; also the usual word for ‘to eat’ of a small child’ make this a likely loan, however. Benedict (1990) compares “CHEW/: P-Austro-Tai *mamaq > Jp. mama~ mamma ‘cooked rice’ (Nelson 1974: a child’s word)”. This word also appears to have been borrowed into Koreanic, NK 밥 *mamma* ‘meal (childish)’ whether through pJ or pAN is uncertain though.

75. pAN *najam ‘accustomed to, familiar with; tame’ > pJ(P) *nara-(a)y (familiar-get) >

WOJ *nare* ‘become familiar and habitual’

pJ(P) *nara-s (familiar-TRZ) > WOJ *naras* ‘train’

Okinawan (Iheya) *narei*_N ‘get familiar with’

Amami (Okinoerabu) *narijun* ‘get familiar with’

Amis (Formosan) *nanam* ‘to be familiar with, accustomed to’

Sound Changes: Coda Loss

Martin (1987. p.732) reconstructs “nareru (B) < nare[y]- < *nara-Ci-. ‘get habituated (accustomed, familiar); become tame, mature’”. This is the only example of pAN *j /gʷ/ > pJ;

rather than positing that pJ borrows this phoneme as *r, several of the target languages change it into pMP *r/l/d, all of which would get borrowed as pJ *r: Yami *ɲaran*, Itbayaten *ɲaran*, Ibaloy *ɲaran*, Bikol *ɲáran*, Hanunóo *ɲáran*, Bontok *ɲádan*, Kankanaey *ɲádan*, Ifugaw *ɲádan*, Tagalog *ɲálan*, Umiray Dumaget *ɲalan*, and Hiligaynon *ɲálan*.

76. pMP *nara/naRah ‘a tree: Pterocarpus indica’ > pJ *nara (type.of.tree) > WOJ *nara* ‘(a kind of) oak tree’

Ilokano (MP) *narrá* ‘leguminous trees with yellow flowers and orbicular pods which yield a valuable timber for construction, etc.: Pterocarpus spp.’

Sound Changes: Exact/Liquid Approximation, Coda Loss

Martin (1987. p.494) reconstructs “nara (2.3). ‘Japanese oak’”. Even though the two types of trees, Pterocarpus indica and oak are different, they look similar, they are both deciduous, and are both used as timber. The use as timber is also emphasized in: Cebuano (MP) *nága* ‘large forest tree yielding a hard, red wood, highly prized for furniture-making: Pterocarpus indicus’ and ‘Palauan (MP) *las* ‘a timber tree: Pterocarpus indicus (L.)’.

77. pMP *ñaman ‘tasty, delicious’ > pJ *nama-su? (fresh.taste-?) > WOJ *namasu* ‘thinly cut meat eaten raw’

NJ *nama* ‘raw, uncooked, fresh’

Malay (MP) *ñaman* ‘healthy feeling; sense of well-being; (Sarawak) tasty, delicious’

Sound Changes: Nasal Approximation, Coda Loss

Martin (1987. p.492) reconstructs “namasu (3.4) (?< *nama_{2.5}su[k[a-C]i]_B ‘raw slice’. ‘(a salad of) thin-sliced raw fish’”. Considering how often raw fish is eaten in Japan, the meaning of ‘raw’ as originating in ‘delicious’ could make sense but is, of course, conjectural.

78. pPh *ɲatu ‘above, on top’ > pJ *natu (summer.solstice)? > WOJ *natu* ‘summer’

Ilokano (MP) *ɲáto* ‘height; upstairs’

Sound Changes: Nasal Approximation

Martin (1987. p.494) reconstructs “natu (2.2a). ‘summer’”. The phonetic closeness here may be accidental, as the semantic drift is strange.

79. pWMP *ɲiaw ‘meow of a cat’ > pJ *nia-ku(r)a (meow-child) > NJ *neko* ‘cat’

Ilokano (MP) *ɲiáw* ‘meow of a cat’

Sound Changes: Nasal Approximation, Coda Loss

Martin (1987. p.495) reconstructs “neko (?2.3, ?2.5) < ne_{mim}kwo_{1.1}. ‘cat’... cf. nyao ‘miaow’”. Considering NJ *nyā*, reconstructing *ne* < *nia is very likely, especially since: pJ *nia > nye# > OJ *ne*, pJ *ne > OJ *ni* (word initial), and pJ *nay > OJ *ne* ‘sound’ is less semantically close.

Francis-Ratte (2016. p.208) reconstructs “pKJ *ko ‘cat’,” through comparison with “MK *kwo:y* ‘cat’. As seen in NJ *hiyoko* (#3.2.100 *piqek), these loans appear to put the pJ affix *ko ‘child after the noun, rather than the typical prefixing *-ko (NJ *koinu* ‘puppy; small dog’ and NJ *koushi* ‘calf; small cow’).

80. pMP *niket ‘sticky, adhesive’ > pJ *nika-(a)p-a? (glue-join/ITER-TR.OBJ) > NJ *nikawa* ‘gelatinous glue (usu. from animal and fish bones, hide, etc.)’

Kankanaey (MP) *niket* ‘resin’

Sound Changes: Schwa Change, Coda Loss

Martin (1987. p.497) reconstructs “*nikawa* (3.1) < *nikafa* < *ni[-Ci]/ni[Ca-Ci]_A*kapa*_{2,3} (‘boil skin’). ‘glue’”. The second element may still be *kapa* if haplology simplified *nika-kapa > *nikapa. A deverbal with transitive object formant *-a is strange here, unless *nika-(a)p was interpreted as a transitive verb.

81. pWMP *padaŋ ‘uncultivated field, open grassland; field covered with sword grass (*Imperata cylindrica*)’ > pJ *para > OJ *para* ‘field, plain’

Miyako (Hirara) *pata-para* ‘the whole of Miyako island (all of its twenty fields)’

Tagalog (MP) *páran* ‘meadow, prairie; a large piece of level or rolling land with grass but not trees; field; uncultivated land’

Sound Changes: Lenition, Coda Loss

Martin (1987. p.399) reconstructs “*hara* (2.3) < *fara* < *para < *paru (the second vowel assimilated to the first) ‘field’... Cf. *haru*, *haruku*, *hareru*.” Francis-Ratte (2016. p.345) reconstructs “pKJ *pəra-a ‘plain’ = ‘that which has been spread out’,” through comparison with MK *pól* ‘plain’”. This reconstruction would yield pJ *pəra, and, as established in #3.1.1 pMP *barbar, OJ *par* is from pJ *par, not *pəra.

This etymon is one of many describing geography (#3.2.110 *sabaŋ/*sabaq, #3.2.113 *sakat, #3.2.171 *waCaN, #3.2.101 *puqun). OJ *para* does not refer to a cultivated field the way OJ *pata* ‘rice paddy’ does, suggesting pWMP as the origin rather than pJ *par-a (clear.land-TR.OBJ). Additionally, several Tungusic words have a final nasal, which *padaŋ could explain that *par-a could not: “Negidal *palan* (< South.), Spoken Manchu *falən*, Literary Manchu *fala(n)*, Ulcha *pala(n)*, Oroch *pālla(n)*, Nanai *palā* (TMC 2, 32, 313).” (Tsintsius et al., 1957-1977. Cited from <https://starlingdb.org/>).

82. pMP *pager ‘enclosure; palisade around a village; fence around a planted tree or cultivated field’ > pJ *paka-ar (enclosed.field-exist) > WOJ *pakar* ‘measure; unit of labor, shift, allotted task’

Okinawan (Iheya) *hakai*/[_N] ‘to measure’

Miyako (Tarama) *pakal* ‘scales’

Ilokano (MP) *pagér* ‘palisade’

Sound Changes: Devoicing, Schwa Assimilation, Coda Loss

Martin (1987. p.683) reconstructs *hakar* (B) < *fakar* < *paka_{2,3}-ra. ‘measure, calculate; consult (with); plan, strive for’”. The Japonic sense of ‘measure and unit of labor’ may be explained as referring to a plot of a set size to be farmed, tilled, harvested, but this is pure speculation.

83. pWMP *pa(m)pet ‘stopped up’ > pJ *pa^mpa-m (block-VBZ) > NJ *habamu* ‘to keep someone from doing, to stop, to prevent, to check, to hinder, to obstruct, to oppose, to thwart’

Javanese *pampet* ‘stop flowing (as blood)’

Sound Changes: Prenasalization, Schwa Assimilation

Martin (1987. p.682) reconstructs “habamu (B) < fabamu < *panpama- (?< *pam[a-C]i pama-, or ?< *pa[ma-Ci] n[i]pama-). ‘obstruct, hinder’”.

84. pAN *panaq ‘throw something at a target; shoot with bow and arrow’ > pJ *pana-r (shoot-V.I) > WOJ *panar* ‘be expelled’

pJ *pana-t? (shoot-TRZ) > WOJ *panat* ‘separate (t)’, NJ *hanatsu* to fire (gun, arrow, questions, etc.), to shoot

EOJ *panar* ‘to be separated’

EOJ *panas* ‘to separate’

Miyako (Hirara) *panari:z* ‘to be separated’

Ilokano (MP) *pána* ‘arrow’

Sound Changes: Coda Loss

Martin (1987. p.387) reconstructs “hanareru (B) < fanare[y]- < *pana-ra-Ci-. ‘get separate(d), distant’”. Francis-Ratte (2016. p.374) reconstructs “pKJ *pə-na- ‘see + go out’,” through comparison with MK *pwonáy-* ‘releases, sends it’”. Since there are several correspondances pKJ *CəC(a) :: pJ *CaC, it might be more sound to say that is how OK borrowed the pJ terms (pJ *par > OK *pəra > MK *polo*, pJ *para > OK *pəl(a) > MK *pól*, and pJ *pana > OK *pəna? > MK *pwonáy*), Comparing the senses in Japonic shows that *panar* and *panat* are likely related, with Ilokano also being closely related in meaning.

85. pWMP *pantek ‘spotted, dappled’ > pJ *paⁿta-ra (spotted-PROP) > WOJ *padara/podoro/madara* ‘light and scattered falling of snow; lightly fallen and scattered snow or frost; colourful, with deep colours in spots’

Kankanaey (MP) *panték* ‘spotted, speckled; marked with spots; stained’

Sound Changes: Prenasalization, Schwa Assimilation, Coda Loss

86. pWMP *pa(n)tik ‘sound of ticking, flipping, light tapping’ > pJ *paⁿti-k/paⁿtik (flick(-VBZ)) > NJ *hajiku* ‘to flip, to snap, to flick, to repel, to use (an abacus), to calculate, to strum, to pluck the strings (of a guitar, etc.)’

Malay (MP) *panték* ‘striking together two hard substances, esp. flint and steel’

Sound Changes: Prenasalization, Coda Loss?

Martin (1987. p.686) reconstructs “haziku (B) < faziku < *pansika- (< ?). ‘flip, snap; repel’... Cf. haneru, hazeru”. Since WOJ *sopot* preserved final -t, it may be that the final -k here is original rather than derivative.

87. pWMP *pa(n)tuk ‘apex, peak’ > pJ *paⁿtu ‘furthest point’? > NJ *hazu* ‘nock of a bow/arrow’

Ilokano (MP) *pantók* ‘top, apex, summit (mountains, houses); tip of the nose (when long)’

Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.403) reconstructs “hazu < fazu < *pansu ‘arrow notch; expectation’” and Martin (1987. p.687) reconstructs hazumu (A) < fadumu < *pantu-ma- (< ?). ‘bounce, spring, (re) bound’ (vi.): ‘splurge (spend big) on’ (vt.)’”. The semantic gap here is wide but not impossible, since the knocking point of an arrow and bow is where the string is most pulled back. NJ *hazumu* ‘bounce’ could be related to the meaning of pulling the bowstring ‘to its peak’.

88. pWMP *paŋkal ‘visible tree roots above the ground’ > pJ *paⁿk(a) (expose.roots) > WOJ *pag* ‘peel’

pJ *panka-(a)s (peel-TRZ) > WOJ *pagas* ‘separate (v.t.)’

pJ *panka-(a)y (expose.roots-get) > NJ *hage* ‘to lose hair, to become bald, to become bare (e.g. a mountain becomes bare of trees)’

Tagalog (MP) *paŋkál* ‘protruding or visible above the ground (said of surface roots of trees)’

Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.683) reconstructs “hageru (B) < fage[y]- < ?*pa(n)ka-ci- (sic). ‘get stripped/bald’”. See also OC 剝 *p[ɸ]rok (~ *mə-p^ɸrok) ‘flay, peel’ (Baxter-Sagart, 2014).

89. pWMP *pasan ‘insert, wedge in; prepare something for use’ > pJ *pasa-m (insert.between-VBZ) > OJ *pasam* ‘hold between’

pJ *pas(a)-i (insert.between-NMZ) > WOJ *pasi* ‘bridge’

pJ *pasa-ma (insert.between-space) > WOJ *pasama* ‘(narrow) valley’

Okinawan (Iheya) *ha[samu]*_N ‘hold between’

Miyako (Hirara) *pasam* ‘scissors’ (Jarosz, p.614)

Tagalog (MP) *pásan* ‘a well-adjusted wedge’

Sound Changes: Coda Loss, Sibilant Approximation

Martin (1987. p.685) reconstructs “hasamu (B) < fasamu / ?*fazamu < *pa(n)sa-ma-. ‘insert; put/catch between; nip’... Cf. haseru; hazama”. The pJ meaning can be explained as placing something between two things, such as a wedge, to a semantic shift focusing on the ongoing act. WOJ *pasi* is likely from Ainu *pasuy* ‘chopsticks’ instead.

90. pWMP *pataR ‘level ground; plain’ > pJ *pata (hold.things.flat)? > WOJ *pata* ‘loom’

Miyako (Hirara) *pata* ‘woven textiles’

Ilokano (MP) *pátar* ‘level’

Sound Changes: Coda Loss

Martin (1987. p.401) reconstructs “hata (2.2a) < fata < *pata. ‘flag’. The WOJ word has been compared to proto-Tai *ptaw by Vovin (2014b). This entry may provide a possible Austronesian source for the pT etymon.

91. pWMP *pegeŋ ‘hold firmly, concentrate’ > pJ *pəkə > WOJ *poko* ‘spear’
 Maranao (MP) pegeŋ ‘firmly, hold strongly’
 Sound Changes: Schwa Retention, Schwa Assimilation, Coda Loss
 Martin (1987. p.414) reconstructs “hoko (2.3) < foko < *poko. ‘halberd’”.

92. pWMP *pera ‘dry, grainy, brittle’ > pJ *pərə-mp-as > WOJ *porobos* ‘ruin’
 pJ *pərə-mp-boy? > WOJ *porobwi* ‘go to ruin’
 Javanese (MP) *pera* ‘dry, grainy, gritty’
 Sound Changes: Schwa Retention, Liquid Approximation, Open Assimilation
 Martin (1987. p.693) reconstructs “horobosu < forobosu < *poronpo-sa- < ?*poronpa-sa- (with the third vowel assimilated to the fourth...). ‘ruin, destroy’”. This word seems to have some kind of -b verbal suffix (present in the suffix -bwi and -bos). Consider NJ *horohoro* ‘tears or flower petals falling quietly; falling apart, crumbling’.

93. pWMP *peteŋ ‘dusky, twilight, as near the end of day’ > pJ *pəʔtə > OJ *podo* ‘time’
 (Vovin 2022, p.462), EOJ *poto* ‘time’
 pJ *pətə-pətə (twilight-RED) > WOJ *potopoto* ‘almost, nearly’
 pJ *pəʔtə-rə (twilight-ADIZ) > WOJ *podoro* dawn, daybreak
 pJ *pətə-a? > WOJ *potwo* ‘time, period’
 Malay (MP) *pətaŋ* ‘afternoon and evening’
 Sound Changes: Schwa Retention, Coda Loss
 Martin (1987. p.416) reconstructs “hotondo < fotondo < (Hn)[?] fotowoto/fotofofo < foto-foto < *poto poto (cf. hotori, hodo). ‘almost (all)’”. The sporadic prenasalization in this morpheme is supported by the presence of EOJ *poto*, which explains WOJ *potopoto*, perhaps from a meaning like ‘almost (time)’. WOJ *podoro* appears to have the property suffix *-ra changed to *-rə to account for Arisaka’s Law, which does not allow pJ *ə to appear in the same morpheme as pJ *a, *u, and *o.

94. pWMP *pihak/piqak ‘one side, one half’/‘to split, separate sections’ > pJ *pia ‘side, plane’ > WOJ *pye* ‘counter for layers; place; direction; shore’
 EOJ *pe/pi* ‘layer (pe only)’/‘side, shore’
 Miyako (Hirara) *psitui* “1nmr one layer, a single layer, 2n one-layer outfit” (Jarosz, p.620)
 Iban (MP) *piak* ‘side’
 Sound Changes: Deletion, Coda Loss
 Martin (1987. p.403) reconstructs “he(-) (?1.3x) < fye(-) < *piCa (?= *pina_{2.2} but the register is incongruent). ‘separate, apart’. Cf. heya, hedatu, henaru” and “(-)he (1.x) < fye < *piCa (?= *pira_{2.2b}, see hira). ‘layer’. See hitoe, hutae, nanae, kawabe, yae; ? hadae”. Francis-Ratte (2016. p.304) reconstructs “pKJ *pijər ‘layer, set,’” through comparison with “MK pól ‘layer, set of’”. Comparing WOJ *pye* to *pihak/piqak explains the vowel cluster in pJ *pia, while also referring to the side of something, a flat plane, and hence the meaning of layer.

95. pWMP *pindaq ‘to move, change location’ > pJ *piⁿta-r-i (change.direction-NTR-NMZ) > WOJ *pidari* ‘left’
 Miyako (Hisamatsu) *psdal* ‘left’
 Sound Changes: Prenasalization, Coda Loss
 Martin (1987. p.405) reconstructs “hidari < fyidari < *pinta(-)ri. ‘left’... Cf. migi(ri)”.
96. pMP *piŋkaw ‘bent, crooked’ > pJ *piⁿka-m (bent-VBZ) > NJ *higam* ‘have a warped view’
 pJ *piⁿka > NJ *higa* ‘evil, untrue, erroneous’
 Tagalog (MP) *piŋkaw* ‘crooked or twisted (said of the arm)’
 Sound Changes: Prenasalization, Coda Loss
 Martin (1987. p.688) reconstructs “higamu (B) < figamu < *fyigamu ?< *pina_{2.2b}-ka-ma-; ?< *pinka-ma- (?< *pi-muka-ma-). ‘get biased, prejudiced, (mentally) warped’... Cf. higa < figa ‘wrong, twisted’ (adjectival noun), Korean pi(-s)-”.
97. pWMP *pintal ‘to twist strands together’ > pJ *piⁿta > NJ *hida* ‘fold, pleat, gathers, tuck, folds (e.g. of a mountain), crease’
 Tagalog (MP) *pintál* ‘warp; a bend or twist’
 Sound Changes: Prenasalization, Coda Loss
 Martin (1987. p.405) reconstructs “hida (2.3) < fida < ?*fyida < ?*pinta. ‘pleat’”.
98. pWMP *pi(n)tas ‘to cut across, cut off’ > pJ *pita > WOJ *pita* ‘direct, unwavering, fully’
 NJ *hita* ‘earnestly, immediately, exactly’
 Ilokano (MP) *pisát* ‘piece of cloth, rope, etc.’
 Sound Changes: Coda Loss
 Martin (1987. p.409) reconstructs “hita(-) (?2.4) < fyita(-) < *pita (cf. hisa). ‘straight; unswerving, unceasing’. See hitasura, hitaburu; hitao; hitati; cf. hito(-) ‘one’”
99. pAN *piliq ‘to choose, to select; to pick out’ > pJ *piri-(a)p (pick.out-ITER) > WOJ *pirip* ‘pick up’
 pJ *pir(i)-əp? (pick.out-ITER) > EOJ *pirəp/pirip* ‘to pick up’/‘to gather’
 NJ *hiro(w)* ‘pick up’
 Okinawan (Iheya) [*hirin* ‘pick up’
 Ilokano (MP) *ag-pili* ‘to choose, to select’
 Sound Changes: Liquid Approximation, Coda Loss
 Martin (1987. p.689) reconstructs “hirou, hirau (A) [dissimilation?- or, back-formation? < Tappi hurar- (Clarke 84) (sic)] < fyr²/₁fu- < *pir²/₁(-)pa- ‘pick up’... Cf irou = irau”. Other Austronesian languages support the reconstruction of this term as agricultural, referring to removing non-grain material from rice: Paze *pīi-piri* ‘be choosing (as when picking up pebbles from rice grain),
 Paiwan *ki-piliq* ‘be choosy, pick out best bits’. The vocalism is strange with the alternating forms

pirip/pirəp which could possibly be explained through two different resolutions to affixing the iterative suffix **(a)p/-(ə)p?*.

100. pAN **piluk* ‘crippled; bent or twisted, of the leg or foot’ > pJ **piru-m*
(get.paralyzed-VBZ) > NJ *hirumu* ‘to falter, to flinch (from), to recoil (from), to quail (at),
to be intimidated, to feel daunted’
Tagalog (MP) *pilók* ‘twisted (referring to arm, ankle or foot)’
Sound Changes: Liquid Approximation, Coda Loss

Martin (1987. p.689) reconstructs “*hirumu* (B) < *firumu* (< ?). ‘get paralyzed; lose heart; flinch, wince’”.

101. pMP **piqek* ‘baby chick; peeping of chick’ > pJ **piyə-ko* ‘chick-child’ > NJ *hiyoko*
‘young bird, chick, fledgling’
Ilokano (MP) *piék* ‘chick’
Sound Changes: Deletion, Coda Loss, Schwa Retention?

Martin (1987. p.412) reconstructs “*hiyoko* (3.1) < *fiyoko* < **fyiyo-kwo* < **pido_{mim}kwo_{1.1}*-
‘chicken’”. There is a discrepancy in the schwa not changing here, which could be due to the
guttural deletion.

102. pMP **puqun* ‘base of a tree; source, origin; beginning; foot of a hill or mountain’ > pJ
**pu(u)-mətə* > WOJ *pumoto* ‘the foot of a mountain’
Ilokano (MP) *puón* ‘beginning; origin, source; base; root; trunk; lower part; parentage;
ancestry; unit for counting trees’
Sound Changes: Deletion, Coda Loss

Martin (1987. p.418) reconstructs “*humoto* (?3,1, ?3.4) < *fumoto* < **pu[m[a-C]i] moto* (cf.
Kuranaka 104 (sic)). ‘the foot of a mountain’”. The sense in pMP strongly accounts for the
meaning seen in WOJ, even though *pu* ‘foot of mountain’ does not appear outside this compound.

103. pWMP **qaŋ(e)sej* ‘sweaty armpits, stench of sweaty armpits’ > pJ **asay* > NJ *ase*
‘sweat’
Okinawan (Iheya) *æi*: ‘sweat’
Amami (Okinoerabu) *æi* ‘sweat’
Kapampangan *aŋsad* ‘stench from the armpits’
Sound Changes: Deletion, Nasal Approximation, Schwa Assimilation, Coda Retention

Martin (1987. p.385) does not provide a reconstruction for “*ase* (2.5) (< ?). ‘sweat’”. This
comparison assumes loss of the prenasalization but matches phonetically and semantically.

104. pMP **qaRta* ‘outsiders, alien people’ > pJ **ata* (outsider) > WOJ *ata* ‘enemy’
pJ **ata-si* (outsider-ADJZ) > WOJ *atasi* ‘other’
Tagalog (MP) *agtá* ‘short black people, Negritos’
Sound Changes: Deletion, Liquid Approximation, Cluster Reduction*

Martin (1987. p.377) reconstructs “ada (2.2a) < ata < *a(n)ta ‘foe; empty, vain’”. WOJ *atasi* is clearly derived from WOJ *ata*, reconstructing pJ *ata as ‘outsider’, considering the connection between ‘enemy’ and ‘other’. Japonic consonant clusters take on the second of the two, yielding pJ *t from pMP *Rt.

105. pWMP *qayam ‘to play; plaything, toy; pet’ > pJ *aya-tur (plaything-hang) > NJ
ayaturu ‘to operate, work (a puppet)’
 Ilokano (MP) ay-áyam, recreation, diversion, amusement, play, game, sport, toy,
 plaything’
 Comparisons, Loss, Coda Loss

Martin (1987. p.679) reconstructs “ayatur (?A/B) ?< *aya-t(w)oru < *ada_{2,3}t(w)ora-B.
 ‘manipulate, work’”. This etymon unfortunately does not have an independent basis for
 reconstructing pJ *aya ‘plaything’ but pJ *tur ‘hang’ would accurately explain the meaning in NJ
 referring to puppets.

106. pMP *qilaŋ ‘lose, disappear, be lost (as possessions)’ > pJ *ira-na (loss-EMP) > WOJ
irana ‘sad’
 Malay (MP) *hilang* ‘loss, disappearance, death; of loss generally’
 Sound Changes: Deletion, Liquid Approximation, Coda Loss

107. pMP *qilaw ‘torch; illuminate’ > pJ *ir(a) (light.aflame) > WOJ *ir* ‘scorch’
 pJ *ak(a)-i(r)? (bright-light.aflame) > WOJ *aki* ‘clear, manifest’
 pJ *ak(a)-ir.a-kay (bright-light.aflame-EP-ADJZ) > WOJ *akirake* ‘brilliant’
 pJ *ak(a)-ir.a-m-ay (bright-light.aflame-EP-VBZ-get) > WOJ *akirame* ‘brighten; clear’
 pJ *kag(a)-ir(a) (shade-light.aflame) > WOJ *kagir* ‘radiate’
 Tagalog (MP) *ilaw* ‘light; lamp, torch, i.e. anything that is used to give light’
 Sound Changes: Deletion, Liquid Approximation, Coda Loss

Martin (1987. p.698) reconstructs “ir-u (B) < *ira-. ‘roast’”. The examples included besides *ir* are
 speculative but semantically support -*ir* ‘scorch’ being attached to two words including light:
 WOJ *aka* and WOJ *kage*.

108. pAN *qinep ‘lie down to sleep’ > pJ *i- > OJ *i* ‘sleep (n)’
 pJ *(i)na-(a)y (sleep-get) > WOJ *ine/ne* ‘sleep (v.i)’, EOJ *ne* ‘id.’
 pJ *na-s (sleep-TRZ) > WOJ ‘make sleep’
 pJ (*i > yu)nəp/-p? (sleep/-ITER?) > WOJ *yunop* ‘sleep’
 pJ (*i > yu)n(ə) (sleep) > EOJ *yun* ‘sleep’
 pJ (*i > yu)-may (sleep-eye) > NJ *yume* ‘dream’, WOJ *ime* ‘dream’
 Miyako (Hirara) *niv* ‘to sleep’
 Tagalog (MP) *panag-inip* ‘deep dream (not easily remembered in detail)’
 Sound Changes: Deletion, Schwa Change, Coda Loss?

Martin (1987. p.420) reconstructs “i (1.3a) i. ‘sleep’: see *ineru*; *ime*; *uma-i*”. The example here contains explanatory power for all iterations of the morphemes **ina*, **i*, and **na*. In *yunop*, **qinəp* appears to be borrowed directly (with guttural deletion), or with the iterative suffix *-p*, and *yun* appears to be from coda loss, which would result in deletion of schwa (**qinəp* > **inə* > **yun(ə)*). It is possible that the variant *yunop* should be interpreted as a verbal form of MJ *oyor* ‘to sleep’ with a correspondence between EOJ *-n-* and WOJ *-r-* as Vovin posits (2022. p. 164-166). However, the explanation above explains all the variants through the source **qinep*.

109. pMP **qisaq* ‘interjection of negative emotive force’ > pJ **isa* > WOJ *isa* ‘interjection’
 pJ **isa-m-ay* (scold-VBZ-get) > WOJ *isame* ‘remonstrate’
 pJ **isa-kap-i* (scold-exchange-NMZ) > NJ *isakai* ‘quarrel, dispute, argument’
 Isneg (MP)? *issá* ‘to shoo chickens’
 Sound Changes: Deletion, Coda Loss

Martin (1987. p.699) reconstructs “*isameru* (?A/B) *isabe[y]-*, “*isaf/me[y]-*” **isanpa-Ci-* < **isa-n[a]-pa-Ci-*. ‘admonish’... **isame-* ‘stimulate, spur on’. Nr also *asamey-?* Cf. *isabu. isame.*” Although there is not an example in a target MP language, the sense in pMP and Isneg shows *isa-me* as being segmentable. WOJ *isa* may not be related, but is included because of its interjectionary usage.

110. pMP **rapet* ‘join together, attach tightly’ > pJ **ap(a)* > OJ *ap* ‘meet’
 pJ **apa-(a)y* (join-get) > WOJ *ape* ‘join’
 pJ **apa-s-ay* (join-TRZ-get) > WOJ *apase* ‘join’
 pJ **ku(p)-rap* (eat-join) > WOJ *kurap* ‘eat and drink’
 Okinawan (Iheya) *a:sun* ‘to join’
 Ilokano (MP) *rapét* ‘to cling on to’
 Sound Changes: Liquid Loss, Schwa Assimilation, Coda Loss

Martin (1987. p.679) reconstructs “*au* (B) < *afu* < **apa-*. ‘meet; fit, agree’”. Francis-Ratte (2016. p.315) reconstructs pKJ **ap-* ‘meets, joins,’ through comparison with MK *awól-* / *ewúl* ‘fits together’. Liquid loss has been present for every word-initial example, but the word *kurap* has a peculiar liquid consonant in it. WOJ *kup* ‘eat’ is likely the first part of this compound which does not have the iterative suffix *-(a)p* or it would become *kupap*. Instead, the liquid may have been preserved through an analytically derived **kup-rap* simplifying the cluster **pr* > OJ **r*. See also OC 𐀓𐀓 **m-rʰəp* ‘join, unite’, which also has an initial liquid. (Baxter-Sagart, 2014). Both this OC and pMP example show the directionality: OC < pMP > pJ > pK, since MK *awól-* lacks a liquid initial that was likely lost in Japonic.

111. pMP **sabaŋ* ‘estuary, shore near the mouth of a river’ > pJ **sapa* > WOJ *sapa* ‘mountain stream; swamp’
 Ilokano (MP) *sabáŋ-an* ‘mouth (of a river); harbor, port, haven’
 pWMP **sabaq* ‘irrigated ricefield’

Malay (MP) *sawah* ‘land cultivated with swamp rice, whether the rice is planted in a natural swamp (*sawah ayer*) or on irrigated soil (*sawah bencah*); rice is planted usually in small lots separated from one another by low dykes (*batas*)’

Sound Changes: Devoicing, Coda Loss

Martin (1987. p.520) reconstructs “sawa (2.1) < safa < *sapa. ‘swamp, bog; valley; lots, much/many’”. Although WOJ *sapa* does not refer to rice fields, this may be because WOJ *pata* ‘paddy field’ subsumed the meaning. Seeing as Malay *sawah* shows how rice plots are separated by low dykes (*batas*), which could have come to refer to early natural paddy fields compared to WOJ *ta* ‘field/NJ *ta* ‘rice field’. Benedict (1990. p.251) compares “SWAMP/: P-Austro-Japanese *[ts, tʃ]abaq > Jp. sawa, OJ saFa ‘swamp, marsh’”.

112. pWMP *sabeD ‘barrier, obstacle’ > pJ *sapa-(a)r (barrier-exist) > WOJ *sapar* ‘be hindered’

pJ *sapa-(a)y (barrier-get) > WOJ *sape* ‘plug’, EOJ *sape* ‘to obstruct’

Okinawan (Iheya) [*sawa*]IN ‘to touch’

Tagalog (MP) *sabid* ‘hindrance, obstacle around’

Sound Changes: Devoicing, Schwa Assimilation, Coda Loss

113. pWMP *sadab ‘to singe, burn off’ > pJ *sara-s (expose.to.fire-TRZ) > WOJ *saras* (Ex. *saray > sare) ‘expose (to the elements); dry; bleach; wash’

Ilokano (MP) *sárab* ‘to singe, scorch; put into direct heat’

Sound Changes: Lenition, Coda Loss

Martin (1987. p.747) reconstructs “sarasu (A) < *sara-sa- ‘bleach; expose’”.

114. pAN *sakat ‘rise, climb up’ > pJ *saka > OJ *saka* ‘slope, hill’

pJ *saka-r (rise-exist) > WOJ *sakar* ‘be at peak’

Okinawan (Iheya) [*saka*] ‘slope’

Amis (MP) *cakat* ‘arise, go up’

Sound Changes: Coda Loss (Nominal)

Martin (1987. p.539) reconstructs “saka (2.3). ‘slope; (= sakai) boundary’”.

115. pMP *sakup ‘to catch’ > pJ *saⁿku?-r (catch-NTR) > WOJ *sagur* ‘hunt for’

Cebuano (MP) *sakúp* ‘to catch, capture (as a runaway pig; someone who is cheating); come upon someone doing something’

Sound Changes: Prenasalization?, Coda Loss

Martin (1987. p.745) reconstructs “saguru (A) (< ?). ‘search/grope for’”. This word is poorly attested and lacks a nasal cluster in pMP but otherwise appears related phonetically and semantically, but may just as well be a coincidence.

116. pWMP *sa(ŋ)kuŋ ‘curved inward, sunken’ > pJ *saku-r (hollow.out-NTR) > WOJ *sakur* ‘dig a well’

Cebuano (MP) *sakún* ‘bent downward more than it should be (as a spring pole used with noose traps)’

Sound Changes: Coda Loss

Martin (1987. p.747) reconstructs “sasakuru (B) < *sakura- (?< *sa[ka-Ci]_Bkura-B). ‘scoop up (dirt/water); dig a ditch; get choked up, sob’”.

117. pPh *salét ‘to insert between, intersperse’ > pJ *sara-ni (to.interject-ADVZ) > NJ *sara ni* ‘furthermore, again, after all, more and more, moreover, even more’

pJ *sara-sara (to.interject-RED) > WOJ *sarasara* ‘furthermore; on top of everything else’

Tagalog (MP) *salít* ‘interspersing; alternately putting something in between; alternating’

Sound Changes: Liquid Approximation, Schwa Assimilation, Coda Loss

This match is semantically strange and may be omitted.

118. pWMP *sama ‘mate, companion, fellow; together with; alike, the same as; aligned or in good order’ > pJ *sama (HON.mate) > WOJ *sama* ‘way, means, reason’

Okinawan (Iheya) *zama* ‘way’

Tagalog (MP) *ka-sáma* ‘companion; partner; associate; companion, as on a trip; person living with another in a house; anything included, attached, or enclosed with something’

Sound Changes: Exact

Martin (1987. p.517) reconstructs “sama (?2.2a ?< 2.1) sama. ‘appearance; direction; ...’”.

Francis-Ratte (2016. p.266) reconstructs “pKJ *sama- ‘forms, shapes, makes it into it’,” through comparison with “MK *sa:m-* ‘makes it into’”. The polysemy in pWMP shows how WOJ *sama* appears and how NJ *sama* becomes an honorific suffix, similar to humble NJ *domo* from WOJ *tomo* ‘together’. Compare NJ *you* written with the same character 様 and used following a demonstrative as an adverb (c.f. WOJ *kaku sama* ‘this way’ (MYS 15.3761) and NJ *kono you* ‘id.’).

119. pWMP *sandaR ‘lean on or against’ > pJ *saⁿta-m-ar (lean.on-AGT-exist) > WOJ *sadamar* ‘be fixed’

pJ *santa-m-ay (lean.on-VBZ-get) > OJ *sadame* ‘decide, pacify’

pJ *santa (a.fixed.thing/time) > WOJ *sada* (time, opportunity)

Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.515) reconstructs “sadame (3.4) < sadamey < *santa-ma-Ci[-Ci]_B. ‘decision; rule’”. The semantic change here may be too convoluted, but can be explained as an object that is placed as a diagonal support becoming fixed in its position. C.f. NJ *motaseru* ‘to lean (something) against’ from WOJ *mot* ‘hold’.

120. pPh *saŋkap ‘well-equipped, having what one needs to do live or do a job’ > pJ *saⁿka > NJ *saga* ‘one's nature; custom, tradition, habit, convention’

Tagalog (MP) *saŋkáp* ‘component part; element’

Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.515) reconstructs “saga (?2.4) (?< Ch syang’ or sya(y)ng’). ‘characteristic; fate, ...’”. This word is not attested in the Old Japanese corpus but does have a comparable phonetic structure and a simple semantic shift from describing particular qualities one regularly exhibits to essentializing them as a natural feature. Additionally, this word is written with the character 質, which is also used for qualities and elements.

121. pWMP *sangaq ‘to arrest the progress of something by stopping or catching it; to support, keep from falling’ > pJ *saⁿka-r (go.down.slowly) > WOJ *sagar* ‘go down’
 pJ *sanka-si (go.down-ADJZ) > WOJ *sagasi* ‘steep’
 Okinawan (Iheya) *agisagi* ‘to lower’
 Ilokano (MP) *sanga-en* ‘to fend off; turn aside (with a sword); to shield; deflect’
 Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.745) reconstructs “sagaru (B) < *sanka-ra-. ‘descend, go/come down, sink, abate; hang down’”. The phonetic match present is an exact match, except the consonant coda, however the semantic connection here is obscure. Itbayaten has a descendant, *sanga-en* ‘to catch with cupped hands’ that may explain the proto-Japonic morpheme borrowing the meaning as ‘to fall with resistance/to lower’. The existence of other synonyms WOJ *oti* ‘fall’ and WOJ *ori* ‘go down’ (possibly related, with the latter having t-lenition) and WOJ *kudar* ‘go down’ (likely related to WOJ *kuda* ‘tube, pipe’, as water flowing down a bamboo tube, such as NJ *shishi odoshi* ‘deer-frightener’), suggests pJ *sanka is a borrowing.

122. pPh *sapát ‘sufficient, enough’ > pJ *sapa > WOJ *sapa* ‘many; splendid’
 pJ *sapa-ⁿ-ta (many-GEN-rice.field) > EOJ *sapaⁿda* ‘many’
 Tagalog (MP) *sapát* ‘ample; enough; adequate; decent; sufficient; enough; as much as is needed’
 Sound Changes: Coda Loss
 Synonyms: WOJ *mane* ‘many’

Martin (1987. p.520) reconstructs “sawa (2.1) < safa < *sapa. ‘swamp, bog; valley; lots, much/many’.

123. pAN *sapet ‘grab with the hand, catch’ > pJ *sapa-r (touch-NTR) > NJ *sawaru* ‘to touch, to feel’
 Miyako (Hirara) *sa:z* ‘1. to touch 2. to be tasty 3. to disturb’ (Jarosz, p.625)
 Amis (Formosan) *capet* ‘to grab with the hand; catch’
 Sound Changes: Schwa Assimilation, Coda Loss

Martin (1987. p.748) reconstructs “sawaru (A) < safaru < *sapa-ra-. ‘hinder, hamper, affect; touch, feel (vi.)’”.

124. pWMP *sapiq ‘agreement, harmony’ > pJ *sapi-ⁿ-tur*
 (harmonize-GEN-make.a.resonant.sound) > WOJ *sapidur* ‘chirp’
 Tagalog (MP) *sápi?* ‘membership, act of becoming a member; joining, uniting’

Sound Changes: Coda Loss

See #3.1.55 *tuRtuR for the latter morpheme.

125. pWMP *sarak ‘parting, separation, as in divorce’ > pJ *sar(a) > OJ *sar* ‘move on; leave’

Tagalog (MP) *sálak* ‘(among Tinguians) separation of a married couple as a form of punishment for cause (Panganiban 1966)’

Sound Changes: Coda Loss

Martin (1987. p.747) reconstructs “saru (?A) < *sara- ‘depart’; also (OJ) ‘move (forward)’”.

126. pWMP *sasag ‘wattle, bamboo plaitwork construction for walls’ > pJ *sasa > WOJ *sasa* ‘broad-leaf bamboo’

EOJ *sasa* ‘bamboo grass’

Tagalog (MP) *sásag* ~ *saság* ‘split bamboo used for walls’

Sound Changes: Coda Loss

Martin (1987. p.518) reconstructs “sasa (2.1). ‘bamboo-grass’”.

127. pMP *sepet ‘obstructed, of the flow of water’ > pJ *səpət-i (be.full.of.water-CONN) > *sopot/sopoti* ‘be wet’

Ilokano (MP) *seppét* ‘be choked (plants); be obstructed (liquids, the meshes of a net or strainer being clogged or filled up)’

Sound Changes: Schwa Retention?

Martin (1987. p.754) reconstructs “sobotu (?A) < ‘so[f]odu’ < sofotu - sobotiru < soboti[y]- < sofod-/sofodi[y]- ?< *sopo(-n)tu/oCi-_{*B}. ‘get drenched’”.

Perhaps the most straightforward in both semantics and phonotactics, especially with final /t/ being kept which is peculiar but undeniable. This may be explained as a morphologically preserved fossil, since *-i places /t/ in a medial position, similar to the example with #3.2.109 *rapet preserving the otherwise initial liquid *r in the medial position *ku(p)-rap > WOJ *kurap*.

128. pPh *sibáq ‘gluttony, uncontrolled appetite’ > pJ *sipa-ku (greedy-ADJZ) > NJ *siwai* ‘stingy, cheap, very frugal’

Tagalog (MP) *síba?* ‘gluttony; greediness; voraciousness’

Sound Changes: Devoicing, Coda Loss

129. pWMP *sikanj ‘wide apart, of the legs’ > pJ *si(n)k(a) (spread.apart) > OJ *sik* ‘reach, catch up with; spread; (waves, etc.) repeatedly approach, (plants, etc.) thicken’

WOJ *sige* ‘rife, overgrown, abundant’, EOJ *sike/sikye/sigye* ‘id’

Tagalog (MP) *síkanj* ‘crosspiece between posts of a house; anything like a wedge put crosswise into an opening to prevent it from closing in; keeping the eyes open to prevent falling asleep’

Sound Changes: Coda Loss

Martin (1987. p.751) reconstructs “siku (A) < *sika-. ‘spread it (out); control, rule’”.
This example appears to involve an innovative optional nasal in Japonic, relating WOJ *sik* and WOJ *sige* which shows the nasal is optional in EOJ *sike*.

130. pMP *silaw ‘dazzling, of light’ > pJ *sira-/sir.o? (bright) > OJ *sira/sirwo* ‘white’
Okinawan (Iheya) [*ɛiru*]/*sa*/N ‘white’
Miyako (Hirara) *sira-kumu* ‘white clouds’
Ilokano (MP) *sílaw* ‘light’
Sound Changes: Liquid Approximation, Coda Loss

Martin (1987. p.526) reconstructs “siro (2.5) < sirwo (?< *sir[a-]uA [ir]o_{2.3} but register incongruent)... Cf. sira(-)_{72.5}, siro_{adj B}, sir-u_A; kuro: kura- kuso: kusa-.”

Interestingly, no other color words are comparable: pJ *aka does not resemble a single etymon for ‘red’ in proto-Austronesian languages: pAN-F *taNah, pMP *rara, *pula(q), *deRaŋ, *ma-iRaŋ; pJ *awo ‘blue’ also does not resemble pMP *buluŋ ‘dark green/dark blue’ or *ma-qitəm ‘black, deep blue’. Interestingly though, pAN *pulaq/pMP *pula ‘red’ does highly resemble proto-Tungusic *pula- ‘red’ (Starostin et al., 2003. p.232): e.g. Manchu □□□□□□□□ (*fulgiyan*) ‘red’, Evenki хулама (*hulama*) ‘red’.

131. pMP *simbuR ‘to sprinkle’ > pJ *si^mpu-k? (splash-VRBZ) > WOJ *sibuk* ‘to splash; to spray’
Ilokano (MP) *sibúg* ‘water used for watering plants’
Sound Changes: Coda Loss, Prenasalization

The final verbal consonant here may be an approximation of pMP *R /R/ (c.f. *sibúg*), or attaching the verbal formant *-k.

132. pMP *si(m)pet ‘close, shut off’ > pJ *si^mp(ə)-ar (close.up-NTR) > NJ *sibaru* ‘to tie, to bind, to fasten’
pJ *simpə-(a)r/*simpə-r (close.up-NTR) > NJ *siboru* ‘to wring (towel, rag), to squeeze’
pJ *sima-p-r > pJ *sima-r > WOJ *simar* ‘bind’
Miyako (Hirara) *šibal ‘narrow’ (Jarosz, p. 628)
Miyako (Hirara) *sibu:z* ‘to squeeze, to wring (id., p.635)’
Sasak (MP) *sémpet* ‘close, shut off’
Sound Changes: Prenasalization, Coda Loss, Schwa Retention

Martin (1987. p.749) reconstructs “siboru (B) < *sinpo(-)ra- (< ?). ‘wring, squeeze’”.

133. pWMP *sipak ‘crack apart at the joints’ > pJ *sipa > NJ *siwa* ‘wrinkle, crease, ripple’
pJ *sip(a) > WOJ *sip* ‘go senile, go numb’
pJ *sipa-ⁿ-puk > WOJ *sipabuk* ‘cough’
Okinawan (Iheya) *ɛiwa*: ‘wrinkle’
Tagalog (MP) *sipák* ‘rent; split; cracked’
Sound Changes: Coda Loss

Martin (1987. p.528) reconstructs “siwa (2.1) < *siba. ‘wrinkle’”.

134. pPh *sukáy ‘to part the hair, as when searching for lice’ > pJ *suk(a) > NJ *suk* ‘to comb (out), to card, to untangle (hair), to thin (hair), to reduce in volume’
Bikol (MP) *mag-sukáy* ‘to part the hair, usually by using the fingers; to search for something on the scalp by parting the hair in this way’
Sound Changes: Coda Loss?
135. pAN *suked ‘prop, support; to prop up or support’ > pJ *suka-(a)y (support-get) > WOJ *suke* ‘rescue’, EOJ *suke* ‘assistant’
Amis (Formosan) *coker* ‘to prop up, support, help, as the cross pole in the hitch of a wagon’
Sound Changes: Schwa Change, Coda Loss
136. pWMP *suker ‘trouble, toil, difficulty’ > pJ *suka-na (mind.at.unease-EMP) > WOJ *sukana* ‘depressed, melancholic’
Malay (MP) *sukar* ‘arduous; (of an illness) hard to cure’
Sound Changes: Schwa Change, Coda Loss
137. pMP *sukul ‘cornered, unable to escape’ > pJ *suku-m (trapped.and.afraid-VBZ) > NJ *sukumu* ‘to freeze (from fear, etc.), to be unable to move (e.g. from surprise), to be paralyzed (with horror, etc.)’
Tagalog (MP) *ma-sukól* ‘to be surrounded, cornered, or enclosed; to be driven into a corner’
Sound Changes: Coda Loss
138. pMP *su(ŋ)kab ‘to open, break open’ > pJ *suk(a) > WOJ *suk* ‘to hoe’
Miyako (Hisamatsu) *sukus* ‘spade, plough’
Ilokano (MP) *sukáb* ‘open from below, at the bottom (e.g. when extracting a mollusk from its shell, when a thief breaks in, etc.)’
Sound Changes: Coda Loss
139. pMP *suŋkul ‘poke, prod with the tip of something’ > pJ *suⁿku ‘pass/poke (through)’
> NJ *sugu* immediately, at once, right away
pJ *sunku-r > WOJ *sugur* ‘pass’
pJ *sunku-s/*sunku-as > WOJ *sugus/sugwos* ‘let pass’
pJ *sunku-y > WOJ *sugwi* ‘pass time’
Tagalog (MP) *suŋkól* ‘a stroke, knock or blow with the knuckles’
Sound Changes: Prenasalization, Coda Loss

Martin (1987. p.532) reconstructs “sugu (?2.5) < *sunku. ‘straight; immediate’... Cf. *sugiru*_B, *sugosu*_B; *suzi*_{2,4}*su*^{~?}_{1.3a}, *muku*_A”.

The shift in meaning in NJ can be compared to #3.2.97 *pi(n)tas ‘to cut across, cut off’ becoming NJ *hita* ‘earnestly, immediately, exactly’. Kelabit *ŋ-ukul* ‘poke into a hole with a finger, stick, etc.’ shows how the pMP meaning changed to ‘pass/poke through’.

140. pWMP *su(ŋ)kuŋ ‘stooped, bent (as the back)’ > pJ *suku-m-ay (stoop-VBZ-get) > NJ *sukumeru* ‘to shrug (shoulders), to duck (head), to draw back (body)’
 una-suku-m-ar
 pJ *pinsa-ma-ⁿ-suk (knees-GEN-stoop) > NJ *hizamazuku* ‘to kneel (in respect), to genuflect’
 pJ *tuma-ⁿ-suk (nails-GEN-stoop) > NJ *tumazuku* ‘to trip (over), to stumble, to fail, to suffer a setback’
 pJ *nuka-ⁿ-suk (forehead-GEN-stoop) > NJ *nukazuku* ‘to kowtow (to bow from a kneeling position such that the forehead touches the ground)’ (c.f. NJ *nukagami* ‘forelock, hair on top of the head’)
 Toba Batak (MP) *suhuy* ‘stooped, bent; tie hands to feet’
 Sound Changes: Coda Loss

Martin (1987. p.759) reconstructs “sukumeru (A) < sukumey- < *sukuma-Ci-. ‘crouch (one's body); make (one's body) small, shrink; duck (one's head)’”.

141. pWMP *sulaŋ ‘to pass one another in coming from opposite directions’ > pJ *sur(a) > OJ *sur* ‘rub; scrape’
 Malay (MP) *sulaŋ* ‘to meet and pass over and under one another when coming from all directions’
 Sound Changes: Coda Loss, Liquid Approximation

Martin (1987. p.759) reconstructs “suru (B) < *sura-. ‘rub; grind; print; pick one's pocket’”.

Francis-Ratte (2016. p.368) reconstructs “pKJ *sura- ‘rubs together’,” through comparison with “MK *swolóy* ‘sound’ ?< pre-MK *swolo- + deverbal *-i, which implies a root *swolo- ‘makes a sound’”. Both comparisons are somewhat semantically distant, although a meaning based on friction might be closer than one only based on sound.

142. pWMP *sumpel ‘stopper, plug, cork’ > pJ *su^mpə-m (close.up-VBZ) > NJ *subomu* ‘to get narrow, to contract’
 pJ *sumpə > pJ *tumpə > WOJ *tubo* ‘jar’
 pJ *tumpə-m (tighten-VBZ) > NJ *tubomu* ‘to close (of flowers), to shut, to get narrower’
 Toba Batak (MP) *sumpol* ‘stopper, cork’
 Sound Changes: Prenasalization, Coda Loss, Schwa Retention

Martin (1987. p.757) reconstructs “subomu (A) < *sunpo_{adj B}-ma- ?< *sunpa-ma (with partial vowel assimilation, cf. subaru). ‘get narrow, contract’ (= subomaru)” and (p.552) reconstructs “tubo (2.1a) < tufo (?< *tupo; ?= tuwo < tu[b]o; “also tufu”). ‘jar’”. These two morphemes are likely related considering NJ *tubomu*.

143. pWMP *tabaR ‘tasteless, insipid; powerless; antidote, as against snakebite’ > pJ *tapa > WOJ *tapa* ‘foolish’

pJ *tapa-ⁿ-koto (foolish-GEN-words) > NJ *tawagoto* ‘nonsense, silly talk’

Iban (MP) *tabar* ‘tasteless, insipid, unseasoned’

Sound Changes: Devoicing, Coda Loss

Martin (1987. p.766) reconstructs “tawakeru (?A/B) < tafakey < *tapa-ka-Ci-. ‘fool around; misbehave, engage in adultery’”. The semantic gap here is strange but not absurd, ‘something that is without taste or power’ is ‘useless’ or ‘foolish’.

144. pPh *takéd ‘tie down, fasten by binding’ > pJ *tak(a) > WOJ *tak* ‘comb and bind up’

Ibaloy (MP) *taked* ‘a tie, fastening, binding --- what is wrapped around to hold something in place’

Sound Changes: Schwa Assimilation, Coda Loss

Martin (1987. p.762) reconstructs “taku (?) < *taka-: (1) ‘comb (one's hair) up into a bun’; (2) ‘row hard’; (3) ‘pull a net in/up’; (4) ‘handle a horse with rein’”. These are not all necessarily related to *takéd, but the first sense almost certainly is.

145. pAN *takut ‘fear’ > pJ *taku-m-asi (fear-VBZ-ADJZ) > NJ *takumasii* ‘burly, strong, sturdy, indomitable, indefatigable, strong-willed, resolute, bold, robust’

Malay (MP) *takut* ‘fear’

Sound Changes: Exact

Martin (1987. p.841) reconstructs “takumasi- (B) < takuma-_B (< *ta_{1.3a}kuma-_B)-si-. ‘stout; stalwart, bold’”. NJ *takumasii* implies an unattested verb pJ *taku-m (fear-VBZ) ‘to frighten’ since several adjectives ending in -asi(ku) are derived from verbs (e.g. NJ *yorokobu* ‘to be delighted, to be glad, to be pleased’ and NJ *yorokobasii* ‘joyful (news, event, etc.), joyous, delightful (e.g. result), pleasing (e.g. sight)’”).

146. pWMP *tandes ‘state explicitly, stipulate’ > pJ *taⁿta > WOJ *tada* ‘directly; just, only; precisely’, EOJ *taⁿda* ‘directly’

pJ *tanda-si (direct-ADJZ) > NJ *tadashi* ‘right, correct’

Miyako (Hirara) *tada* ‘only, simply’

Tagalog (MP) *mag-tandis-an* ‘to have a definite understanding about something’

Sound Changes: Prenasalization, Schwa Assimilation, Coda Loss

Martin (1987. p.537) reconstructs “tada (?2.4, ?2.5) < *tanta. ‘direct[...]’”.

147. pWMP *ta(m)benj ‘block, obstruct’ > pJ *tap-ay (block-get) > WOJ *tape* ‘withstand’

Sasak (MP) *tabəŋ* ‘to dam off’

Sound Changes: Devoicing, Schwa Assimilation, Coda Loss

Martin (1987. p.761) reconstructs “taeru (B) < tafe[y]- < *tapa-Ci-. ‘endure, withstand; block, intercept’” and (p.767) “(-)tawaru (B) < (-)tafaru < *tapa-ra- ‘be blocked’... Cf. taeru.”

148. pWMP *ta(m)pas ‘lop off, cut off’ > pJ *tampa > NJ *taba* ‘bundle, bunch, sheaf’
Tagalog (MP) *tápas* ‘dehusking of coconuts; lopping of trees’

Sound Changes: Coda Loss, Prenasalization

Martin (1987. p.536) reconstructs “*taba* (2.4) < *tanpa. ‘bunch’”. This word is only attested in modern Japanese and has a very limited distribution but appears phonetically sound and semantically close. Finding a dialect distribution across Japonic may assist in finding the routes this lemma may have taken.

149. pAN *tama ‘appropriate, suitable, right; fit together; hit the mark’ > pJ *tama > NJ *tama ni* ‘occasional, infrequent, rare’

Ilokano (MP) *t<um>áma* ‘to hit with a missile; hit a target’

Sound Changes: Exact

Martin (1987. p.539) does not provide a reconstruction for “*tama* (ni/no) (2.1). ‘occasional’”. The semantic shift observed in *tama* is observable in the use of the word *occasionally* as *sometimes*.

150. pMP *taŋad ‘look upward’ > pJ *tana (higher.area) > WOJ *tana* ‘shelf, plank in boat’

pJ *ta-tana-ⁿ-tuk (RED-higher.area-GEN-attach) > WOJ *tatanaduk* ‘pile up (i)’

pJ *tana-ⁿ-pik (higher.area-GEN-pull) > WOJ *tanabik*, EOJ *tana^mbik* ‘to trail’

Okinawan (Hirara) [*tana*] ‘shelf’

Miyako (Hirara) *tana* ‘shelf’

Ilokano (MP) *ag-taŋ-táŋad* ‘to walk with the head up; to be looking up’

Sound Changes: Coda Loss, Nasal Approximation

Martin (1987. p.541) does not include a reconstruction for “*tana* (2.1). ‘shelf’”.

151. pWMP *taŋgap ‘receive, accept’ > pJ *tanka-(a)y (receive-get) > WOJ *tage* ‘eat/drink’

pJ *tank(a) > EOJ *taⁿg* ‘to eat’

Tagalog (MP) *taŋgáp* ‘accepted, admitted, received’

Sound Changes: Coda Loss

Martin (1987. p.761) reconstructs “[*tageru*] (B) < tagey- < *tanka-Ci- (< ?; cf. *taberu*, *tamau*). ‘(let one) eat or drink’”. The relation between ‘receive’ and ‘eat’ has long been related in Japonic. Compare WOJ *tab* ‘give’ WOJ *tabar*, EOJ *ta^mbar* ‘receive (humble)’ > NJ *taberu* ‘eat’ (*tab-e* ‘give-get’) and NJ *itadaku* ‘receive (humble)’ used before eating a meal. Vovin compares this word to a possible word in Fangyan OC *dzak (2014) / LH *dzak, which may be related but is phonetically distant.

152. pWMP *tayug ‘swing, oscillate’ > pJ *tayu-tap (swing-be.close.together) > WOJ *tayutap* ‘be unsettled by movement’

Mansaka (MP) *tayug* ‘to swing (as in a hammock)’

Sound Changes: Coda Loss

Synonyms: NJ *yusur* ‘shake (t.)’

This cognate only appears in combination with 3.2.31 pMP *dapet/dápat but is clearly phonetically and semantically related.

153. pWMP *teraŋ ‘clear, bright’ > pJ *ter(a)- > OJ *ter* ‘shine’ (Ex. *tar*) EOJ *tor* ‘id’

*tera-s (be.bright-TRZ) > *teras* ‘shine’

*tera-p (be.bright-ITER) > *terap* ‘show’

Okinawan (Iheya) [ti]:[N ‘to shine’

Miyako (Hirara) *tirasi* ‘to illuminate’

Malay (MP) *təraŋ* ‘clear; bright; obvious’

Sound Changes: Schwa Change, Liquid Approximation, Coda Loss

Martin (1987. p.767) reconstructs “ter-u (B) < *tera- ?< *to(-)ra- (cf. tomosu, tomoru) ‘shine’”.

The phonetic and semantic correspondence is nearly identical, however the schwa in pWMP *teraŋ is expected to be borrowed as *tar(a) not *ter(a). This also appears to be the case in #3.1.42 *seksek ‘to stuff, cram in; be crowded’ and #3.1.44 *selsel ‘insert, cram in’ which are borrowed as *sek* and *se*, respectively. Both *sye* and *tye* are not permissible phonotactics in Japonic, but otherwise there is no readily apparent rule here.

154. pMP *ti(ŋ)kas ‘swift, energetic’ > pJ *tika-ra (strong-PROP) > WOJ *tikara* ‘strength’

Remontado (MP) *ma-tikas* ‘strong’

Sound Changes: Coda Loss

Martin (1987. p.546) reconstructs “tikara (3.3) ?< *ti_{71.1}kara_{2.2b} (‘spirit characteristic’).

‘strength’”. The example in Remontado supports the semantic shift from ‘energetic’ to ‘strong’, becoming a noun ‘strength’ with the property suffix pJ *-ra.

155. pAN *tukad ‘sloping upward; to climb a mountain’ > WOJ *tuka* ‘mound’

Bikol (MP) *túkad* ‘a slope, incline’

Sound Changes: Coda Loss (Nominal)

Martin (1987. p.554) reconstructs “tuka (2.2b) (?< *tuku-A < ?*tuka-A ‘build’). ‘mound’”.

156. pAN *tukəd ‘prop, support’ > pJ *tuka ‘clutch’ > WOJ *tuka* ‘handle’

pJ *tuka-m (clutch-VBZ) > NJ *tukamu* ‘to seize, to catch, to grasp, to grip, to grab, to clutch’

Tagalog (MP) *túkod* ‘support; bracket; brace; prop; prop or support in the figurative sense’

Sound Changes: Schwa Change, Coda Loss (Nominal)

Martin (1987. p.554) reconstructs “tuka (2.3) (?< *tuka-B ‘attach’, cf. *tukamu* < *tuka-m[a-]_B ‘grasp’). ‘hilt’”. Francis-Ratte (2016. p.275) reconstructs “pKJ *cukəm- ‘grasps it’,” through comparison with “MK *chóm-* ‘bears it, withstands it’”. The presence of a final -m in this and MK *tatóm* may suggest they were borrowed with a reanalysis, pJ *tuka-m > OK *cukəm- and pJ *tata-(a)m > OK *tatəm-.

157. pAN *tukul ‘to sprout, begin to grow’ > pJ *tuku-r (to make grow-NTR) > OJ *tukur* ‘make’

Okinawan (Iheya) [*cukuin* ‘make’]

Miyako (Hirara) *cī fu:z* ‘make’

Tagalog (MP) *tukól* ‘overripe rice grains that are still unharvested’

Sound Changes: Coda Loss

Martin (1987. p.774) reconstructs “tukuru (B) < *tuku-ra-. ‘make’”. The semantic change here supports the repeated use of metaphor found in the agricultural vocabulary (#3.1.5 *budbud ‘sprinkle seed, sow seed in dibble holes’ > OJ *pur* ‘rain, fall (precipitation)’, WOJ *pur* ‘shake’ and #3.2.8 *ataq ‘unhusked rice grain left among the husked grains’ > WOJ *atara* ‘in vain, be a waste’). Francis-Ratte (2016. p.312) reconstructs “pKJ *cukir- ‘makes, prepares’,” through comparison with “MK *choli*- ‘recovers consciousness, steels oneself,’ MK *cholGwo*- ‘makes, prepares,’ ENK *cholhi*- / *cholho*- ‘makes, prepares’”. See also OC 築 *truk ‘pound earth, build’ (Baxter-Sagart, 2014) as an alternative possibility.

158. pMP *tulak ‘push something away, shove, repel; push off, as a boat leaving shore or being poled through shallow water’ > pJ *pik.əⁿ-tur(a) (pull.EP-GEN-cast.off) > WOJ *pikodur* ‘pull’ (‘pull in a docking rope and push away from the shore’?)

pJ *ipiⁿ-tur(a) > WOJ *ipidur* [unknown] (‘say that one is going to sea’?)

Tagalog (MP) *túlak* ‘push, shove; forced by circumstances’

Sound Changes: Coda Loss, Liquid Approximation

The reconstructability of *tulak as boat terminology may explain two other curious trisyllables with rendaku as a compound, *pikodur* and *ipidur*. The former is simply translated as ‘pull’ in the ONCOJ, same as *pik* ‘pull’, and both are present in the same poem relating to red boats *sopobune*. The schwa between *pik* and *tur* to resolve the phonotactics is also present in examples like *kikose* (< *kik.ə-s-ay), *kikoye* (< *kik.ə-y-ay).

159. pWMP *tuluy ‘caught up in, carried along’ > pJ *tur(u) > WOJ *tur* ‘fish, angle’

Cebuano (MP) *túluy* ‘be caught in a net’

Sound Changes: Liquid Approximation, Coda Loss?

Martin (1987. p.775) reconstructs “turu (A) < *tura-. ‘hang; suspend; fish for; entice’” and (p.762) reconstructs “ture < *ture[y] < *tura-Ci[-Ci]_A. ‘companion’.

The final -y appears to be dropped in this other item as well (#3.2.172 *waray), likely to create the monosyllabic verb morpheme *tur(u), similar to coda loss in other entries creating nominal morphemes.

160. pWMP *tumpaq ‘to spill out, pour out’ > pJ *tu^mpa-ra (overflow-PROP) > WOJ *tubara* ‘plentiful, complete’

pJ *tumpa-ra-ka (overflow-PROP-ADJZ?) > WOJ *tubaraka* ‘plentiful, complete’

Malay (MP) *tumpah* ‘spilling; overflowing in some quantity’

Sound Changes: Prenasalization, Coda Loss

161. pMP *tu(m)pus ‘to complete, finish’ > pJ *tup(u)-i > WOJ *tupi* ‘final, ultimate’
 Kayan (MP)? *tupuh* ‘be completed (polite term)’

Sound Changes: Coda Loss

Martin (1987. p.554) reconstructs “tui(-ni) < ‘tufi’ (= tuwi ?< *tu[w]i < *tu[k]iy < *tuku-Ci[-Ci]_B ‘exhaust’ -- but M 4508 has tufyi, therefore < *tupi? -- cf. tubi_{2.3}). ‘(at the) end. final(ly)’”. The optional nasal here is not present in the Kayan example either, lending support to this otherwise poorly attested example.

162. pWMP *tunay ‘real, genuine, true, sincere’ > pJ *tunay > WOJ *tune* ‘normalcy; eternity’

Tagalog (MP) *túnay* ‘true; correct; exact; meaningful; rightful; lawful; real; genuine; substantial; actual; factual; positive’

Sound Changes: Coda Retention, Exact

Martin (1987. p.556) reconstructs “tune (2.5) (?< *tu[ra]ne < *tura-na-Ci[-Ci]_A but register incongruent; ?< *tuna-Ci_{2.3} ‘rope’... ‘longlasting, always’”.

Similar to #3.2.43 *gunay, *tune* does not seem to have an exposed counterpart *tuna-*, so WOJ *e* here may not have come from suffixing *(a)y. The semantic relation is reality as what is normal, enduring, actual.

163. pWMP *tunjkup ‘cover’ > pJ *tunku-m (cover.body-VBZ) > NJ *tugumu* ‘to shut one’s mouth, to hold one’s tongue, to keep silent’

Old Javanese (MP) *a-tukup* ‘to cover (with the hands)’

Sound Changes: Prenasalization, Coda Loss

This example is more semantically speculative, perhaps coming from the narrowing of a longer collocation, such as *kuti wo tugumu* ‘to cover the mouth’ to merely *tugumu* ‘to cover the mouth’ > ‘to keep quiet’, although there is no textual evidence for this original form.

164. pAN *tutuh ‘beat, pound, bump, thump’ > pJ *tut(u)-i/*tutuh-i (pound-NMZ) > WOJ *tuti/tutui* ‘hammer; mallet’

Old Javanese (MP) *tulun tutu* ‘help in pounding rice’

Sound Changes: Coda Loss?

Martin (1987. p.557) reconstructs “[tuti] (2.4 ?< *3.5) < *tuti[y] < *tutu-Ci. ‘Hammer’”.

Francis-Ratte (2016. p.313) reconstructs “pKJ *tocuŋ ‘mallet’,” through comparison with “MK *two:skwáy*, *two:chwóy*, *two:chóy* ‘axe’”. This comparison is both phonetically and semantically further than pAN *tutuh from pJ *tutu(h). WOJ *tutui* with its vowel cluster looks like an attempt at a closer pronunciation of /tutuh/ with final *h* between *u* and the nominalizer *-i, *uhi* > *ui*. WOJ *tuti* shows a reduction of *ui* > *wi*, but the coronal consonant blocks OJ *twi* resulting in OJ *ti*. The Old Javanese descendant shows its use in hulling rice. See also OC 擣 *tʰu? ‘pound (v.)’ (Baxter-Sagart, 2014)

165. pMP *tutur ‘to speak clearly and deliberately’ > pJ *tutu-a? > WOJ *tutwo* ‘present; gift’

pJ *tut(u)-ay > WOJ *tute* ‘communicate’

pJ *tut(u)-ap > OJ *tutap* ‘be communicated; to pass along (EOJ)’

pJ *tut(u)-ap-ay > WOJ *tutape* ‘communicate’

Ilokano (MP) *ag-tutor* ‘to take one’s time; be patient’

Sound Changes: Coda Loss

Martin (1987. p.776) reconstructs “tutaeru (A) < tutafoy < *tuta-pa-Ci-. ‘transmit, convey, tell’”.

Francis-Ratte (2016. p.223) reconstructs “pKJ *tuta- ‘is conveyed’,” through comparison with “MK *tut-* / *tul-* ‘hears’ < pre-MK *tutu-”. The pre-MK parallel interestingly shows a form closer to pMP *tutur than pJ *tut(u), perhaps showing it was loaned to Korean and then borrowed into Japonic with the simpler MK *tut-* > pJ *tut-ap. The word *tutwo* does appear to be related by the first syllable and meaning of passing something onto someone else, but the expected form is *tuta* (< tut-a).

166. pWMP *tumpuk ‘small heap, rounded mass’ > pJ *tu^mpu-r-ay (round.mass-NTR-get) > WOJ *tubure* ‘rounded, worn smooth’

pJ *tumpu-ra (round.mass-PROP) > NJ *tsubura* ‘round and cute (esp. of eyes)’

Tagalog (MP) *tumpók* ‘heap; mound; mass (as of rubbish)’

Sound Changes: Coda Loss, Prenasalization

Martin (1987. p.553) reconstructs “tubura (3.6) < *tunpu-ra < *tunpu_{2,4}[a]ra_{2,3} ‘rotund’ and (p.553) does not provide a reconstruction for “tubu (2.4) < *tunpu. ‘grain, granule’”.

167. pWMP *ubal ‘rise to the surface of water’ > pJ *upa- (top.surface) > WOJ *upa* ‘top’

pJ *upa-(a)y (top-get) > OJ *upe* ‘above’

Miyako (Hirara) *upa:m* ‘a high priestess’

Sundanese (MP) *ubal* ‘tie pieces of banana stalk or other buoyant material to a sunken object in order to make it float to the surface’

Sound Changes: Devoicing, Coda Loss

Martin (1987. p.560) reconstructs “ue (2.1) < ufey < *upa-Ci. ‘above, up’”.

168. pWMP *ukanj ‘slow and tentative movement’ > pJ *uka-r-ay

(slow.movement-NTR-get) > WOJ *ukare* ‘amble’

Tagalog (MP) *úkanj* ‘slow movements peculiar to the aged and the infirm’

Sound Changes: Coda Loss

It is worth highlighting how the meanings are nearly identical, despite not being a common word (in Japonic, it is a hapax legomenon).

169. pWMP *u(ŋ)kanj ‘spread open, as the legs’ > pJ *uka-(a)y ((spread.)open-get) > WOJ *uke* ‘receive’

pJ *uka-(u)ka-(a)p > WOJ *ukagap/ukakap* ‘be open, gape; spy on’

Ilokano (MP) *ukkáy* ‘to open (a razor, pocket knife, calipers, the legs, etc.)’

Sound Changes: Coda Loss

Martin (1987. p.778) reconstructs “ukeru (B) < ukey- < *uka-Ci-. ‘receive’”. Francis-Ratte (2016. p.356) reconstructs “pKJ *uku- ‘receives, gets it’,” through comparison with “MK *wuhwúm* ‘a fistful, handful’”. The reconstruction suggested above for *ukagap* is probable based on the optional *rendaku* present and repeating syllable. pJ *uka would also better explain WOJ *uke* than pKJ *uku.

170. pMP *umpu ‘grandparent/grandchild (recipr.); ancestor’ > pJ *u^mpu ‘newborn’ > NJ *ubu* ‘innocent, naive; birth, newborn’

Cebuano (MP) *umpu-?* ‘grandparent or a grandparent's sibling or cousin of the same generation’

Sound Changes: Prenasalization, Exact

Martin (1987. p.559) reconstructs “ubu (?2.5, ?2.1) < *unpu (?< um[a]-p[a]-u_A, ? variant of *um[a]-u_A). ‘(by) birth; naive’”. Although this word is poorly attested, it has no changes phonetically and is clearly semantically related. Considering the presence of other kinship terms (#3.2.3 *amax ‘father’ and #3.2.72 *mamah ‘parent’s younger sibling, junior uncle; a young child’s term of address for his father, vocative of *ama’), this word appears to be part of a set of borrowings.

171. pPh *úŋur ‘moan, growl’ > pJ *un(u)-ar (moan-exist) > NJ *unar* ‘to groan, to moan’
pJ *un(u)-as-ar-ay (moan-TRZ-exist-get) > NJ *unasare* ‘to cry out in one's sleep (e.g. from having a nightmare)’

Ilokano (MP) *úŋor* ‘to moan with closed mouth (when hurt or dying)’

Sound Changes: Nasal Approximation, Coda Loss

Martin (1987. p.779) reconstructs “unaru (B) < *u_{mim}na(-)ra-_A. ‘groan’”.

172. pWMP *utus ‘send on a mission, command someone to do something’ > pJ *utu-r (change.location-NTR) > WOJ *utur* ‘change (i)’

pJ *utu-s (change.location-TRZ) > WOJ *utus* ‘transfer’

Tagalog (MP) *útos* ‘command; order; mandate; edict; ordinance; law’

Sound Changes: Coda Loss

Martin (1987. p.781) reconstructs “utusu (B) < *utu-sa-. ‘move/transfer/divert it; reflect it, copy/photograph it’”. Francis-Ratte (2016. p.224) reconstructs “pKJ *icu-* ‘conveys, moves it across space’,” through comparison with “MK *cwú-* ‘gives it’. The semantic change here is explained as being generalized to a change of location (c.f. NJ *utsuru* ‘to move (house), to transfer (department), to change the target of interest or concern’), hence the meaning of transfer.

173. pAN *waCaN ‘deep pool or pond’ > pJ *wata > WOJ *wata* ‘sea, ocean’

pJ *wata-r > OJ *watar* ‘to cross’

Okinawan (Iheya) [*watai*]_N ‘to cross’

Miyako (Hirara) *watal* ‘to cross, to pass through’
 Paiwan (Formosan) *vatsal* ‘a deep pool; lake (Western Paiwan)’
 Sound Changes: Affricate Approximation, Coda Loss
 Synonyms: OJ *umi* ‘sea’

Martin (1987. p.569) reconstructs “wata (?2.1) < *bata. ‘sea’”. The word *wata* is normally compared to Korean (Francis-Ratte, 2016. p. 368) as a cognate, but the Korean form may have been borrowed from Japonic. Korean lacks initial *w* (c.f. WOJ *wase* > Late Old Korean *psar* (Vovin 2015)), and as Francis-Ratte (id.) explains, the final *-h* in *patah* could be from a lenited locative suffix **-k*. This connection to Austronesian should be considered due the significance of the god *watatumi*, which have been long posited to relate to the *hayato/kumaso* people. A mountainous source for the god *opoyamatumi* could be related to a lost northern strata with a more inland-focused mythology. Benedict (1990. p.228) compares “SEA: P-Austro-Japanese **wacal* > Jp. *wata*”.

174. pAN **waray* ‘to separate, as two people’ > pJ **war(a)* > WOJ *war* ‘divide (v.t.)’
 pJ **wara-(a)y/waray* > WOJ *ware* ‘be split’
 Okinawan (Iheya) [*wain* ‘divide’
 Tagalog (MP) *wálay* ‘separate from; wean (a baby) from mother's breast’
 Sound Changes: Coda Loss?

Martin (1987. p.783) reconstructs “waru (A) < *bara-. (vt.) ‘split, break, halve’”. This word has a final palatal consonant that seems to be lost, since there is both WOJ *war* and *ware*.

175. pMP **zaqet* ‘bad, evil; ugly’ > pJ **a(a)-si* (bad-ADJZ) > OJ *asi* ‘bad’
 Yami (MP) *rahet* ‘bad’

Sound Changes: Liquid Loss, Schwa Assimilation, Coda Loss
 Francis-Ratte (2016. p.185) reconstructs “pKJ **asi-* ‘undesirable, bad’,” through comparison with “MK *askí-* ‘grudges it,’ *askaW-* ‘is unwilling to let go’ < **ask-* ‘unwilling’”. This comparison both phonetically and semantically competes with **zaqet*. This is the only example of pMP **z* compared to pJ/pJ, but similarly to #3.2.74 **najam*, it has several descendants with initial *r* which would be dropped: Itbayaten *raet* ‘defect, deficiency, flaw’ and pMP **zaŋan* ‘handspan’ Yami *raŋan*, Ivatan *raŋan*.

3.3 Monosyllables

This section interestingly contains several glide initials derived from lenition: the vowel cluster with an initial labial conditions the labial pair pAN/pPh **buV* to lenite into a single labial glide initial pJ **wV*; and pAN/pWMP **iV* lenites into a palatal glide pJ **(i)yV*, as was also seen in #3.2.100 pMP **piqek* > **piek* /*piək*/ > pJ **piyə(k)-ko*.

1. pAN **buak* ‘bubbling up (as spring water)’ > pJ **wak-as/.a-s* (boil(.EP)-TRZ) > WOJ *wakas* ‘boil (v.t.)’
 Okinawan (Iheya) [*wakasu*]*n* ‘boil’

Malay (MP) *buak* ‘bubbling up, as water in a spring; spouting out in jets, as blood from a severed artery’

Sound Changes: Lenition, Epenthesis

The lenition here is due to the vowel cluster with the rounded *u* following the labial stop, triggering a shift to a labial glide (bu > pu, bua > w(u)a).

2. pPh *buás ‘tomorrow’ > pJ *was.a- (early.EP) > WOJ *wasá* ‘early’

pJ *was-ay (early-get.NMZ) > WOJ *wase* ‘early rice’

Kankanaey (MP) *buás* ‘morning (used only in tales)’

Sound Changes: Lenition, Epenthesis

Francis-Ratte (2016. p.243) reconstructs “pKJ *wasər ‘early stage of plant growth’,” through comparison with “MK *psól* ‘hulled rice grain’”. Although he cites developments for pKJ *w > pK *p, pJ *w, it is possible instead that, lacking initial /w/, OK borrowed pJ *w as *p.

‘Tomorrow’ and ‘early’ are semantically close and provide one more example for pAN/pMP/pPh *buV to pJ(P) *wV. WOJ *wase* meaning ‘early rice’ might be a deverbal noun from the connective of WOJ -e ‘get’. There may be an echo vowel to resolve the consonant final monosyllable to get a disyllabic morpheme.

3. pAN *ian ‘dwell, reside, live in a place’ > pJ *ya > OJ *ya* ‘house’

Okinawan (Iheya) *ja*/: ‘house’

Common Ryukyuan *ja*: (Jarosz, p.191)

Ilokano (MP) *yan* ‘place, location, situation, locality, spot, position’

Sound Changes: Lenition, Coda Loss (Nominal)

Francis-Ratte (2016) reconstructs “pKJ *i- ‘roofs, places over the head’,” through comparison with “MK *i-* ‘to place over the head,’ *ni-* ‘to thatch a roof, to put on a roof’”. Vovin (2014b) compares pT *ʔyaaw (C1) ‘house’, which may be related to pAN *ian. The descendant Ilokano *yan*, shows how pAN *ia to *ya occurred in Ilokano (Itbayaten also has this change) and was then borrowed. Japonic also regularly uses *ya* for place as a suffix (c.f. *heya* ‘room’, *chaya* ‘teahouse’, *naya* ‘shed’).

4. pAN *kæn ‘eat’ > pJ *ka-(a)y (eat-get.NMZ) > OJ *ke* ‘food (container)’

Tagalog (MP) *káʔin* ‘consumption of food; consumption of any supply, e.g. fuel’

Sound Changes: Schwa Change, Coda Loss

The vowel *æ* here appears to be the only example in pAN, but a correspondence with pJ *a makes sense as the closest back vowel. This comparison was also explored by Murayama (1988. p.179).

5. pMP *kiak ‘to squawk, as a frightened chicken’ > pJ *ka(k)-kia (cluck-squack) > WOJ *kakye* ‘chicken’

Ilokano (MP) *kiák* ‘cry of a chicken’

Sound Changes: Coda Loss

See #3.1.12 *ke(k)kek for the first part of this compound.

6. pWMP *qiəm ‘shade, darkness, gloom’ > pJ *yəm-oy(?) (dark(ness).EP-?) > WOJ *yomwi* ‘the world of the dead’, WOJ *yamwi* ‘dark(ness); the underworld’
Ilokano (MP) *yúyem* ‘overcast, clouded over, gloomy’
Sound Changes: Deletion, Lenition, Epenthesis?

This loan may be a coincidence since the second segment *-oy(?) is difficult to explain but *yomwi/yamwi* is likely not *-ay ‘get’, which would result in *yome*. The WOJ pair does look like it shares an initial morpheme *yəm/yam ‘darkness’ though, so this set is included since the pWMP *ə could have lead to pJ *y.əm or pJ *y.am, considering pAN *kəmkəm.

7. pWMP *uit ‘bringing, carrying’ > pJ *wi-(ay)? (bring(-get)) > WOJ/EOJ *wi* ‘lead, guide, accompany, bring’/‘bring; take along’
Kelabit(MP) *uit* ‘way of bringing’
Sound Changes: Lenition, Coda Loss

Martin (1987. p.698) reconstructs "iru (A) < wi- < *wi[y]- < *bu/o-Ci. ‘lead’”

3.4 Trisyllables

These last few words help to reinforce that the previous entries cannot be simple phonological coincidence, as they follow the same rules including schwa retention and Arisaka’s Law changes, while being much less statistically likely at 6 phonemes (from 7 minus the coda).

1. pMP *bəkəlaɟ ‘spread out, unroll (mats, etc.), open out, unfold (as the hand); wide’ > pJ *pəkərə-mp-as > NJ *hokorobas* ‘to rip up (a seam), to burst, to tear, to break out (e.g. into a smile)’
pJ *pəkərə-mp-oy? > NJ *hokorobi* ‘to come apart at the seams, to be ripped, to be torn, to begin to open, to begin to bloom, to smile broadly, to break into a smile’
Cebuano (MP) *buklád* spread something (as a mat) out flat; for a girl to grow to maidenhood (poetic); for a blossom to open
Sound Changes: Devoicing, Schwa Retention, Open Assimilation, Coda Loss

Martin (1987. p.692) reconstructs “hokorobu (B) = hokorobiru < fokorobi [y]- < fokorob- (Nr consonant-stem) ?< *fok^o/rab- < *po(n)ku-ra-n[a]-pa- (?< *panka-..). ‘come unsewn, come apart at the seams’”. The medial schwa is not optionally marked in the pMP example, suggesting that pJ borrowed the term with it. The schwas in the word likely triggered assimilation in the final *a*, yielding *pəkərə. Both NJ and Cebuano even use their respective words to refer to blooming.

2. pPh *buk(ə)law ‘gluttonous, greedy with food’ > pJ *puk.ura-(a)y (full.belly-EP-get) > WOJ *pukure* ‘fat person’
pJ *pukur.o? (full.belly)? > WOJ *pukurwo* ‘bag’
Ilokano (MP) *bukláv* ‘gluttonous, greedy with food’
Sound Changes: Devoicing, Coda Loss, Epenthesis

Martin (1987. p.694) reconstructs “hukureru (A) < fukure[y]- < *puku-ra-Ci-. ‘swell, bulge (= hukuramu); sulk’”. Seeing as the Ilokano example lacks a medial schwa, the medial *-u-* in *pukure* is likely an echo vowel; Modern Japanese phonotactics would likely also borrow *bukláv* as *bukura*. The semantic relation is self-explanatory and unlikely to be mere coincidence. The final syllable *-law* is likely borrowed rather than being the *-ra* suffix.

3. pPh *daq(ə)taR ‘floor’ > pJ *tata-am-i (floor-weave-NMZ) > WOJ *tatami* ‘tatami mat’
EOJ *tatami* ‘rice straw mat’
Okinawan (Iheya) [*tatamun*] ‘fold (clothes)’
Miyako (Hirara) *tatam* ‘a mat, tatami’
Ilokano (MP) *datár* ‘floor, bamboo flooring’
Sound Changes: Devoicing, Deletion, Coda Loss

Francis-Ratte (2016. p.261) reconstructs “pKJ *tatəm ‘folds, flattens it’,” through comparison with “MK *tatóm-* ‘smooths cloth, trims it, rubs it together’”. Although *tatəm would actually become pJ *tata(m), the sense of ‘bamboo flooring’ in Ilokano is notable considering the abundance of bamboo tatami mats in Japan. Deletion of the guttural would leave *dataR, which is seen in Ilokano *datár*, resulting in pJ *tata. The /m/ in the word may be from WOJ *am* or the verbalizer pJ *-m.

4. pMP *taq(ə)pil ‘in layers, adjacent to one another’ > pJ *tapi-ra (flat-PROP) > NJ *taira* ‘flat, level, even, smooth, calm’
pJ *tapi-ra-ankay (level-ADJZ-raise) > WOJ *tapirage* ‘quell, rule’
pJ *tapi-ra-kay? (level-PROP-ADJZ) > WOJ *tapirake* ‘calm, quiet, peaceful’
Ilokano (MP) *ag-tapʔil* ‘to be adjacent to each other; to touch each other’
Sound Changes: Deletion, Coda Loss

The reasoning here may be far-fetched, but layers would be placed flat if placed adjacently, like floor tiles perhaps. The phonetic comparison is likely, with the final consonant possibly influencing the borrowing with the *-ra* suffix.

5. pMP *udəhi ‘last; come after or behind; late, later’ > pJ *ura(h)i > OJ *ura(/ure* WOJ)
‘inside, underside; (plant) growth, tip’
EOJ *ura* ‘inside’, ‘back’, ‘heart’
Common Ryukyuan *ura* ‘the rear, the back side’ (Jarosz, p.660)
Ilokano (MP) *údi* ‘last, latest, rear, hindmost’
Sound Changes: Lenition, Deletion, Coda Retention

Compare the meanings in Ryukyuan and Ilokano, showing a clear semantic connection. The final vowel in pMP seems to explain the exposed form *ure*.

3.5 Grammatical Parallels

The following list is the most striking, albeit lacking several fundamental morphemes present in pJ. For example, pJ *-i ‘deverbal formant/continuative suffix’, pJ *-u ‘nonpast suffix’, pJ *-k-

‘come’, pJ *-(a)ku ‘nominalizer’, pJ *-ra ‘property suffix’, etc. are missing. Discovering the origin of these words may reveal clues towards the other stratum/strata present in Japonic and their relative time and location. The verbs used in the examples are templates for the conjugation and segmentation.

1. pAN *-a ‘imperative suffix’ > pJ *agar-i-a (rise-CONN-IMP) > WOJ *agarye* ‘rise !
(command)’

Kanakanabu (MP) -a ‘imperative suffix’

Sound Changes: Exact

It is very interesting that this morpheme was already suffixal in pAN, making it likely to have been preserved in Japonic’s agglutinative morphology, which it shares in common with Korean. It should also be noted that bigrade verbs use the suffix WOJ -yo ‘imperative suffix’. I believe it developed from a case agreement with the interjection particle WOJ *yo* which developed a vocative-like use, something like *nare-yo tabe-yo* ‘hey you! eat!’. The imperative of WOJ *ko* ‘come’, *koi* < pJ *kə-i (come-CONN), shows how the suffix *-i is likely involved in the surfacing of the imperative as WOJ -ye < pJ *i-a.

2. pAN *-a ‘subjunctive suffix’ > pJ *agar-a-m (rise-SUBJ-VBZ?) > WOJ *agaram* ‘may rise’
Atayal (Formosan) *a* ‘suffix forming active subjunctive from the reduced stem, sometimes preceded by the prefix m-’

Sound Changes: Exact

This form is bound in most Japonic languages, however, in Miyako, the “[b]are irrealis base form seems to indicate a future action or state, or an estimation that a given action or state will take place[...] It also appears that irrealis can express conjecture referring to the present by itself” (Jarosz, 2015. p.259). The suffixing, use, and form match one to one, despite being another monophonemic morpheme.

3. pMP *-a ‘3sg.object’ > pJ *pər-a (scoop.out-TR.OBJ) > NJ *hora* ‘hollow, cavity, hole, cave’

Balinese (MP) -a ‘suffix indicating the third person (singular or plural) as a pronoun expressing the genitive, or the agent of a passive verb, or the direct object of a transitive verb’

Sound Changes: Exact

The use in Balinese for the direct object of a transitive verb is notable because it demonstrates why the object suffix *-a largely appears after transitive verbs (e.g. WOJ *abur* ‘broil’, WOJ *abura* ‘oil’, NJ *nau* ‘twine, twist’, WOJ *napa* ‘rope’). The existence of three homophonous suffixes with identical uses as their Japanese parallels should not be written off as mere coincidence.

4. pAN *-Cu ‘2pl deixis and spatio-temporal reference: that; there, then’

Paiwan (Formosan) *tu-tsu* ‘now’

Sound Changes: Affricate Approximation

The word *tu-tsu* in Paiwan is very remarkable, since it shows a clear parallelism with NJ *tsutsu* ‘while, even though, despite’ used to mark ongoing simultaneous actions by the same agent (e.g. NJ *terebi mitutu sara wo arau* ‘while watching TV, (I) also wash dishes’). This morpheme may also be compared with WOJ *tu* ‘genitive particle’, showing up often in place names such as *oki tu nami* ‘waves (at the) offing’ (MYS 2.222) *oki tu sima* ‘island (at the) offing’ (MYS 6.917) *kami tu se* ‘rapids (of the) gods’ (MYS 9.1764), and the name *watatumi* ‘the ocean god’ (< pJ *wata-tu-muy (sea-LOC.GEN-body)).

5. pMP *ias ‘particle of doubt or interrogation’ > pJ *ya > WOJ *ya* ‘open question marker’
Kankanaey (MP) *ías* ‘true? yes? indeed? sure? An interjection used when doubting’
Sound Changes: Lenition, Coda Loss

The same explanation of vowel cluster lenition here would explain the glide initial here, as well as offering a source and possibly entering a paradigm with WOJ *ka* ‘yes/no question marker’.

6. pAN *-in- ‘perfective marker’ > pJ *-in > WOJ *-in* ‘perfective verbal suffix’
Kankanaey (MP) *in-* ‘a prefix of -en and -an verbs and many complex verbs with these endings’
Sound Changes: Exact

Note how in Kankanaey, this infix has shifted towards being a prefix instead. This may have been an intermediate role before pJ as well, making it easier to borrow over as a head-final suffix, as with pJ *-ka ‘adjectivalizer’.

7. pMP *ha ‘marker of a tag question’ > pJ *ka? > WOJ *ka* ‘yes/no question marker’
Tagalog (MP) *ha* ‘interrogative particle meaning: what is it? or what do you mean?; interrogative particle meaning: do you understand?; exclamatory interrogative particle with the element of surprise or reproach: is that so?’
Sound Changes: Fortition?

This parallel is of special interest, as it offers the only other possible example of initial pMP *h > pJ *k, although this would eliminate the entry 3.2.47 pAN *heban > WOJ *op*, with h-deletion.

8. pAN *ka- ‘stative marker in irrealis constructions, counterpart to *ma- in realis constructions’ > pJ *s-i-ka-ⁿpa (do-CONT-IRR-PROV) > WOJ *sikaba* ‘provisional simple past auxiliary’ (Frellesvig, 2010. p.62)
pJ *s-i-ka-ⁿto (do-CONT-IRR-CONC) > WOJ *sikado* ‘concessive simple past auxiliary’ (id.)
Bontokka (MP) *ka-* ‘a stative prefix occurring only on verbs preceded by the auxiliary daan (‘yet, still’): daan ka-loto ‘not yet cooked’ (cp. *loto* ‘to cook’)
Sound Changes: Exact

The morpheme here WOJ *ka* is opaque but clearly segmentable, as WOJ *-si* ‘simple past suffix’ and WOJ *-(e)ba* ‘provisional suffix’, WOJ *-(e)do* ‘concessive suffix’. Strangely, it seems to fill the place of a realis suffix, even in the exclamatory simple past auxiliary WOJ *sika*, so this entry is more questionable. There is the conditional simple past auxiliary WOJ *seba* which does not

include this unit, so the pAN affix may have offered a suppletive form for the simple past forms, *sikaba* and *sikado*.

9. pAN *ka- ‘formative for abstract nouns of quality’ > pJ *pənə-ka (faint-ADJZ) > WOJ *pono/ponoka* ‘faint’

Itbayaten (MP) *ka-* ‘creates abstract nouns of quality: ka-vayah ‘redness’ (cp. vayah ‘red color’)’

Sound Changes: Exact

Although modern, NJ *kabosoi* ‘thin, skinny, delicate, fragile, feeble’, NJ *kayowai* ‘weak, feeble, delicate, frail, helpless’, and NJ *kaguroi* ‘deep black’ may be fossilized examples of a prefixal adjectivalizer. This prefix was likely turned into a suffix to agree with proto-Japonic’s head-finality.

10. pAN *ma- ‘stative prefix’ > pJ *ma-sirwo (ADJZ-white) > WOJ *masirwo* ‘completely white’

pJ *ma-kanay (PROP?-metal) > WOJ *makane* ‘pure gold’

Tagalog (MP) *ma-* ‘prefix forming adjectives denoting (a) a certain quality, e.g. magandá ‘beautiful’, (b) having much of something, e.g. mabató ‘stony (full of stones)’

Sound Changes: Exact

This WOJ example shows how emphatic gemination is innovative, e.g. NJ *masshiro* ‘pure white, blank (e.g. mind, paper)’. The use in Tagalog illustrates how this prefix was used with both adjectives of a certain quality and with nouns.

11. pAN *ni ‘genitive case marker for singular personal names and pronouns; marker of possession, part-to-whole relationships, and agency of a non-actor voice verb’

Tagalog (MP) *ni* ‘preposition used before a person’s name or names to connote possessive case; may also indicate the doer of a certain action [after a non-actor voice verb]’

Sound Changes: Exact

This morpheme would explain and offer a separate source for WOJ *ni* ‘agent marker’ in passive constructions. The use of the passive may have shifted verbs towards a non-actor voice, marking the agency of the non-subject syntactically. For example,

- | | | | | | | | | | | | |
|-----|-----------------------------|-----------|------------|-----------|----------------|---|-------------------------------------|-----------|--------------|-----------|-------------------|
| 1.) | <i>tutui</i> | <i>ga</i> | <i>isi</i> | <i>wo</i> | <i>tubus-u</i> | | <i>isi</i> | <i>pa</i> | <i>tutui</i> | <i>ni</i> | <i>tubus-ar-u</i> |
| | hammer | SUBJ | rock | ACC | crush-CONC | > | | | person | DAT | crush-PASS-CONC |
| | “(A) hammer crushes a rock” | | | | | | “(A) rock is crushed by (a) hammer” | | | | |

12. pAN *-ni ‘proximal spatio-temporal deixis: this, here; now’ > pJ *-ni > WOJ *ni* ‘dative case particle’

Seediq (Truku) (Formosan) *nii* ‘here, now, this’

Sound Changes: Exact

Although the pAN etyma is more generalized, I believe the locative use of *ni developed first, with the dative case forming as an extension of the locative case. Following a location, this word would likely be interpreted as ‘here’, then as a locative. Hypothetical pre-proto-Japonic and WOJ sentences are offered to visualize the change from ‘here’ to a locative, dative, and allative case.

2.)	<i>yama</i>	<i>ni</i>	<i>ari</i>		<i>hito</i>	<i>pa</i>	<i>yama</i>	<i>ni</i>	<i>ari</i>
	mountain	here	exists-CONC	>	person	TOP	mountain	DAT	exists-CONC
	n								
	“(There is) a mountain here”				“(A) person is (here) on the mountain”				

These two uses would explain why the collocation *ni ari* is used as a copular, ‘(there is) a mountain here’ > ‘(it) is a mountain’, from a declarative use being reanalyzed as copular. For example, how in modern Japanese one might see a mountain and tell someone, *mite, yama da* ‘look, (it is) a mountain’. Being a suffix of proximal deixis, context would also clarify the use of the locative as a dative with non-location nouns, and as allative with verbs of motion.

3a.)	<i>hito</i>	<i>ni</i>	<i>tamapu</i>		<i>hito</i>	<i>ni</i>	<i>tamapu</i>
	person	here	give-CONC	>	person	DAT	give-CONC
	“Give here (to) someone”				“Give to someone”		

3b.)	<i>yama</i>	<i>ni</i>	<i>aruku</i>		<i>yama</i>	<i>ni</i>	<i>aruku</i>
	mountain	here	walk-CONC	>	mountain	ALL	walk-CONC
	“Walk here (to the) mountain”				“Walk to (the) mountain”		

The fact that this is a suffix in pAN once more makes it easy to have been borrowed into a head-final language.

13. pPh *sa ‘that (indicating something close to the hearer)’ > pJ *sa > MJ *sa* ‘archaic 2nd person pronoun’ (<https://kobun.weblio.jp/content/さ>)

pJ *sa-tay (that-GER) > WOJ *sate* ‘that way’

pJ *sa-(a)ru (that-exist) > MJ *saru* ‘that kind of’ (<https://kobun.weblio.jp/content/然る>)

Bontok (MP) *sa* ‘that, close to hearer; there, close to hearer’

Sound Changes: Exact

The Japonic morpheme here *sa* may be related as an allophonic variant of the distal demonstrative *sə* instead, if it is not the source of it instead. Note that this pPh morpheme is not affixal, showing why it can appear in isolation.

14. pAN *Sa- ‘stative prefix’ > pJ *-sa > WOJ -sa ‘exclamatory form of adjectival copula’
Cebuano (MP) *ha-* ‘affix added to adjectives referring to degree’

Sound Changes: Exact

The inclusion of this affix creates a paradigm with the adjectival suffixes, WOJ -*ka* ‘adjectivalizer’, WOJ *ma-* ‘pure, perfect, outstanding’. This paradigm shows that the two types of adjectives, nominal (‘na-adjectives’) and copular (‘i/shii adjectives’) are probably a result of a mixed strata.

15. pMP *ta- ‘prefix marking spontaneous or involuntary action’ > pJ *ta-ⁿ-pasir
(PREF-GEN-run) > WOJ *tabasir* ‘splash violently’
pJ *ta-wasuray (PREF-forget) > WOJ *tawasure* ‘forget’

Tagalog (MP) *ta-pilók* ‘to stumble or trip, thereby causing one to twist or sprain the foot or ankle (pilók ‘twisted, referring to arm, ankle or foot’)

Sound Changes: Exact

This morpheme is poorly attested, therefore it may be coincidental but appears similar in its unique prefixal usage, which is unexpected of the highly head-final Japonic.

16. pAN *ua ‘exclamation of surprise, etc.’ > pJ *wa > NJ *wa* ‘indicates emotion or admiration, indicates emphasis’
Paiwan (Formosan) *ua* ‘exclamation of pain or surprise’

Sound Changes: Lenition

The explanation that has been offered before for vowel cluster lenition would perfectly explain why Japonic exhibits a glide initial here, as well as providing a source for the exclamatory sense of *wa*.

17. pMP *wada ‘be, exist, have; wealthy; not exist, not have’ > pJ *ar(a)? > WOJ *ar* ‘exist, live’

Malay (MP) *ada* ‘existence; to be; present; to appertain to; (vulg.) to have; *ber-ada* ‘have; rather wealthy, not lacking in means’

Sound Changes: W-loss?

This etymology is perhaps amongst the most important in determining whether there is an adstratum of pAN/pMP in proto-Japonic, or simply a large number of borrowings. However, there is the obvious issue of why pMP *w does correspond to pJ *w in 3.2.171 pAN *waCaN > pJ *wata or 3.2.172 pAN *waray > pJ *war. The Malay example shows initial w-loss so this change could have happened before contact with proto-Japonic, but Malay appears to have lost w-initials altogether.

18. pMP *mai ‘and’ > pMiyako? *mai > Miyako -mai ‘inclusive marker’ (Jarosz, 2015. p.294)

Hanunóo (MP) *may* ‘and (only when indicating the existence of a numerable entity in excess of what has been enumerated previously)’

Sound Changes: Exact

The inclusive marker -mai is rather rare, and may only be an allophonic variant of WOJ *mo* ‘id.’ However, what kind of alteration would lead to WOJ *o* becoming Miyako *ai* is uncertain, and both the sound and use of the pMP etyma matches the Miyako marker rather well. It may be treated as a fossil of Austronesian language(s) spoken in the Southern Ryūkyū islands rather than being part of the Austronesian adstratum present in the rest of proto-Japonic and Japanese.

19. pAN *<um> ‘verbal infix marking actor voice’ > pR *kak-i-um (write-CONN-CONC) > Okinawan *kachuN* (Vovin, 2014)

Tagalog (MP) -um- ‘verbal infix “used to express simple, single or general acts. (a) simple or single performance[...] (b) general acts, like the functions of nature and the natural acts of the human body[...] Many of the bases that occur in intransitive -um- verbs also occur as the bases of ma- adjectives and the verbs, which may be called -um- VERBS OF ‘BECOMING,’ denote the acquisition of the quality expressed by the adjective.”’

(Schachter and Otnes 1972:310).

Similar to how the pAN infix *-in- becomes a suffix in Japonic, it is not too farfetched that this infix would also become a suffix to fit Japonic’s syntax. Vovin reconstructs the conclusive non-past suffix as pJ *-um, which would suggest the final nasal was preserved in Ryūkyūan languages and that the suffix itself does extend to proto-Japonic.

20. pMP *na ‘already’ > pR? *na > Okinawan *naa* ‘already more, again, further’ (Read, 2011), Miyako (unspecified, most likely Hirara) *nn’a*: ‘now, already’ (Jarosz, 2015. p.603)

pR *na-ma (now-space) > Okinawan *nama* ‘now; at the present moment; currently’
Miyako (Irabu) *nnama* ‘id.’ (Read, 2025)

Itbayaten (MP) *na* ‘now, already’

The Japanese branch of proto-Japonic uses *ima* for the sense ‘now’, while many Ryūkyūan languages have some form of *nama*, which may be better explained as using a morpheme *na ‘already’ in Okinawan *naa*, with the noun *ma* ‘space’ which appears in many compounds.

Taken in combination with the grammatical morphemes Vovin (2021) found: OJ -sa ‘dative-locative case marker’ < pAN *sa ‘locative marker’ and OJ *i-* ‘directive-locative focus verbal prefix *i-*’ < Amis *i-* locative (pAN *i ‘generic marker of location in space or time’). There are about twenty secure grammatical reconstructions present within Japonic. While many are monosyllabic open syllables, hence the phonetic shape may be more likely to correlate by accident, the semantic and syntactic uses of these comparisons strongly support viewing these similarities as *consilience*, not mere *coincidence*.

3.6 Noteworthy Exceptions

pMP *hemay ‘cooked rice’ > pJ *kəmay (?) ‘dehusked rice’ > WOJ *kome* ‘rice’

Amis (MP) *hmay* ‘cooked rice’

Considering the significance of rice to the Austronesian people and comparisons that have been made between *hokora* ‘shrines’ (Vovin, 2014b) and early Japanese architecture (Arbi et al, 2013) with Austronesian houses built on piles, based on elevated rice-granaries. Many comparisons have been proposed for rice. Sagart (2011, p.15) and Robbeets (2017) compare WOJ *kome* ‘rice’ to pMP *həmay ‘cooked rice’ but this should become pJ *əmay, if entry 3.2.47 pAN *heban > pJ *əp(ə) is correct, since it is the only word with an initial /h/. Since it appears possible that /h/ was temporarily preserved within 3.2.162 pAN *tutuh ‘beat, pound, bump, thump’ > pJ

*tut(u)-i/*tutuh-i, it may be that OJ *op* used to have an initial /h/, although there is no current evidence of this. It may instead be the case that Francis-Ratte’s (2016, p.207) is correct that pKJ *əp- is the true source of OJ *op*, allowing pMP *hemay to perhaps be the only example with an /h/ initial that could have gone through fortition to yield pJ həmay# > pJ *həməy > *kəməy > OJ *kome*.

Elmer (2019a, p.67-69) notes about the transcription of the pre-OJ word *pimiko?* that, “[t]he consonant of the graph 呼 *hə>hə is very challenging to interpret, because it is thought that neither OJ nor Proto-Japonic had the fricative /h/. It is also unlikely that it is a transcriptional error, because this graph is used in two titles (卑彌呼 *pie-mie-hə (queen of Wa), 卑彌弓呼 *piemie-kuŋ-hə King (king of Kona)) and two toponyms (不呼 *pu-hə, 呼邑 *hə-ʔip)[...]Considering also the modern Japanese Sino-Japanese readings of the characters as 呼 *ko* and 好 *kō*, these graphs may have been used to transcribe some sort of velar quality[...] It is still unlikely that we should interpret 呼 LHC *hə as *kə, because there were graphs for transcribing *k in the Gishi-Wajinden, which the scribes should have then used”.

Therefore, pre-OJ *həmay/həməy may have become *həmay through velar approximation word-initially while it was lost in final and intervocalic positions, losing the phoneme /h/ altogether before Old Japanese.

Alternatively, Vovin (1998) has noticed a comparison to proto-Viet-Muong *kēm ‘cooked rice’ (Matsumoto, 1928), but Blench (2005, p.44) points out how this term is isolated in Austroasiatic, which may suggest that it was instead borrowed into Austroasiatic, not from it. Therefore the word may have come from proto-Viet-Muong rather than pMP. Jarosz (2021) observes that *kome ‘rice grain’ is not reconstructable for proto-Ryukyuan and was borrowed into the Northern branch after it split with the Southern branch. Note that Kavalan, another East Formosan language like Amis *hmay* ‘cooked rice’, has *m:ay* ‘cooked rice’. The common word for cooked rice in Ryūkyūan languages appears to reflect pR *mai: Yaeyama *mai*, Central Southern Okinawan *mè:*, Miyako *-maz*, Yonaguni *furumai*. pMP *hemay may instead be the source for Ryukyuan rice vocabulary but not Japanese if Vovin and Jarosz’s hypothesis stand true. I believe pMP to be the source for proto-Ryūkyūan *mai, and Vietnamese *com* /kə:mɦ/ (Ferlus, 1996).

p.64) (proto-Viet-Muong *kēm) as the source for proto-Japanese *kəmōy (< *kəm-ay (cooked.rice-get.NMZ)).

Note that Vietnamese *ngoām* ‘bite’ was also a closer match for WOJ *kam* ‘bite’ than 3.1.13 pAN *kemkem. Furthermore, see Vietnamese *cây* /kəjH/ ‘plant with a hard trunk or stem, such as a woody tree or bamboo’ which looks remarkably like proto-Japonic *kiy, with a slightly different central vowel;

These two next comparisons were listed in the ACD as “noise”, cognate sets that cannot be securely reconstructed but may yet be related. The former of these was also observed by Murayama (1988. p.198).

pMP ‘accustomed to’ > pJ *təmā > WOJ *tomo* ‘together; companion, friend; fellows, guys [pluralizing noun]’

Iban (MP) *teman* ‘attendant on a superior, escort; (hence polite for) companion, partner’

Malay (MP) *təman* ‘attendance on a person of higher rank than oneself; friend’

Karo Batak (MP) *teman* ‘comrade, friend’

Toba Batak (MP) *toman* ‘decent, chaste, reserved’

Sound Changes: Coda Loss, Open Assimilation

pMP ‘stand apart’ > pJ *pampa > NJ *haba* ‘width, breadth, freedom (e.g. of thought), latitude, gap, difference (e.g. in price), range (e.g. of voice)’

Ngaju Dayak (MP) *pampanj* ‘the ends of the antlers of deer; more generally the ends of anything similar; raised part, excrescence on something else’

Malay (MP) *pampanj* ‘kemudi fork-support for paddle-rudder

Bare'e (MP) *pampa* ‘stand at right angles to one another, as pieces of wood that are tied into a cross; stand out straight, as buffalo horns that do not curve’

Sound Changes: Coda Loss, Prenasalization

4 Results

4.1 Phonetic

Type				
Approximations	Nasal: 13/242 (5.4%)	Liquid: 26/242 (10.7%)	Affricate: 12/242 (5.0%)	Sibilant: 4/242 (1.7%)
Loss	Deletion: 17/242 (7.0%)	Liquid Loss: 13/242 (5.4%)	Coda Loss: 171/242 (70.7%)	Lenition: 13/242 (5.4%)

Schwa Change	Schwa Retention: 26/242 (10.7%)	Schwa Assimilation: 27/242 (11.2%)		
Voicing	Prenasalization: 36/242 (14.9%)	Devoicing: 55/242 (22.7%)		

Table 1: Amount of sound changes per total comparisons within the lexical dataset (excluding grammatical and exceptions)

The most notable fact about Japonic, its total exclusion of closed syllables (excluding modern h / N /), is also the most notable change present in the data, with 70.3% of the comparanda exhibiting coda loss. Interestingly, this change occurs frequently and seemingly independently across Austronesian languages. (X). The next most common change, devoicing, shows initial and medial voiced stops $\text{pAN}^* \text{b}$ / b / and $\text{pAN}^* \text{g}$ / g / become $\text{pJ}^* \text{p}$ / p / outside of nasal clusters ($\text{pAN}^* \text{mp} > \text{pJ}^* \text{m} \text{p} > \text{OJ} \text{b}$, $\text{pAN}^* \text{ŋk} > \text{pJ}^* \text{n} \text{k} > \text{OJ}^* \text{g}$). This and the large number of approximation rules are a clear sign that the Japonic parallels are not from evolution as an Austronesian branch, hence genetic, but from a language attempting to mimic foreign phonemes.

Both Old Chinese and Tungusic possessed a distinction between voiced and unvoiced stops, ruling them out as recipients; however proto-Koreanic did not even have voiced stops or fricatives: / b /, / d /, / g /, or / z /, like proto-Japonic. Comparing the consonant inventories of proto-Koreanic, proto-Japonic, and proto-Austronesian shows that proto-Koreanic is the most closely related language to the recipient language (not that very language itself).

proto-Austronesian		Labial	Alveolar	Retroflex	Palatal	Velar	Uvular	Glottal
Stop	voiceless	p /p/	t /t/			k /k/	q /q/	
	voiced	b /b/	d /d/	D /d/	j /g ⁱ /	g /g/		
Affricate	voiceless		C /ts/		c /tʃ/			
	voiced				z /dʒ/			
Nasal		m /m/	n /n/		ñ /ɲ/	ŋ /ŋ/		
Fricative			S /s/		s /ʃ/			h /h/
Lateral			l /l/*		N /ʎ/			
Tap			r /ɾ/					
Trill			R /r/					
Approximant		w /w/			y /j/			

Table 2: proto-Austronesian consonant inventory with phonemes common to pK in blue, pJ in red, and both in purple.

As can be seen clearly, all those consonants in gray change by the time of Old Japanese (/d/ has no pJ parallel):

Deletion: pAN /q/ > pJ $\emptyset$; which likely occurred before proto-Japonic since Austronesian descendants near universally delete /q/ (pAN *qabaŋ ‘boat, canoe’ > Formosan: Kanakanabu *abaŋu*, Siraya *avaŋ*, proto-Rukai *avaŋə; WMP: Yami *avaŋ*, Itbayaten *avaŋ*, Arta *abaŋ*, Gaddang *abaŋ*, etc.)

Devoicing: pAN /b/ > pJ /p/, pAN /g/ > pJ /k/; which may be due to the aforementioned lack of voiced stops in pK.

Affricate approximation: pAN /tʃ/ > pJ /t/; I suspect /tʃ/, a palatal affricate, may have been borrowed as /ts/, an alveolar affricate, through the recipient language, placing it one step closer to becoming /t/, an alveolar stop.

Nasal approximation: /ɲ/ > pJ /n/; proto-Koreanic did not differentiate palatalized consonants, hence borrowing it as /n/.

Sibilant approximation: /ʃ/ > pJ /s/; proto-Koreanic would borrow the palatal fricative as an alveolar fricative for the same reason.

Liquid approximation: /r/ > pJ *r; the alveolar trill is borrowed as a tap, the only liquid in proto-Koreanic. The lack of initial r- in native Korean words would also motivate the initial liquid loss seen in proto-Japonic.

Unique changes: /d/, /gi/, /dʒ/ > /ɾ/; several pMP languages turn /d/ into a liquid /ɾ/ or /l/ (ex. pMP *badas ‘grit, coarse sand, gravel’ > Hanunóo *barás*, Bukat *bara*, Ngaju Dayak *baras*, Cebuano *balás*, Ayta Maganchi *balah*, Kapampangan *balás*, Tagalog *balás*. 3.2.173 *zaqet > Yami *rahet*, Itbayaten *raet*, *zaŋan > Yami *raŋan*, Ivatan *raŋan*, show how /gi/ already evolves into /ɾ/ independently outside pJ. The same is true for /dʒ/ > /ɾ/, 3.2.74 *najam > Yami *ŋaran*, Itbayaten *ŋaran*, Ibaloy *ŋaran*, Bikol *ŋáran*).

Since there are some examples of allophony between pJ *t and *s, it could be possible that /ts/ once existed in proto-Japonic and then broke down into this allophony, motivated by reanalysis as a cluster pJ ts# becoming a simple pJ *t, pJ *s. 3.2.162 *tutuh > pJ *tutu-i/*tutuh-i > WOJ *tuti/tutui*, show that pJ *h may have existed for a while before being deleted.

proto-Koreanic did not have /l/ so it is likely this is why pAN /l/ is approximated by pJ /ɾ/. pJ /ɲ/ could have existed but would have merged with pJ /n/ according to the pAN parallels if it did. Glide consonants are the only consonants in proto-Japonic but not Old Korean, which may have been motivated by vowel clusters with pJ *i or pJ *u becoming pJ *y and pJ *w (3.2.100 *piqek > NJ *hiyoko*, 3.3.1 *buak > WOJ *wakas*, 3.3.2 *buás > WOJ *wasá*, 3.3.3 *ian > WOJ *ya*, 3.3.6 *qiəm > WOJ *yomwi/yamwi*, 3.3.7 *uit > WOJ/EOJ *wi*).

The only change that remains unexplained is why Arisaka’s Law occurs, although it does reveal that the conditioning is based on the first syllable of a word, explaining 3.4.1. *bəkəlaɟ > pJ *pəkəɾə. This may rule out reconstructions such as “pJ *kətə ‘word, thing’ ~ pJ *kətə-ar- ‘have words’ > *kətar- > OJ *katar*- ‘tells’” (Francis-Ratte, 2016. p.66). Whether it does or not depends if all examples of affixed pJ *-r are actually pJ *-ar, since the schwa is preserved in examples like

#3.2.37 *gebáq > pJ *kəpə/kombə-(a)r?-ay > WOJ *kopore/kobore*, which would mean the *a is deleted during affixing. This is likely true since the suffixes *-(a)p, *-(a)y, also come from verbs starting in *a, rather than coming from epenthesis or the morphemes they attach to. Additionally, OC 曰 *[ɕ]ʷat ‘say’ and OC 謂 *[ɕ]ʷə[t]-s ‘say, tell, call’ (Baxter-Sagart, 2014) provide new sources for pJ *kat-ar > WOJ *katar* ‘tell, relate’ and pJ *kət.ə > WOJ *koto* ‘word’. The most likely explanation may be that epenthesis primes a proto-Japonic speaker to match vowels across syllables, as in the cases above *katar/koto*. For example, CaC > CaC.a-, CəC > CəC.ə-, then CaCə > CaCa, CəCa > CəCə, which may have resulted in back assimilation of /ə/: CəCu/CuCa, CəCo/CoCə > CaCu/CuCa, CaCo/CoCa. This change would appear as a prohibition but would actually be intramorphemic vowel assimilation resulting from epenthesis motivating intramorphemic vowel agreement between /a/ and /ə/. This last step is purely conjectural but would have significant explanatory power for the reasoning behind Arisaka’s Law.

All of these steps demonstrate a strong link between proto-Korean phonology and proto-Austronesian phonotactics with proto-Japonic phonology and phonotactics. This may be taken as a sign that a proto-Koreanic language was the likely recipient of the proto-Austronesian loans here. This fact would explain other parallels observed between proto-Koreanic and proto-Japonic, such as their SOV typology and agglutination.

4.2 Semantic Fields

There are two large sets relating several words by semantic grouping: farming, and cooking/eating, which are of course intimately related to each other. An additional small set of family-related vocabulary is also notable, possibly providing some hints, albeit unclear, regarding how these words may have been borrowed. Collating these as an index should help to illustrate the importance of farming in this comparanda, as well as an aid to navigate the data for their explanations in their full entries.

4.2.1 Farming Vocabulary

The following list shows a rather broad list of agricultural activity, which taken altogether illustrates a quite descriptive picture. Uncultivated fields (*para*) must have been cut short (*par*), possibly even burned and slashed (*hatake*), for upland (dry) agriculture; wet fields were set up in swamps (*sapa*) using boundaries (*pata*) to hold soaked soil (*sopoti*) and plant (*par*, *pur*) rice seeds (*tane/sane*), sprinkling (*sibuk*) the sprouts (*tukur*) with water; crops swayed (*yugameru*) before being reaped (*kar*) for their grains, which were beaten (*tuk*) and crushed (*pispis*) in a mortar (*usu*), with a hammer (*tuti/tutui*); separating pebbles from the rice (*piliq*) to be served for consumption.

3.1.1 pMP *barbar ‘spread out, unfurl’ > WOJ *par* ‘stretch’

3.1.5 pMP *budbud ‘sprinkle seed, sow seed in dibble holes’ > WOJ *pur* ‘rain, fall (precipitation)’

3.1.30 pWMP *palpal ‘trimmed short, as grass; bare (of a field)’ > WOJ *par* ‘open land’

3.1.34 pWMP *patpat ‘cutting, slashing, as in clearing land’

- 3.1.38 pMP *pispis ‘to squeeze’ > pJ *pis.i-nk? (press.between-EP-VBZ) > NJ *hisigu* ‘to crush, to squash, to overpower, to beat, to crush’
- 3.1.41 pWMP *punpun ‘bunch, cluster, as of fruit’
- 3.1.54 pAN *tuktuk ‘to knock, pound, beat; crush’ > WOJ *tuk* ‘pound; hull’
- 3.2.8 pWMP *ataq ‘unhusked rice grain left among the husked grains’ > WOJ *atara* ‘in vain, be a waste’; although this word no longer relates to agriculture, it also shows rice related vocabulary is especially commonly borrowed.
- 3.2.16 *pWMP *bates ‘border, boundary’ > WOJ *pata* ‘side, edge’
- 3.2.26 pAN *CaNem ‘grave; to bury’ > WOJ *tane* ‘seed’, WOJ *sane* ‘seed, pit, stone’
- 3.2.28 pAN *CekeS ‘kind of very slender bamboo’ > WOJ *taka/take* ‘bamboo’
- 3.2.58 pPh *keRaw ‘dry up’ > OJ *kare* ‘wither’
- 3.2.59 pAN *keRet ‘cut something slender or soft; reap crops’ > WOJ *kar* ‘cut; harvest; split’
- 3.2.63 pMP *lebet ‘set close together, dense, luxuriant’ > pJ *əpə-ku (dense, big, many-ADJZ) > > WOJ *opo(-ku)* ‘big, grand; plentiful, great in amount or degree’, WOJ *opos* ‘cultivate’, WOJ *opwi* ‘grow’
- 3.2.67 pWMP *lingan ‘to sway back and forth, as plant stems in the wind’ > NJ *yugameru/igameru* ‘to bend, to curve, to warp, to distort, to disfigure’
- 3.2.70 pMP *lumen ‘smeared, smudged, dirty with’ NJ *umeru* ‘to bury (e.g. in the ground)’
- 3.2.71 *pAN/pMP *lusun ‘rice mortar’ > WOJ *usu* ‘mortar’
- 3.2.80 pWMP *padan ‘uncultivated field, open grassland; field covered with sword grass (*Imperata cylindrica*)’ > OJ *para* ‘field, plain’
- 3.2.87 pWMP *pankal ‘visible tree roots above the ground’ > WOJ *pag* ‘peel’
- 3.2.98 pAN *piliq ‘to choose, to select; to pick out’ > WOJ *pirip* ‘pick up’; Pazeh (MP) *pii-piri* ‘be choosing (as when picking up pebbles from rice grain)’
- 3.2.101 pMP *puqun ‘base of a tree; source, origin; beginning; foot of a hill or mountain’ > WOJ *pumoto* ‘the foot of a mountain’
- 3.2.110 pMP *saban ‘estuary, shore near the mouth of a river’ > WOJ *sapa* ‘mountain stream; swamp’
- 3.2.125 pWMP *sasag ‘wattle, bamboo plaitwork construction for walls’ > WOJ *sasa* ‘broad-leaf bamboo’
- 3.2.126 pMP *sepet ‘obstructed, of the flow of water’ > pJ *səpət-i (be.full.of.water-CONN) > *sopot/sopoti* ‘be wet’; Ilokano (MP) *seppét* ‘be choked (plants); be obstructed (liquids)’
- 3.2.130 pMP *simbuR ‘to sprinkle’ > WOJ *sibuk* ‘to splash; to spray’; Ilokano (MP) *sibúg* ‘water used for watering plants’
- 3.2.137 pMP *su(ŋ)kab ‘to open, break open’ > WOJ *suk* ‘to hoe’
- 3.2.147 pWMP *ta(m)pas ‘lop off, cut off’ > NJ *taba* ‘bundle, bunch, sheaf’
- 3.2.151 pWMP *tayug ‘swing, oscillate’ > WOJ *tayutap* ‘be unsettled by movement’
- 3.2.156 pAN *tukul ‘to sprout, begin to grow’ > OJ *tukur* ‘make’
- 3.2.162 pAN *tutuh ‘beat, pound, bump, thump’ > WOJ *tuti/tutui* ‘hammer; mallet’
- 3.3.2 pPh *buás ‘tomorrow’ > WOJ *wasá* ‘early’, WOJ *wase* ‘early rice’

4.2.2 Cooking/Eating Vocabulary

Kindling (*hota*) or other fuel (*atu-*) had to be kindled/burned/scorched (*tak*, *tomos*, *ir*) to start the fire, boiling (*wak*) rice grains to soften them; the water bubbling (*fuku*) and steaming (*kusuburu*). Fish (*asar*) that had been hooked (*tur*) and the chickens (*kakye*) they raised (*hiyoko*), were also roasted (*saras*); it was then chewed (*paqpaq*) for small children (*mamaq*) before being received (*tage*) as a meal (*ke*). Some meals tasted stronger (*kara*), others blander (*tapa*), while the better ones (*nama-*) were voraciously eaten (*pukure*).

3.1.6 pMP *bukbuk ‘bubbling, boiling’ > NJ *fuku* ‘to emit, to spout, to spurt, to boil over’

3.1.10 pPh *gidgid ‘rub something off’ > pJ *kir- (use.friction) > MJ? *kir* ‘to make fire’

3.1.12 pMP *ke(k)kek ‘to cluck’ > WOJ *kakye* ‘chicken’

3.1.13 pAN *kemkem ‘bite down hard’ > > WOJ *kam* ‘to chew (rice to make sake)’

3.1.15 pMP *kitkit ‘to bite, gnaw’ > WOJ *ki* ‘fangs, teeth’, NJ *kiba* ‘tusk, fang’

3.1.21 pWMP *kuskus ‘rising of smoke or steam’ > NJ *kusuburu* ‘to smoke, to smoulder, to smolder, to sputter, to be sooty’

3.1.25 pAN *ñamñam ‘tasty, delicious (?)’ > EOJ *namwi* ‘offering’, NJ *name* ‘to lick, to lap, to suck, to taste’

3.1.32 pMP *paqpaq ‘premasticate food for infant’, ‘chew’ (PAN) > OJ *pa* ‘tooth’

3.1.35 pPh *petpet ‘piece of something that has been cut’ > NJ *hota* ‘kindling, small bits of scrap wood used to fuel a stove’

3.1.45 pAN *supsup ‘sip; suck’ > NJ *su(w)* ‘to smoke, to breathe in, to inhale; to suck, to sip, to slurp’

3.1.49 pPh *temtem ‘to burn something in a bonfire’ > WOJ *tomos* ‘kindle’

3.2.31 pWMP *deket ‘ignite, set aflame’ > OJ *tak* ‘kindle’

3.2.35 pAN *garan ‘fierce, intense, as of appetite or temper’ > WOJ *kara-ku* ‘salty’

3.2.45 PMP *hasan ‘gills’ > pJ *asa-r > WOJ *asar* ‘look for; hunt for food’, NJ *isar* ‘id.’

PWMP *isan ‘gills’ > pJ *i(n)sa-r > WOJ *izar* ‘to fish’

3.2.46 PPh *hátun ‘fuel for fire, anything used to start a fire’ > WOJ *atu* ‘warm, hot; thick; deep, extreme’

3.2.73 pMP *mamaq ‘chew without intending to swallow, as betel nut; premasticate food to give to an infant; premasticated food’ > NJ *mama* ‘cooked rice (children’s word)’

3.2.76 pMP *ñaman ‘tasty, delicious’ > WOJ *namasu* ‘thinly cut meat eaten raw’

3.2.100 pMP *piqek ‘baby chick; peeping of chick’ > NJ *hiyoko* ‘young bird, chick, fledgling’

3.2.106 pMP *qilaw ‘torch; illuminate’ > WOJ *ir* ‘scorch’

3.2.112 pWMP *sadab ‘to singe, burn off’ > WOJ *saras* (Ex. *saray > sare) ‘expose (to the elements); dry; bleach; wash’

3.2.142 pWMP *tabar ‘tasteless, insipid; powerless; antidote, as against snakebite’ > WOJ *tapa* ‘foolish’

3.2.150 pWMP *tanqap ‘receive, accept’ > WOJ *tage* ‘eat/drink’

3.2.158 pWMP *tuluy ‘caught up in, carried along’ > WOJ *tur* ‘fish, angle’

3.3.1 pAN *buak ‘bubbling up (as spring water)’ > WOJ *wakas* ‘boil (v.t)’

3.3.4 pAN *kæn ‘eat’ > OJ *ke* ‘food (container)’

3.4.1 pPh *buk(ə)law ‘gluttonous, greedy with food’ > pJ *puk.ura-(a)y (full.belly-EP-get) > WOJ *pukure* ‘fat person’

4.2.3 Family Vocabulary

This last set, including Vovin’s pJ *tina- ‘mother’ reconstruction demonstrates a tight-knit kinship consisting of a mother, father, and grandchildren, with marriage extending the family. This likely suggests Austronesian was spoken as a home language before being publicly displaced.

Interestingly, Manchu has the words □□□ /əmə/ ‘mother’ and □□□ /ä.mä/ ‘father’, the former likely cognate with WOJ *omo* ‘mother’ < pJ *əmə, and possible NK 엄마 *eomma*. The migration of Austronesian speakers to Korea was likely male-biased (Jin et al., 2009), so the women they married on the peninsula may have been Tungusic before Koreanic speakers arrived. The pAN sense of *duma ‘someone; other people; be different’ lends credibility to this being parallel with pJ *tuma ‘wife’, as a woman from the peninsula who was not also Austronesian. This change could have originally referred to non-hostile non-Austronesians, unlike WOJ *ata* ‘enemy’ < pMP *qaRta ‘outsiders, alien people’.

Vovin (2014b) reconstructs pJ *mia, comparing it with pT *mia (A2) ‘wife’; Pittayaporn (2009) reconstructs pT *miä (A), though either would yield WOJ *mye* ‘woman’. Pittayaporn, however, also reconstructs several other family terms which I believe to have been loaned into proto-Japonic: pT *me: (B) ‘mother; woman’ > , pJ *-mi ‘female’ > pJ *wo-mi-na (small-female-name) > WOJ *omina* ‘old woman’; pT *pa: (C) ‘parent’s older sister’ > WOJ *papa* ‘mother’; pT *bi: (B) ‘older sibling’ > pJ *miä-pi (woman-older.sibling), pJ *wo-pi (man-older.sibling) > NJ *mei* ‘niece’, NJ *oi* ‘nephew’; pT *bo: (B) ‘father, man’ > pJ *wo > WOJ *wo* ‘male; man; husband’. How ‘older sibling’ could come to mean ‘niece/nephew’ is admittedly difficult to unravel, however it would explain the -i in NJ *mei*, *oi* which is otherwise etymologically opaque. My conjecture is that this developed from an unattested reduction of *miä-pi-no-ko and *wo-pi-no-ko, ‘(my) older woman/man-sibling’s child’, which is a rather long phrase with no homophones /miäpi/ or /wopi/, making the truncation easy to understand. The shift from pT ‘parent’s older sister’ to pJ ‘mother’ suggests that it was one’s aunt who used to rear the child, later using pJ *wo-ⁿ-pa (small-GEN-aunt) > NJ *oba* ‘aunt’ when *papa* came to refer to one’s birth mother, rendering the use of *taratine* ‘birth mother’ unnecessary. Further proof of pJ *pi ‘older sibling’ exists in WOJ *pikwo* ‘grandchild’ which also has an uncertain *pi that could be linked to NJ -i in *mei/oi*. This word would have rendered the original sense of pMP *umpu ‘grandchild’ as redundant as well, with the meaning narrowing into ‘newborn’.

3.2.3 pAN *amax ‘father’ > WOJ *tama*, *tamasipi* ‘soul’

3.2.32 pAN *duma ‘someone; other people; be different’ > OJ *tuma* ‘spouse’

3.2.72 pAN *mamah ‘parent’s younger sibling, junior uncle; a young child’s term of address for his father, vocative of *ama’ > NJ *mamako* ‘stepchild’

3.2.168 pMP *umpu ‘grandparent/grandchild (recipr.); ancestor’ > NJ *ubu* ‘innocent, naive; birth, newborn’

4.3 *Lexical*

The following words are selected from the Leipzig-Jakarta word list, which identifies basic/common words that are resistant to borrowing (Tadmor et. al, 2010). Therefore, these may have a higher probability towards being from a re-lexified Austronesian adstratum rather than merely being creolized or nativized by a pre-Japonic language coming into contact with Austronesian speakers.

4.3.1. *Body Vocabulary*

3.1.4 pMP *biRbiR ‘lip’ > NJ *kuchibiru* ‘lips’

3.1.15 pMP *kitkit ‘to bite, gnaw’ > WOJ *ki* ‘fangs, teeth’, NJ *kiba* ‘tusk, fang’

3.1.32 pMP *paqpaq ‘premasticate food for infant’, ‘chew’ (PAN) > OJ *pa* ‘tooth’

3.1.52 pMP *titi ‘breast; suck the breast’ > WOJ *ti* ‘milk, breast’, ‘blood’?

2.3.42 pWMP *gu(m)pit ‘pinch together; narrow’ > WOJ *kubi* ‘neck; collar’

3.2.53 pWMP *kabal ‘invulnerable, invulnerability’ > OJ *kapa* ‘pelt, hide’

4.3.2 *Verbs*

3.1.22 pPh *layláy ‘to wilt, wither, droop, as plants; dangling, hanging loosely’ > WOJ *aye* ‘fall off’

3.1.26 pMP *ɲakɲak ‘raucous sound’ > OJ *nak* ‘to cry’

3.1.33 pPh *paspás ‘do something quickly, in a hurry’ > WOJ *pasir* ‘run’

3.1.38 pMP *pispis ‘to squeeze’ > NJ *hisigu* ‘to crush, to squash, to overpower, to beat, to crush’

3.1.47 pAN *taktak ‘clatter, clack’ > WOJ *tatak/tadak* ‘hit, strike’, WOJ *tatakap* ‘fight’

3.2.31 pWMP *deket ‘ignite, set aflame’ > OJ *tak* ‘kindle’

3.2.47 pAN *heban ‘carry a child with a carrying cloth’ > OJ *op* ‘bear, carry; cover’

4.3.3 *Miscellaneous*

3.2.52 pAN *isa ‘one’ > WOJ *isasaka* ‘be for a little while’, WOJ *sasara* ‘small, minutely’, WOJ *isasameni* ‘perfunctorily, a little while’, MJ? *isara/isasa* ‘a little/small’

3.2.60 PPh *kulap ‘dim, not well-lit’ > WOJ *kura-ku* ‘dark’, WOJ *kurwo* ‘black’, OJ *kure* ‘get dark’, WOJ *kuras* ‘let time pass’

3.2.129 pMP *silaw ‘dazzling, of light’ > pJ *sira-/sir.o? (bright) > OJ *sira/sirwo* ‘white’

3.2.21. pWMP *bidan ‘unit of measure for cloth, sails, etc.’ > WOJ *piro* ‘wide; unit of measurement (length)’, EOJ *pirə* ‘to be wide’, WOJ *piror* ‘be widespread’, WOJ *pira* ‘flat’ (EOJ ‘flat, ordinary’)

3.3.3. pAN *ian ‘dwell, reside, live in a place’ > OJ *ya* ‘house’

5 **Discussion**

5.1 *Period and Point of Contact*

The absolute lack of proto-Japonic > proto-Austronesian loans can be explained by a migration of the latter into Japan, rather than Japonic peoples living in Taiwan or Austronesia. The Austronesian stratum matches the earlier period of wet rice cultivation which, according to Whitman (2011) “reaches northern Kyūshū by 950 BCE, and the western end of the Inland Sea by 600 BCE” (id.). The fact that wet rice cultivation arrived in the western end explains “[t]he relative importance of wet rice cultivation show[ing] regional variation throughout the Mumun period: wet rice is dominant in the central and southwest regions, while millet and other dry field crops are relatively more important in the southeast (Ahn 2010: 93)”. He also cites “According to Ahn (2003: 81–81; 2010: 91), rice farming settlements disappear from the archaeological record during the third century BCE, reappearing again in the first century CE[...] The hiatus in wet rice sites coincides with the emergence of the Korean-style bronze dagger culture in the third century BCE.” Therefore, this was likely the period that Koreans migrated south into the Korean peninsula, resulting in the phonological and phonotactic changes observed in 4.4.1.

Whitman observes several possible cognates relating proto-Korean to proto-Japonic agriculture: pK *patəh > LMK path (L) ‘dry field’ > pJ *pata (2.4), *patakaC (3.7a/b) ‘id.’, pK *matəh > LMK math (L) ‘garden plot’ > pJ *mati (2.2a) ‘id.’, pK *nat > LMK nat (H) ‘sickle’ > EMJ nata (2.1) ‘hatchet’, -na ‘blade’, pK *non > LMK non (H) ‘rice paddy’ > pJ *nuu (13.b) ‘marsh’, pK *pasi/*pəsi > LMK psi ‘seed’ > pJ *pisi (2.1) ‘water caltrop (Trapa)’ (id.). pK *patəh may be a loan from pJ *pata instead; pK *nat is likely related to OC 刈 *ŋa[t]-s ‘mow, cut (v.)’, possibly borrowed from proto-Japonic which would have merged initial /ŋ/ with /n/ and resolved through epenthesis OC *ŋa[t]-s > pJ *nat.a > NJ nata ‘nata, sturdy broad-bladed knife, used in woodcraft and hunting’. Another Old Chinese parallel appears in pK *non, likely from OC 農 *n[ʰ]oŋ ‘agriculture; to farm’ which also becomes pJ *no, no-ma (wet.land-space) > WOJ nu, numa ‘swamp’. pK *pasi/*pəsi may ultimately be from Tungusic: Manchu *fisihe* ~ *fisike* ‘broomcorn millet’, Ulch *pikse* ‘id.’ and Manchu *fisi-* ‘to sprinkle with the hands’ (Robbeets, 2017. p.223). pWMP *pispis ‘to sprinkle, shake off water’ and pMP *picik splash, spray, sprinkle must be related as well, perhaps showing one of few possible examples of a northern loan in proto-Austronesian.

I believe the arrival of Korean at this time gave Japonic several agricultural etyma: pJ *pwo (1.3a) ‘ear of grain’; pJ *ta (1.3a) ‘ricefield’ (Vovin, 1998. p.368); pJ *mʷnki ‘wheat’. OC 飯 *bo[n]ʔ-s ‘cooked rice or millet’ (Baxter-Sagart, 2014) appears rather similar to Francis-Ratte’s (2014. p.242) reconstruction “EAR OF GRAIN: MK pyé ‘rice plant, kernel of rice,’ psól ‘hulled rice grain’ pKJ *pə ‘ear of grain’[...] MK pyé and psól [are] morphologically complex, from pre-MK *po-ye and *po-sol respectively. OC 麥 *m-rək appears rather similar as well to MK *milh* 밀 ㅎ ‘wheat’ (Lee & Ramsey, 2011. p.183). Finally, the *gukja* character 畚, read 답 *dap* meaning ‘rice paddy’ clearly talks about wet rice fields, whereas the *kokuji* character 畑 *hatake* means ‘dry field’, both specifying types of rice farming, hence 畚 may be the source for pJ *ta < OK? *tap?. Francis-Ratte (2014. p.317) also relates MK *pap* ‘cooked rice; food’ with OJ *apa* ‘millet’, although I believe OC 粒 *p.rəp ‘grains of rice’ (Baxter-Sagart, 2014) is the source for MK *pap*, as it matches the sense and phonetic shape better. Interestingly, this word becomes

Middle Chinese *lip* (id.) which I believe to be the source for OJ *ipi* ‘cooked rice’ < pre-OJ *(l)ip.i < MC *lip*. OJ *apa* may have been introduced from outside Koreanic, such as Hungarian *árpa* ‘barley’ and Mongolian *apᠪᠠᠭ* (arvaj) ‘id.’, both likely coming ultimately from 13th c. Turkish *arpa* ‘id.’ (Clauson, G. 1891. p.198). The existence of a separate word for millet, WOJ *piye* ‘barnyard millet (*Echinochloa esculenta*)’, was observed by Robbeets (2017. p.224) to be related to “MK *phi*, NK *phi* ‘(Japanese) barnyard millet (*Echinochloa esculenta*)’.

All of these words support Whitman’s conclusion that “Japonic arrives in the Korean peninsula around 1500 BCE and is brought to the Japanese archipelago by the Yayoi expansion around 950 BCE,” and “Koreanic arrives in the south-central part of the Korean peninsula around 300 BCE with the advent of the Korean-style bronze dagger culture”. They also reveal a mixed strata regarding agricultural synonyms, such as WOJ *ta* ‘paddy field’ vs WOJ *pata* ‘paddy field’.

5.2 *Interdisciplinary Evidence*

5.2.1 *Genetic*

Haplogroups are passed down, either paternally down a male lineage, or maternally through mtDNA through all descendants. “Interestingly, a high frequency (31.4%) of haplogroup O2b-SRY465 [now named O1b2] (and its sublineage) is characteristic of male Koreans[...]the coalescence time (~4,400 years) for the age of haplogroup O2b1-47z [O1b2], and its Y-STR diversity, suggest that this lineage probably originated in Korea” (Soo Jin Kim et al., 2011). Therefore, it is more likely that this lineage is more closely related to other northeast-Asian peoples rather than southeast-Asian peoples, despite being a so-called para-Austroasiatic clade. (Van Driem, 2009). “Both mtDNA haplogroup D4 and Y chromosome haplogroup O1b2 (formerly O2b) are common lineages not only among the Japanese, but also among continental East Asians, including Koreans, Chinese, and Mongolians” (Kim et al., 2026). This fact is sometimes taken as evidence that O1b2 is a marker of proto-Japonic speakers who came from the Korean peninsula during the Yayoi period. “The analysis of admixture modeling for Yayoi individuals, Kofun individuals, and modern Japanese respectively supported a two-way admixture model assuming Jōmon-related and Korean-related ancestries. These results suggest that between the Yayoi and Kofun periods, the majority of immigrants to the Japanese Archipelago originated primarily from the Korean Peninsula” (Kim et al., 2024). While this is likely true correlatively, this haplogroup “does not appear to be correlated with any extant linguistic group today” (Van Driem, 2009). I believe this paternal haplogroup represents the likely Koreanic-related adstratum that absorbed the Austronesian-speaking populations during the Yayoi period. It is necessary to reiterate that whether that incoming language was specifically proto-Koreanic or another related family is unknown and that Austronesian-speakers on the peninsula are likely distantly related to the original pAN/pMP populations, rather than having come directly from Indonesia or South China. This population may have been Northern Han related and part of a larger Austronesian cultural sphere encompassing the Chinese seaboard, hence being Austronesian-speaking but not possessing the common genetic markers of Taiwanese and Malayo-Polynesian peoples.

“[T]he current genetic foundation of Koreans may have been established through a rapid admixture with ancient Southern Chinese populations associated with Iron Age Cambodians. We speculate that this admixing trend initially occurred mostly outside the Korean peninsula followed by continuous spread and localization in Korea, corresponding to the general admixture trend of East Asia. Over 70% of extant Korean genetic diversity is explained to be derived from such a recent population expansion and admixture from the South” (Kim et al., 2020). Needless to say, over two-thirds of the diversity is rather significant and may be evidence of a significant Austronesian-speaking population becoming part of the Korean genetic profile. Similar to how the Japanese genetic profile has been described as having formed from a dual model of Jōmon-period indigenous peoples and Yayoi-period immigrants, the Yayoi-period immigrants may themselves have been composed of both Koreanic-related immigrants from the north with Austronesian-speaking peoples who had migrated to the south of the Korean peninsula during the late-Jōmon period. O1b2 is the dominant marker of the former, while the latter I believe to be a subclade of O2-M122, “Recent molecular genetic analyses and the geographical distribution of haplogroup O-M122 [O2] lineages, found widely throughout East Asia at high frequencies (especially in southern populations and China), have suggested a link between these Y-chromosome expansions and the spread of rice agriculture in East Asia. In general, Y-chromosomes might be spread via a process of demic diffusion during the early agricultural expansion period” (Jin et al., 2009).

The subclade that I believe to have been highly related to the Austronesian-speaking adstratum present in the extant diversity of Korean genetic profile is O2a2b2-N6, a subclade of haplogroup O2a2b-P164. “Previous studies suggested that the Y-chromosome haplogroups O2-N6-B451-AM01756 and O1a-M119 are two founder lineages of proto-Austronesians at about five thousand years ago.” (Yu et al., 2023) In a study that analyzed “seventeen Y-chromosome sequences of haplogroup O3a2b*-P164(xM134) [now O2a2b-P164] and generated a revised phylogenetic tree of this lineage based on 310 non-private Y-chromosome polymorphisms. [It was] discovered that all available O3a2b*-P164(xM134) [now O2a2b-P164] samples belong to the newly defined haplogroup O3a2b2-N6 [now O2a2b2-N6].” O2a2b-P164 is present at elevated rates within the Amis of Taiwan in comparison with other indigenous Formosan groups,

“Just over a decade ago sub-haplogroup O2a2b-P164 (formerly O3a2c-P164) was described as a link between the Ami aborigines of Taiwan and the populations of Tonga and Samoa suggesting that this native community from Formosa was a source population of the Austronesian speakers that settled West Polynesia. Subsequently, it was established that O2a2b-P164 is widely distributed in the eastern coastal regions of Asia, from Korea to Vietnam and signals a genetic connection between proto-Austronesians-speaking populations of MSEA and the early Austronesian-speaking populations that crossed the Taiwan strait ~4800 ya and then spread into Oceania. According to this study, these Austronesian migrants were the ancestors of the Neolithic farmers that cultivated rice and foxtail millet in northern Taiwan ~4500 ya” (Luis et al., 2024).

The subclade of O2a2b-P164, “[t]he Austronesian-associated haplogroup O2a2b2-N6[,] was observed in both Dawenkou and Hanben individuals and is prevalent among modern Austronesian speakers, suggesting a direct genetic link between Neolithic northern and southern Chinese farmers and contemporary Austronesian groups in Southeast Asia.” (Wang et al., 2026). Curiously, “[e]xcept for eleven males from Hunan, all O3a2b2*-N6(xB451) [O2a2b2-N6] samples were from North China and the eastern coastal regions of East Asia,” (Wei et al., 2017) which shows that while “O3a2b2-N6 [O2a2b2-N6] samples are widely distributed on the eastern coastal regions of Asia, from Korea to Vietnam,” (id.) O2a2b2-N6 was more common along the northern coast than the far south. Therefore, it may be determined that this subclade represents the pre-Yayoi Austronesian-speaking adstratum present in the southern Korean peninsula. That said, the role of this subclade within the history of Northeast Asia and especially within Korean populations merits further examination and scrutiny.

O2a2b2-N6 may have spread from the Yangtze river then to Taiwan, since Wang et al. (2021) observe “gene flow from Neolithic northern China into Taiwan, which [they] estimate at ~25% if [modeled] as derived from one of the two source lineages of Yellow River farmers. This ancestry does not fit as coming from Yellow River farmers themselves, suggesting a north-to-south migration not associated with expansions of these farmers. A speculative possibility is that this ancestry was carried by cultivators of foxtail millet which was domesticated in the north by ~8000 BCE, and which in the south appears relatively early in the Taiwan Neolithic Tapenkeng culture (~3000–2500 BCE)”. The relation of the Amis people of Taiwan with this northern ancestry is explored in the following section.

5.2.2 *Archaeological/Cultural*

According to Hudson (2012), “[t]he Sakishima islands of the southern Ryukyus were settled around 4300 years ago by a quite different group of people(s) who seem to have come not from Japan, but from somewhere in Taiwan and/or Southeast Asia[...], the archaeological record offers no evidence for the movement of people or artefacts across the gap between Okinawa and Miyako Islands, and it is widely assumed that this marks the boundary between two different cultural zones” (Hudson 2012: 258–259) [cited from Elmer, 2019b. p.215]. The lack of evidence of Austronesian settlers may present some curious problems at first, but this is remedied by positing that Austronesian-speaking peoples rather than direct genetic descendants arrived sometime during the Jōmon period. Hudson (1999) claims “the presence of Austronesian (or an ancestral form of Austronesian) in Jōmon Japan is the easiest way to account for links with Japanese/Ryukyuan. I mostly agree with Robbeets’ claim that,

“[F]arming can also explain prehistorical connections between Japonic (sic) and Austronesian, albeit within a borrowing context[...], a para-Austronesian language was spoken on the Shandong Peninsula, while the homeland of Japonic was situated on the Liaodong Peninsula between the third and second millennium BC, with its speakers adopting rice agriculture and Austronesian borrowings within the Liaodong-Shandong interaction sphere.” Robbeets (2017)

Rather than para-Austronesian, which Robbeets suggests would be a proto-Austronesian sister language, I believe it is possible that proto-Malayo-Polynesian was spoken along the Chinese coast adjacent to Taiwan. The name Amis is said to mean northerner, possibly coming from Malayo-Polynesian peoples referring to their northern origin in Taiwan, although there is no current evidence of this (Ross, 1995). I suggest that the name must be pAN in origin, since pMP descendants all come from pMP *qamihan ‘north; northern wind’ < pAN *qamiS-an, with Amis and Puyuma peoples keeping the unaltered original *qamiS ‘north wind’. This would mean the Amis would have come from north of Taiwan, not the Philippines, perhaps from the Yangtze river. Placing the origin of the Amis closer to Northeast China may explain their elevated rates of O2a2b-P164, seeing as the descendant subclade O2a2b2-N6 was present from Korea to Vietnam, especially around coastal east Asia and northern China. This would also make the substratum present in Japonic a close relative of the “Austronesian Ancestor of Tai-Kadai” (Sagart, 2005. p.161), which may explain etyma that only have a close match with proto-Tai but not pAN/pMP, as both could have inhabited the coast together.

The significance of millet rather than rice to the Amis may also suggest an origin north of Taiwan. According to Lee (2024), “[f]or a long period of time, the Ami people sustained themselves by hunting, fishing, foraging wild vegetables, and cultivating millet in horticulture as their major food sources. The traditional life style changed greatly during the Japanese colonial period (1895-1945) when the Amis were forced to learn rice cultivation in paddy fields”. The cultivation of millet rather than rice demonstrates how Amis are also more closely related to millet farmers from the north prior to, ironically, being forced to use wet rice cultivation by the Japanese. Fortunately, “millet still plays an important role in traditional activities of modern Taiwan aborigines, such as festivals, ceremonies, and sacrifices to their ancestors” (Luis et al., 2024).

Besides millet, several Jōmon period customs appear very culturally Austronesian: tooth blackening (NJ *ohaguro*), tooth ablation, cranial deformation, tattoos, jar burials, stilt homes, and curved jade beads (*magatama*). For the first, Chen (1968. p.256) says “Tooth-blackening was also common among the Paiwan and Ami”. Next, “[t]he sites from Late and Final Jomon periods have much higher frequencies of ablation,” (Harvey et al., 2017) the estimated time that pAN/pMP peoples could have arrived in Kyūsyū. Seguchi et al. (2023) “concluded that Hirota site crania were intentionally modified. Although the motivation of the practice is unclear, the Hirota people³ may have deformed their crania to preserve group identity and possibly aid in the long-distance trade of shellfish, as seen archaeologically”. Meanwhile, Visayan peoples of the Philippines were found to have a large amount of ancient samples with artificially modified crania (Clark, 2013), although this similarity could have been convergent.

Third, Whitman (2011) notes from the Wei Shu that ‘Chinhan men and women are close to Wa (男女近倭),’ the ethyonym for the contemporary inhabitants of the Japanese archipelago, and like the Wa tattoo their bodies. Peters (1990. p.16) cites from the Zhuangzi that “the Yueh

³ The Hirota site is found in Kagoshima, Japan

people cut their hair short and tattoo their bodies”. The ancient Yue (‘Baiyue’) who inhabited the southern coast of China have been thought to be Tai-speaking, but that remains far from certain, so they could have been Austronesian speaking instead. Elmer (2019b, p.222) mentions how “Some Chinese sources record that the Wa people consider themselves to be descendants of Tàibó[...] Tàibó and Zhòngyōng moved to an area in the lower valley of the Yangtze[...] the area of the indigenous Jīng Mán 荊蠻 people[...]These ‘Jing Man’ natives cut their hair short and tattooed their skins” Clay figures depicting tattooed people have been found in the Batanes islands dated around 3kya (Bellwood et al., 2013. p.90)

Fourth, the Sannai Maruyama site in Aomori, Japan, the northmost prefecture on the mainland, has as many as 800 jar burials for children (Habu et al., 2011). “[T]he practice of secondary burial in jars possibly had its origins in Borneo, eventually spreading throughout the Philippines” (Pineda & Matthews, 2025). Lastly, the curved jade pendants of historic Japan, *magatama*, highly resemble some curved jade beads traded across Austronesia, *lingling-o*.



Figure 1: Wikipedia contributors. (2012). Magatama from the ancient Kofun period. Tokyo National Museum. In Wikipedia, The Free Encyclopedia.
<https://commons.wikimedia.org/wiki/File:Magatamas.JPG>



Lingling-o penannular earrings. Fig. 4. Hung Hung, H. 2016. Jade in Southeast Asia. Encyclopaedia of the History of Science, Technology, and Medicine in Non-Western Cultures.

5.2.3 Historical

5.2.3.1 Korean Peninsula

The migration of these Koreanic peoples south likely began due to the war between Gojoseon, the first Korean state located near the Liadong peninsula, and Yan, a Chinese state east of Gojoseon, during the Warring States Period in 323 BCE. (Lee, S., 2016). The war would have motivated Liadong residents to migrate south along the coast, eventually meeting and admixing with Austronesian speakers to become the first proto-Japonic speakers who started the Yayoi period.

Concerning the Samhan period, Whitman cites “The Wei shu gives phonogrammatic spellings for 54 Mahan settlement names. As pointed out by Toh (2008: 234–5), most are disyllabic: 34 are transcribed with two syllables, 10 more with two syllables and a suffix, one of which is identifiable as *-pieliai 卑離, usually related to the Paekche word *puri* < 夫里 > “town,” itself typically compared to Late Middle Korean -βir ‘town.’ All but one of the 12 Chinhan names are disyllabic (like the Mahan names). Five of the 12 Pyōnhan names have three or more syllables, and three of these appear to involve a suffix, *-mieton 彌凍 and *-jama? 邪馬. The first of these suffixes also occurs in the only Chinhan polysyllabic settlement name. The second is identical to the transcription given in the Wei zhi for the first two syllables for the contemporary *Jama?dǝ “Yamato” grouping in the Japanese archipelago. Scholars generally interpret these as involving a morpheme cognate with proto-Japonic *jama “mountain” (Bentley 2008: 14).”

The tendency towards bisyllabism may have been a remnant of widespread Austronesian-like phonotactics, likely from the peninsular para-Japonic (a lost stratum of Japonic preserved in some former place names in Korea) population. However, unless pJ *yama ‘mountain’ comes from pJ *ya ‘house’ and *ma ‘space’, neither of these words has an Austronesian equivalent.⁴ Therefore, it may be the case that Jōmon period Austronesian speakers were largely confined to the Japanese archipelago, with bisyllabism emerging as an areal trait. Several place names recorded by Koguryo (Lee & Ramsey, 2011. p.41-43) have several Japonic parallels such as: Kog. 旦 *tan, 頓 *twon, 吞 *thon ‘valley’ :: WOJ *tani* ‘valley’; Kog. 烏斯含 *wosaham ‘rabbit’ :: WOJ *usagi* ‘rabbit’; Kog. 那勿 *namwul ‘lead’ WOJ *namari* ‘lead’ and Koreanic parallels: Kog. 於斯 *esa ‘horizontal’ :: MK es- 엇- ‘crosswise’; Kog. 也次 *yacho ‘mother’ :: MK *ezi* 어 ‘mother, parent’; Kog. 波兮 *phahyey, 波衣 *phauy, 巴衣 *phauy ‘boulder’ :: MK *pahwoy* 바회 ‘boulder’; Kog. 別 *pyel ‘pile’ :: NK *pol* ‘pile, layer’; Kog. 首 *sywu ‘ox’ :: MK *sywo* 쇼 ‘ox’. However, none of these have Austronesian parallels, suggesting the lexicon here represents the stratum that underlies proto-Japonic. The dominant language of Koguryeo itself remains uncertain, however Janhunen (2010, p.295) suggests it may have been Tungusic or even Amuric. Whereas, Vovin (2014a) says “it is possible that Paekche, like Silla, had a Japonic substratum,” and “we can safely arrive at the conclusion that a Japonic substratum language in the Southeastern area of the Korean peninsula was pretty much marginalized by this time, probably surviving only in some rural communities”. There are some words recorded by Koguryeo with Tungusic parallels, observed by Pauwels (2024), that are markedly Austronesian: Kog. *kar (加尸) “to plough (犁) :: OJ *kar* ‘to trim, to prune, to mow, to cut’ :: Manchu *χalχan* ‘plowshare’ :: MK *kalay* ‘plow’, *kal-* ‘to plow’ (耕), NK *kalay* ‘three-man plow; spade’; Kog. *patan (波旦) ‘ocean, sea (海)’ :: OJ *wata* ‘sea, ocean’ :: MK *pàtáh* ‘sea, ocean’ NK *pata* ‘sea, ocean’ :: Evenki *ōta* ‘wave, billow, roller’, Negidal *wāta* ‘(large) wave’ Nanai *wāta* ‘wave’; Kog. *sirap/*širap (尸臘) ‘white (白)’ :: OJ *sirwo* ‘white’ :: MK *syēy-* ‘to gray, to turn gray (said of hair)’, MK *hoy-*, NK *huy-ta* ‘to be

⁴ Consider Kunigami *yama* ‘mountain; forest’ (Nakasone, 1983, p. 580), although the latter sense could be innovative.

white, to be fair; to be gray, to be hoary' :: Orok (Uilta): *silei* 'pink, rose-colored; orange', *siloi* 'not very dark color'. These may be the few remaining linguistic traces of Austronesian speakers within the Korean peninsula.

Whitman (2011) cites the Wei shu as describing “the language of Chinhan [later, Silla] as “not the same as that of Mahan [later, Paekche]” (其言語不與馬韓同)[...], planting the five crops with the addition of rice (宜種五穀及稻), suggesting that the latter may have been something of an add-on.” Therefore, Wa and Kaya, both of which allied with Paekche in a war against Silla and Koguryeo, may have originally had an Austronesian cultural stratum, pitting them against the incoming northern Korean and Tungusic immigrants.

5.2.3.2 Hayato and Kumaso

The Hayato and Kumaso people were pre-Yamato tribes present in Kyūshū, likely Japonic in origin, according to Elmer (2019b),

“The most promising material for verifying the ancestry of the Hayato people is supplied by DNA data. Mitochondrial DNA of skeletons associated with the Hayato people has been analysed by Saiki Kazunobu 佐伯和信 and Wakebe Tetsuaki 分部 哲秋 [...]Skeletons have been unearthed from six burial sites from the mountain area of south Kyūshū around Ebino えびの City, Miyazaki 宮崎 Prefecture [...]After analysing thirty samples, Saiki and Wakebe concluded that the Hayato people share almost no characteristics with Jōmon populations and seem to be closer to people from the continent and thus Yayoi DNA[...]This study indicates that the Hayato were early Yayoi immigrants. It remains to be seen if future research can confirm such a proposition” (Elmer, 2019b. p.229).

Unfortunately, “[o]nly two words survive from the Hayato language” (Vovin 2024): Hayato 久西良 *kusera* ‘comb’, WOJ *kusi*, Ryukyuan **kusi*; and Hayato 必志 *pisi* ‘shoal’, EOJ *pīʼzi*, Ryukyuan **pisi*, Ainu *pís* ‘shore’ “The second word, *pisi* ‘shoal’ is probably a native Hayato word, although its relationship with similar forms in Ryukyuan and Ainu opens a host of interesting problems, unless it is accidental.” (id.). The second word is likely a bisyllable derived from epenthesis, while the former is clearly related to 3.1.20 **kuskus*. Ilokano *ag-k<an>uskús* ‘to rub the head with something rough’ shows this phonetic and semantic parallel with the above terms. The addition of the diminutive suffix pJ *-ra, that Vovin notes, shows *kusera* is not an Austronesian loan directly, instead more closely related to Japonic. Therefore, Hayato were probably insular Japonic peoples admixed with Ainu or Ainu-related peoples.

Vovin (2021) demonstrates how a rare spelling of *kumaso*, WOJ *Kumasə*, 熊襲, preserves pAN **Cau* ‘person, human being’, likely through Tagalog *taʔó* ‘mortal’, “with a stress on the last syllable. It is quite possible that WOJ *səo* (贈於, 贈於) represents an attempt to transcribe *səʔo*”. WOJ *-so/to* may have been an independent word before it became a bound morpheme, being replaced by WOJ *pito*, possibly a compound of pJ **pi* and pJ **cə* /*tsə*/ which was favored for its bisyllabism. “The Hayato of Ata was an ethnic group that brought tribute to the court [...] and had a role in the ‘great tasting’ (*dajōsai*) celebrated at the Yamato Court[...] Kojiki tells that Ninigi’s

oldest son Hoderi was the ancestor of the Hayato's Ata no kimi" (Domenig, 2023). Hoderi is the older of two brothers in a tale perhaps meant to illustrate Japonic mixed origins. The younger brother, Howori, was especially skilled at hunting with a bow, while Hoderi was skilled at fishing. Hoderi is eventually enslaved after a dispute between the two leads to Howori winning the favor of Watatsumi, the sea god. Howori was likely meant to represent peninsular Japonics that became the Yamato who kept the Hayato, descendants of Hoderi, as servants.

6 Conclusion

Agriculture and rice were of great importance according to the large amount of words relating to them, though only one that may dubitably relate to wet rice. This may reflect that wet rice was cultivated but not at the scale that it was after the pause in wet rice farming, which may have arisen from cooperation between peninsular Japonic peoples, Yamato, with insular Japonic peoples, the so-called Hayato and Kumaso peoples. This event may be preserved within the story of Howori making Hoderi subservient to him. Yamato Japanese then begins to displace the existing genetic landscape of the Japanese archipelago, creating a founder's effect that makes people with O1b2 appear to be the main carriers of proto-Japonic while O2a2-P201/O2a2b-N6 get(s) absorbed within the Koreanic genetic profile as Koreanic languages begin to dominate the peninsula and replace the remnant para-Japonic outside Japan. This migration forms the basis for proto-Kyūsyū-Ryūkyūan to develop from insular-Japonic while peninsular Japonic develops into proto-Japanese. Along with the Japanese branch, villages sustained by wet rice farms begin to spread across the peninsula, pushing Jōmon-related peoples further north and south in the archipelago.

In sum, the data collected suggests that Austronesian-speaking peoples, most likely those from the initial wave of the Austronesian expansion related to Amis and Batanic languages, arrived in the Japanese archipelago during the late Jōmon period. The language branch of these people was likely closely related to pMP and Amis people, and possibly spoken along the Chinese coast alongside the Austronesian Ancestor of Tai Kadai. Several shared cultural and archeological traits with Austronesians may be taken as evidence, despite an archaeological gap between them and Japonics. During the Yayoi period, a proto-Koreanic related language heavily absorbed the lexicon and grammar of pAN/pMP while retaining its own phonology and head-final agglutinative typology. The proto-Japonic family was likely a result of the demic diffusion of Koreanic related peoples admixing with Austronesian rice farmers, which can be observed genetically as well.

Out of the entire ONCOJ, several words do not appear to have reliable or any Koreanic nor Austronesian parallels, which leaves them open to be compared to other languages or to represent an original common adstratum. For example the words: WOJ *mimi* 'ear', *yuki* 'snow', *patwo* 'pidgeon', could all have different sources outside Japonic or all come from the same unattested language. Possible external comparisons or internal reconstructions for these kinds of words should be investigated, as they would help further elucidate Japonic pre-history.

Additionally, isoglosses in Japonic dialects should be collated to reconstruct the local linguistic histories of those respective regions.

As Orlandi (2020, p.287-288) points out “[A] mixed language generally involves a language with the grammar of another language x, but the lexicon of a language y. It must be distinguished, however, from those cases where we have registers of a language x, characterised by large numbers of loanwords from a second language y.” The degree to which the Koreanic and Austronesian language families contributed to the formation of proto-Japonic should continue to be examined further. As Vovin (2021) had said about Kupchik’s (2021) revolutionary paper, “a new subfield of Japonic historical linguistics is born,” to which I hope this is a worthy contribution.

Appendix

Abbreviations Used

Languages

pAN- proto-Austronesian

pMP- proto-Malayo-Polynesian

MP- Malayo Polynesian

pT- proto-Tai

pAA- proto-Austroasiatic

pJ- proto-Japonic

pR- proto-Ryukyuan

OR- Old Ryukyuan

OJ- Old Japanese

WOJ- Western Old Japanese

EOJ- Eastern Old Japanese

NJ- Modern Japanese

pK- proto-Koreanic

OK- Old Korean

MK- Middle Korean

LMK- Late Middle Korean

NK- Modern Korean

OC- Old Chinese

pIE- proto-Indo-European

Kog. - Koguryeo

Glosses

ADJZ - Adjectivalizer

ADVZ - Adverbializer

ALL - Allative

CONC - Conclusive

CONN - Connective

DAT - Dative

EP - Epenthetic/Echo Vowel

EMP - Emphatic Suffix

GEN - Genitive

ITER - Iterative

IRR - Irrealis

NTR - Intransitive Suffix

PROP - Property Suffix

RED - Reduplication

SUBJ - Subject

TRZ - Transitive Suffix

TR.OBJ - Transitive Object Nominalizer

VBZ - Verbalizer

Phonetic Transcription

Proto Austronesian

*C: voiceless alveolar affricate /t͡s/

*c: voiceless palatal affricate /t͡ɕ/

*q: uvular /q/

*z: voiced palatal affricate /d͡ʒ/

*D: voiced retroflex stop /ɖ/

*j: palatalized voiced velar stop /gʲ/

*S: voiceless alveolar fricative /s/

*N: palatalized alveolar lateral /ɳ/

*ñ: voiced palatal nasal /ɲ/

*ŋ: voiced velar nasal /ŋ/

*r: alveolar trill /r/

*R: uvular trill /ʀ/

Proto Japonic

*r: alveolar tap

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